



Text Entry Methods

**Mobile QWERTY Keyboard Best Practice
Final Research Report &
Physical Design Variables Recommendations**

10-16-2009

Design | User Experience & Strategic Marketing

Samsung Telecommunications of America

Human Factors Lab

Mobile Design Group

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Introduction

Synopsis of Design & Research

The QWERTY Best Practice design recommendations are built upon the analyses and results from the QWERTY Keyboard Best Practice Keyboard Research and Analysis as well as current knowledge and research of human perception, psychophysics, physical ergonomics and hand functions in relation to mobile QWERTY text entry:

QWERTY Keyboard Design Research:

1. A consumer interactive usability research study comprising of individualized usability evaluation of 11 QWERTY devices and post-evaluation focus groups discussion (3 groups total 27 subjects)
2. A consumer QWERTY usability performance study of 12 QWERTY devices (10 groups total 119 subjects)
3. Human factors and physical ergonomics knowledge and research of hand function and mobile devices.
4. Competitive insight from STA Design and User Experience.



Introduction

QWERTY devices used in various studies

A: Palm Pre



B: HTC Fuze



C: HTC G1



D: LG Xenon



E: Samsung Exclaim



H: Nokia N97



K: Samsung Impression*



M: Samsung Jack*



P: Blackberry Bold



S: Samsung Gravity



T: Samsung XO



W: LG Lotus



X : Samsung Reclaim



Y : Samsung Alias 2



Z: LG Rumor



Devices X, Y, Z were used in first interactive usability evaluation/focus group research were not used in performance study because of time limit of research sessions. Additional devices used to gain insight but did not participate in the full-scale performance study: Samsung Rogue, Samsung Magnet, Blackberry Tour.

Introduction

Synopsis of Recommendations

Variables	Lower bound	Recommended Range	Notes
Spacebar width	14mm	14mm – 21mm	Recommend to span 3 regular keys
Travel (0.001")	0.011"	0.014"	Best performing QWERTY devices
Gap Height	0 mm	0 mm – 1.0 mm	Lower is better
Gap Width	0 mm	0 mm – 1.2 mm	Lower is better
Center-to-Center Vertical Key Distance	4 mm	6 mm – 7 mm	4mm based human psychophysics
# of rows	3	4 - 5	5-row layout is more "design-fault tolerant "
Fmax (gf)	-	under 200gf	200gf from both primary & secondary research
Curvature	Moderate	Moderate to high	Moderate to high for Landscape QWERTY High for Portrait QWERTY
Tactile Response Slope	-	980-1100 gf/mm	Newly defined variable from ASTM standards

Introduction

Synopsis of Recommendations (Landscape)

Variables	Lower bound	Recommended Range	Notes
Spacebar width	14mm	>14mm	Recommend to span at least 2 regular keys
Travel (0.001")	0.011"	0.014"	Best performing QWERTY devices
Gap Height	0 mm	0 mm – 1.0 mm	Lower is better
Gap Width	0 mm	0 mm – 1.2 mm	Lower is better
Center-to-Center Vertical Key Distance	4 mm	6 mm – 7 mm	4mm based human psychophysics
# of rows	3	4 - 5	5-row layout is more "design-fault tolerant "
Fmax (gf)	-	under 200gf	200gf from both primary & secondary research
Curvature	Moderate	Moderate to high	Moderate to high for Landscape QWERTY
Tactile Response Slope	-	980-1100 gf/mm	Newly defined variable from ASTM standards
Number of Keys	Varies	Varies	Depending on number of rows and keyboard width
Key Height	6.5mm	>6.5mm	Diagonal > 8mm
Key Width	5.5mm	>7.5mm	Diagonal > 8mm

Introduction

Synopsis of Recommendations (Portrait)

Variables	Lower bound	Recommended Range	Notes
Spacebar width	14mm	14.8mm – 21mm	Based on Blackberry Tour & LG Lotus
Travel (0.001")	0.011"	0.014"	Best performing QWERTY devices
Gap Height	0 mm	0 mm – 1 mm	Curvilinear keyboard rows may need visual row separator to guide spatial location of keys
Gap Width	0 mm	0 mm – 1.2 mm	Lower is better to increase size of key
Center-to-Center Vertical Key Distance	4 mm	6 mm – 7 mm	4mm based human psychophysics
# of rows	4	4	4 rows standard design
Fmax (gf)	-	under 200gf	200gf from both primary & secondary research
Curvature	High	High	High for Portrait QWERTY
Tactile Response Slope	-	980-1100 gf/mm	Newly defined variable from ASTM standards
Number of Keys	35	35 - 37	35 keys recommended
Key Height	6mm	>6.5mm	Diagonal>7mm
Key Width	5mm	>5 mm	Diagonal>7mm

Research

Physical Design Variables

A comparative study of different analyzes were used to determine the final set of pertinent physical variables impacting the design.

QWERTY Keyboard Design Research:

1. Multiple Regression Studies - performance vs. physical variables + demographic variables); ease vs. physical variables + demographic variables.
2. Factor Analysis and Principal Component Analysis – identification of groups of related variables.
3. Partial Least Square (PLS) Regression (performance, ease and satisfaction vs. physical variables)
4. Linear Regression Analysis with separate observations by Portrait and Landscape-type QWERTY keyboard (performance, ease and satisfaction vs. physical variables)*



* Since only 12 devices (4 Portrait and 8 Landscape) were used in the study and the co-variations of numerous physical design variables, the separate linear regression models should be interpreted with extreme caution. Both STA UX and Strategic Marketing Research recommended that linear regression with separate observations by Portrait and Landscape should be used with caution as basis for recommendation. In case of non-compliant design and the variables meet the minimum lower bound from human perceptual and motor characteristics, usability study can be conducted to decide upon the acceptance of design.

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Research Results

Physical Design Variables

The baseline analysis used is a conservative PLS regression on 12 QWERTY devices used in the performance study with the following:

QWERTY Keyboard Design Research:

1. Dependent Variables: Speed, Accuracy, Ease of Typing and Satisfaction.
2. The following baseline physical design variables have been identified in decreasing order of importance
 - a.) Spacebar width
 - b.) Travel
 - c.) Gap Height
 - d.) Gap Width
 - e.) CCD Key Vertical
 - f.) # of rows
 - g.) FMax
 - h.) Curvature
 - i.) Tactile Slope

Physical Variables	Speed	Accuracy	Ease	Satisfaction	Rank
FMax(grams)	-0.027	0.101	-0.122	-0.107	7
Tallness(.001)					
Travel(.001)	-0.165	0.018	-0.282	-0.258	2
TactileSlope	0.046	0.142	-0.035	-0.025	9
TactileRatio			-0.155	-0.141	
Letter Key Width			0.141	0.13	
Letter Key Height					
CCD Key horizontal					
CCD Key vertical	-0.153	-0.11	-0.163	-0.155	5
Keyboard Width					
Keyboard Height					
Spacebar Width	0.266	0.16	0.308	0.29	1
Gap Width	-0.193	-0.164	-0.186	-0.178	4
Gap Height	-0.206	-0.174	-0.199	-0.191	3
Curvature	0.098	0.148	0.045	0.048	8
# of keys					
# of rows	0.131	0.098	0.138	0.131	6

Standardized coefficients of physical variables from PLS Regression (2 components)

* Selection of variables with VIP > 0.8.

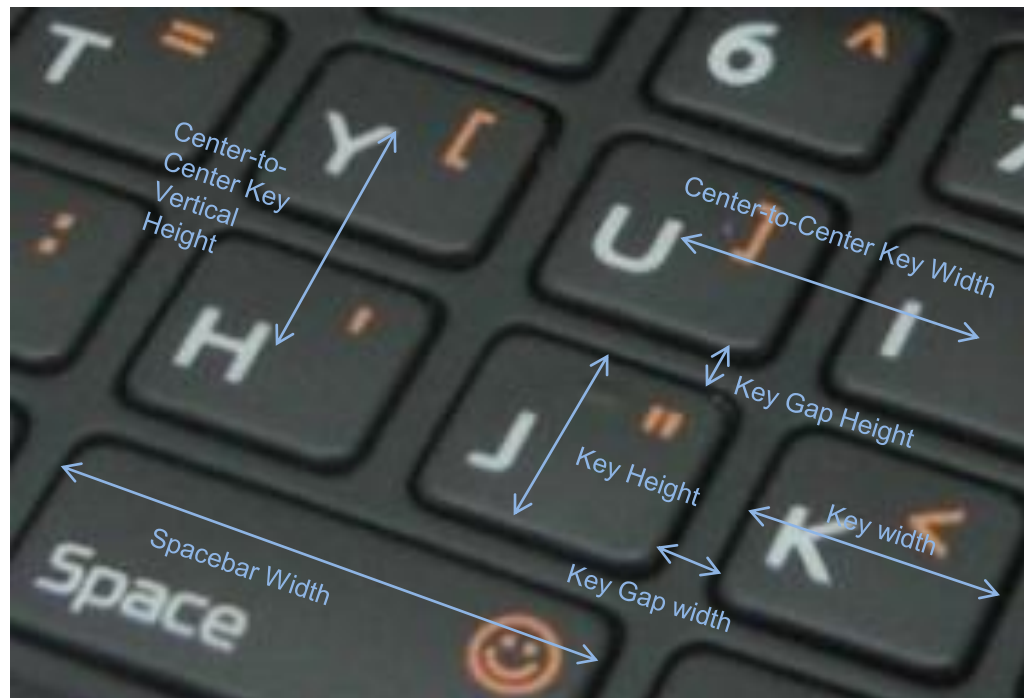
* Ranking with double weight on ease and satisfaction.

Research Results

Physical Variables Definitions

The following describes the definitions of the physical variables of QWERTY keyboard:

1. Spacebar width: see figure.
2. Travel: the downward displacement of a key from its normal height to actuating the button
3. Gap height: see figure
4. Gap width: see figure
5. CCD Key Vertical: see figure
6. # of rows: number of rows of keys containing alphabets, symbols, numbers and space, shift and Fn keys.
7. Fmax: the maximum force required to collapse the dome and actuate the key
8. Curvature: flat (0), moderate (1), high (2)
9. Tactile Response Slope: the ratio of the rate of force applied to displacement (as a measurement of "tactile response").



QWERTY Human Performance Study – Overview of Results

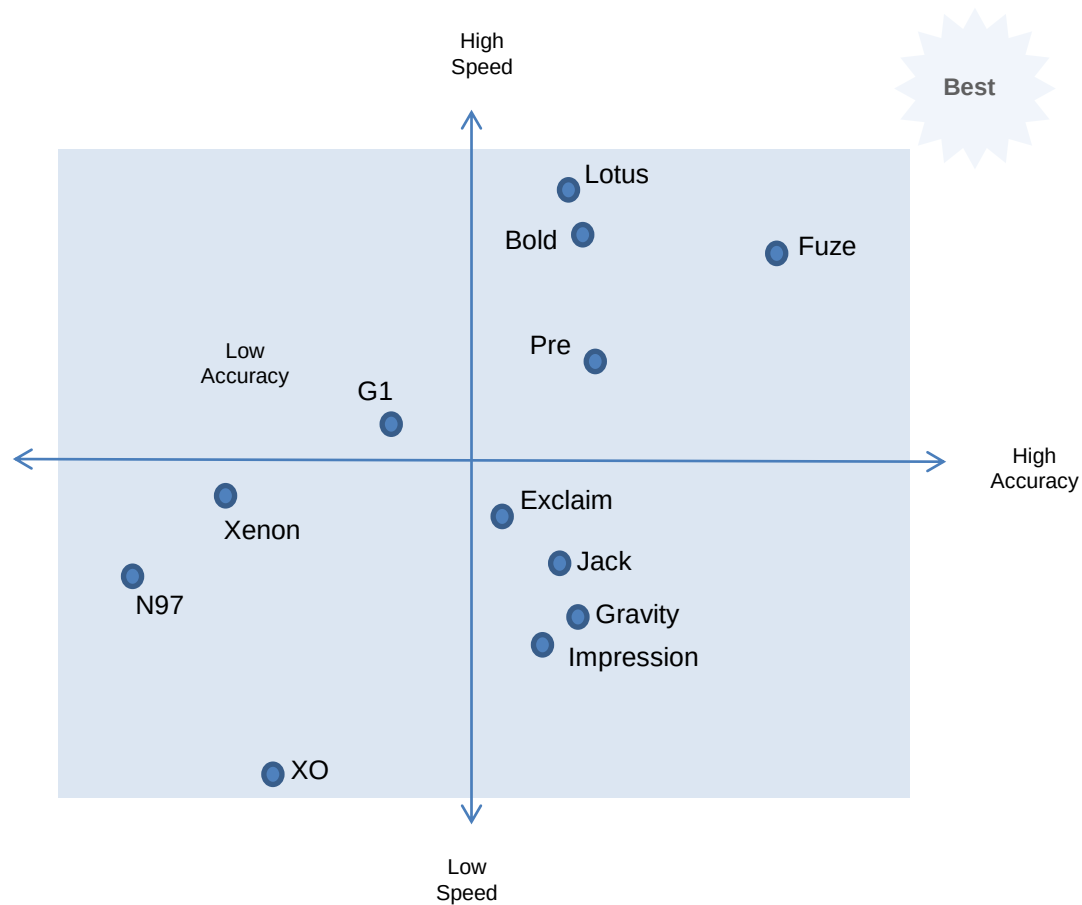
Speed / Accuracy vs. Satisfaction



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QWERTY Human Performance Study - Overview of Results

Speed vs. Accuracy



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QWERTY Design Style Guide

Human Factors Lab

Design Group Style guide



Portrait - Key Size/Shape Guideline

Key size: requirement of minimum key size. Key width > 5mm, Key height > 6.5mm

Key height should be larger than key width.

Key gap: minimize gap between the keys to increase key size. Recommendation of no-gap shape.

Curvature: curvature to enhance tactile feedback differentiation between keys

eg. bottom-side convex & top-side (left or right) curved or cut surface

Space bar: requirement of minimum key size. Key width > 14mm

※ Click force
-Minimize max force
: $F_{max} \leq 200gf$

SG_GL_KM001

QWERTY > 세로

Curvature

(example 1)



(example 2)



Design Style Guide based on joint data analysis and interpretations from this report.

QWERTY Design Style Guide

Human Factors Lab

Design Group Style guide



Landscape - Key Size/Shape Guideline

Key size: requirement of minimum key size. Key width should be larger than key height.

Key gap: minimize gap between the keys to increase key size.

> with Gap: key width > 7.5mm, key height > 6.5mm

> with no Gap: key width > 7.0mm, key height > 5.5mm (Gap: width max 0.5mm, height max 1.0mm)

Curvature: a little curvature to enhance tactile feedback differentiation between keys.

(Flat shape is allowed with enough large key size and gap existence)

Space bar: requirement of minimum key size. Key width > 14mm

※ Click force
-Minimize max force
: Fmax ≤ 200gf

SG_GL_KM002

QWERTY > 가로

Curvature

(example 1)



(example 2)



Design Style Guide based on joint data analysis and interpretations from this report.

Physical variables driving QWERTY Performance & Satisfaction

Spacebar

Analysis & Recommendations

Spacebar width has been identified as the most important physical design variables affecting performance as well as users' satisfaction (from both quantitative and qualitative analyses).

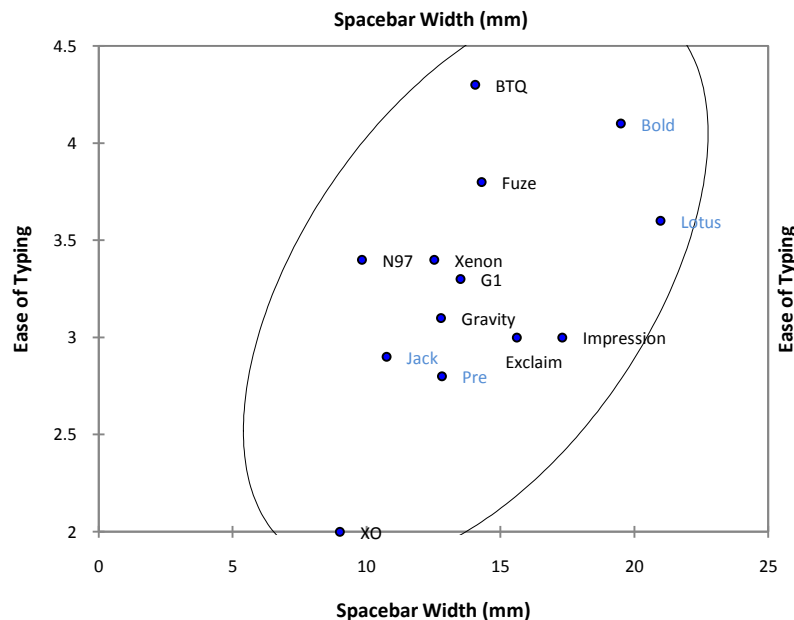
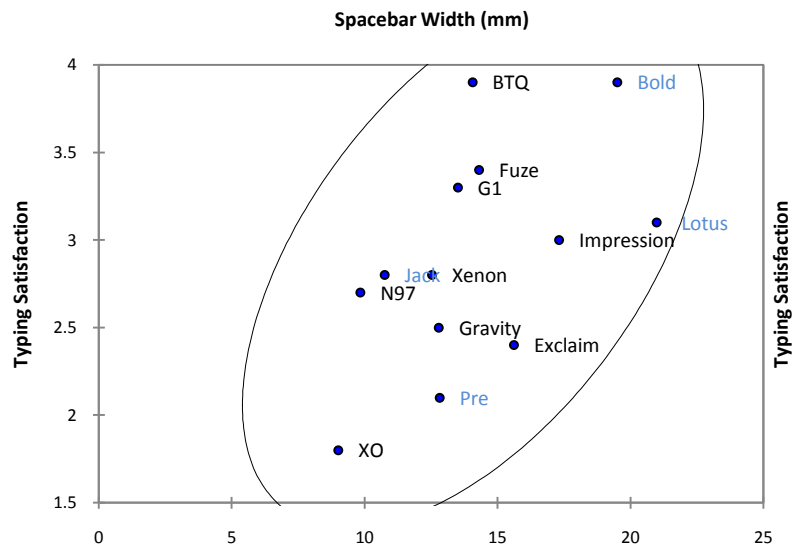
1. Spacebar must be substantially wider than regular QWERTY keys.
2. Using the best performing devices as best practice criteria, the *minimum* width of spacebar width (Portrait or Landscape) is **14mm**
3. Spacebar Location must be centered on the QWERTY keyboard of the device. (qualitative analysis of LG Env2 and Nokia N97)
4. The line of symmetry of the device should pass through the center of spacebar.

Landscape

>14mm Spans at least 2 regular keys

Portrait

>14mm Spans 3 regular keys



Spacebar

Analysis and Recommendation

5. Spacebar is located at the center line of the device to provide a spatial reference anchor optimized for typing experience.
6. The spacebar is oversized to reduce the attention effort (and hence the time) to locate the “target” when typing (Fitts’ Law). The width should span at least 2 regular physical keys. STA Product Design and User Experience recommends the spacebar spans the width of approximately 3 physical keys for perceived usability and symmetry.
7. The background color should be subtle but distinct from other regular keys if the width spans only 2 keys.
8. Pressing the spacebar twice in succession will add a period, move 2 spaces and automatically capitalize the next letter typed.
9. It should provide consistently crisp tactile feedback when pressed with either left or right thumb at left, center or right part of the key.

Center line split

- Symmetrically mirrored on both sides.
- You want your brain to automatically know which key to use.
- As finishing one stroke, you are starting the next.

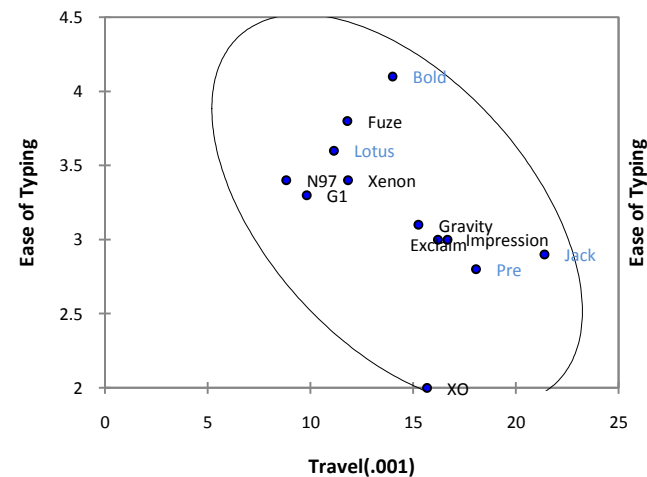
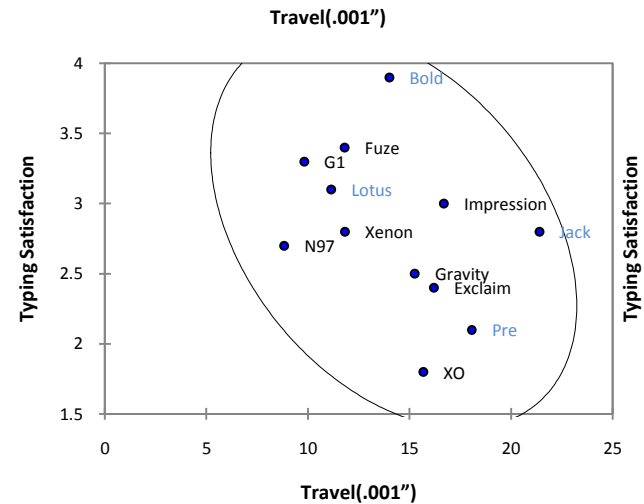


Travel

Analysis and Recommendations

Travel was found to have a relatively large negative impact on satisfaction and ease of typing. It also impacted negatively on speed of text entry.

1. The explanation was relatively straightforward as the *work* done is *force x displacement*. More effort will be needed to activate buttons with larger travel distance.
2. Travel forms part of the measurement of tactile response slope (the rate of change of applied force with respect to displacement) which was found to be related to accuracy of text input (N97 and G1 had relatively low travel and activation force and the accuracy of text input was directionally under average).
3. With the above analysis, the recommended range of travel should not be too high to impact on satisfaction and ease and should not too low for inaccurate typing. The range recommended is between 0.011" and 0.014" (between that of Lotus and Bold).



Gap Height

Analysis and Recommendations

Gap height is negatively related to speed, accuracy, ease and satisfaction. The implications for design are:

1. Gap heights should be minimized so that the size of a physical key can be maximized. This follows from Fitts' Law:

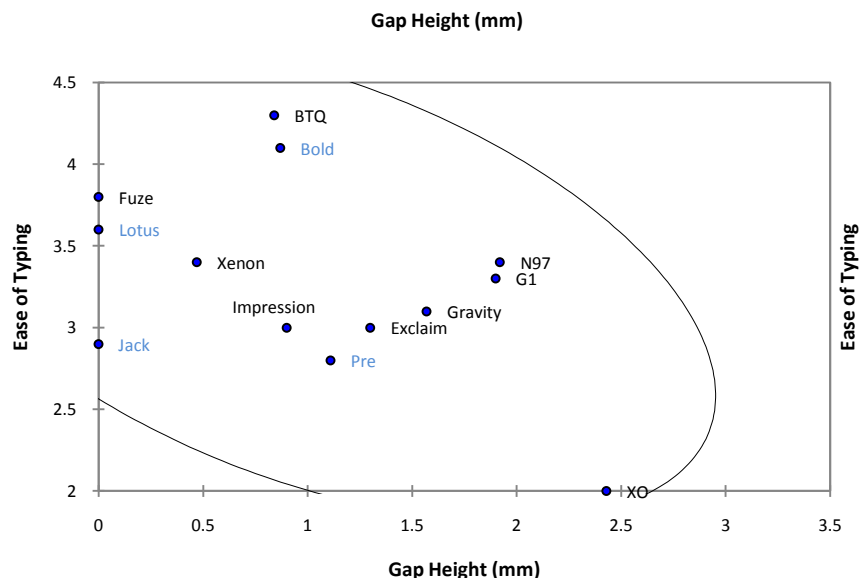
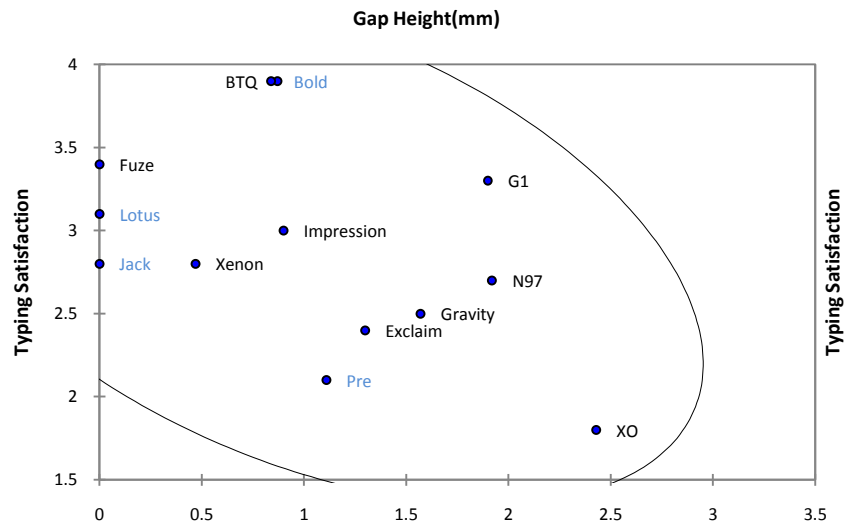
$$T = a + b \log_2 \left(1 + \frac{D}{W} \right)$$

T – time to acquire target

D – distance from starting point to target

W – width of target (the diagonal size of button is used which depends on both key width and height)

2. Although BTQ has high typing satisfaction, performance (speed and accuracy) was directionally negative compared to Bold, Fuze and Lotus. This might have been part of the reason of lowered performance (BTQ was included as additional data point collected in separate study and scaled accordingly).



Gap Height

Analysis and Recommendations

The best performing devices are Bold, Fuze, Lotus and BTQ. Both BTQ and Bold have moderate gap heights.

1. For landscape type QWERTY devices, physical keys can be designed with or without vertical gaps in between without loss of perceived usability.
2. From the theoretical perspective (Fitts' Law) and the empirical performance results of the studies, it is recommended that the vertical gap should be kept small to maximize the height of the physical keys.
3. For portrait type QWERTY devices, the gap height of the keys should be kept close to 0 to maximize the height of physical keys. Exceptions may be observed in curvilinear keyboard arrangements where visual separators /references and vertical gap is added to guide eye-hand coordination in locating the appropriate key rows.



Curvilinear row separators add vertical gaps between the rows but serve as visual reference to the spatial locations of the rows of keys (the rows are arranged with increasing curvature towards the screen).

Landscape

0 – 1.0mm (lower is better)

Portrait

0 mm (exception for curvilinear key rows)

Gap Width

Analysis and Recommendations

Gap width was negatively related to both performance and satisfaction. Similar analysis to gap height applied.

- Gap width should be minimized so that the diagonal size of the target keys can be maximized. This follows from Fitts' Law:

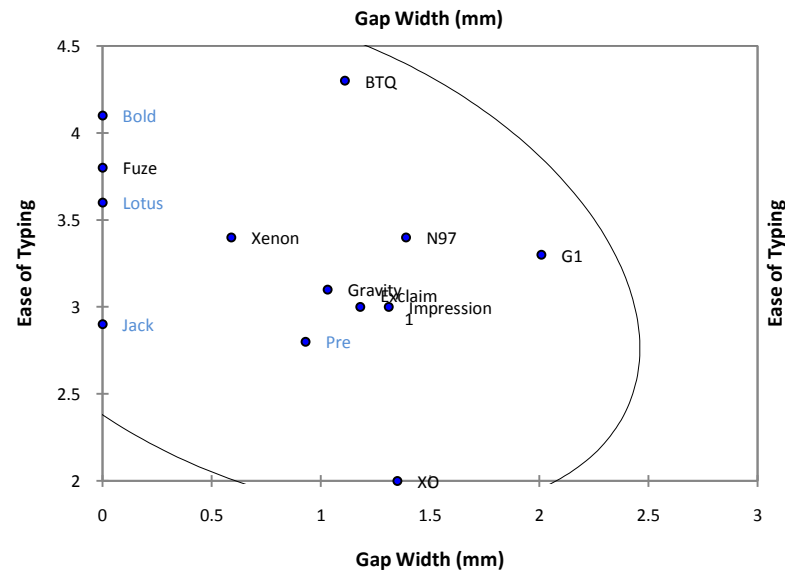
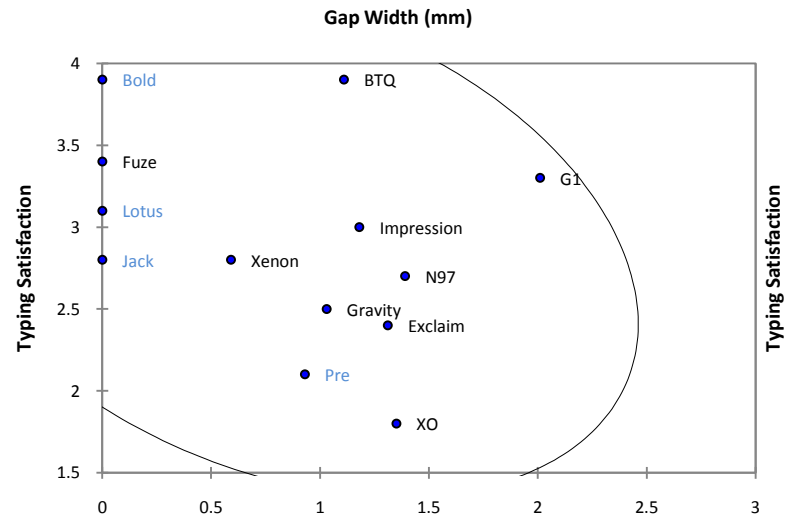
$$T = a + b \log_2 \left(1 + \frac{D}{W} \right)$$

T – time to acquire target

D – distance from starting point to target

W – width of target (diagonal size of button)

- Although BTQ has high typing satisfaction, performance (both speed and accuracy) was directionally less when compared to Bold, Fuze and Lotus. (Note that the differences in performance were not significant).



Gap Width

Analysis and Recommendations

The common devices are Bold, Fuze, Lotus and BTQ, 3 of which have gap widths zero or close to zero except BTQ. Note that Fuze and BTQ are landscape type.

1. For landscape type QWERTY devices, physical keys can be designed with horizontal gaps in between without loss of perceived usability. The maximum recommended gap width for regular (non-staggered) keys is 1.2mm.
2. From the theoretical perspective (Fitts' Law) and the empirical performance results of the studies, it is recommended that the horizontal gap should be kept small to maximize the width of the physical keys, which in turn contributes to the diagonal "size" of the key.
3. For portrait type QWERTY devices, the gap width of the keys should be kept as close to 0 as possible because of the narrower width of the device.



Although the gaps are very small (0.2mm) between keys, the surface contours of the keys make it easy to actuate key either using the device with one-hand or two-hand.

Landscape

0 – 1.2mm (lower is better)

Portrait

0 mm (or close to 0)

C-C Key Vertical Distance

Analysis and Recommendations

Center-to-Center key vertical distance was negatively related to both performance and satisfaction. However, XO was a more extreme outlier and the results should be re-interpreted with caution as smaller Center-to-Center distance might have the opposite effect. The shape of the scatter plot (excluding the extreme point XO) did not seem to show much linear relationship within the range of devices observed.

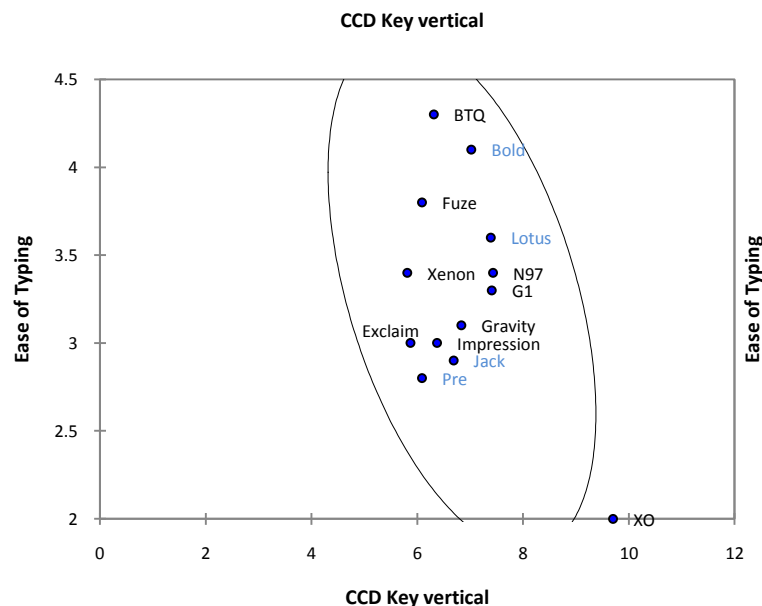
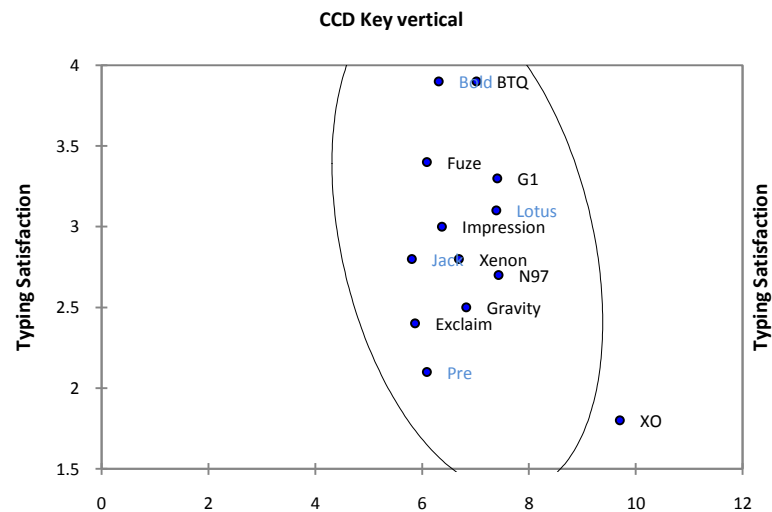
With the consideration of the human perceptual system and neurophysiology of fingertips and exemplary QWERTY devices in the market, the recommended range for Center-to-Center Vertical Key distance is 6 mm to 7mm. The lower bound is 4mm which is the two point threshold of fingertip (see analysis in later section)

Landscape

6 – 7 mm

Portrait

6 – 7 mm



Number of Rows

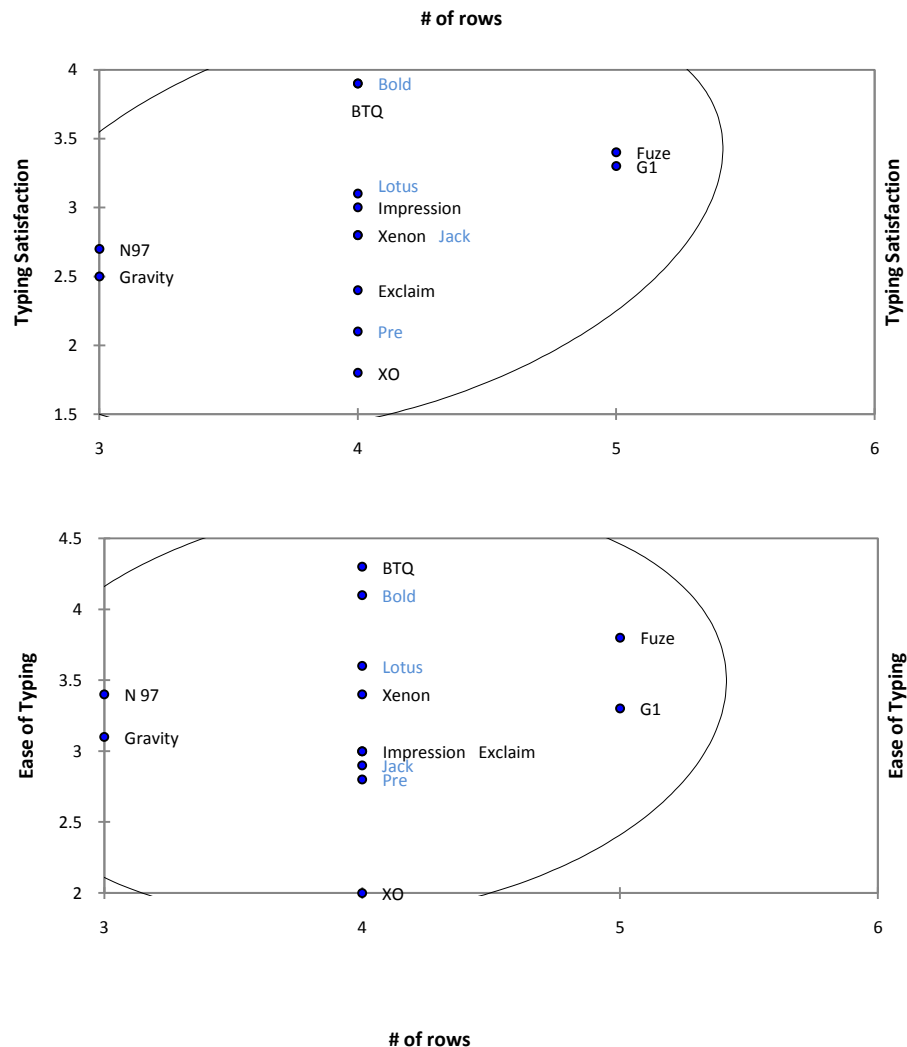
Analysis and Recommendations

The number of rows affect both typing performance, ease and satisfaction.

1. For the best QWERTY typing experience, 4 and 5 rows are recommended.
2. 5-row layout s are “fault-tolerant” and are relatively safe to use even with some known design defects or issues (e.g., Fuze has problematic symbols key-map layout and G1 has a staggered keyboard with flat keys with part of chassis obstructing right hand typing).
3. 3-row QWERTY does not promote typing satisfaction though the ease of typing does not suffer as much. Directionally, 3-row QWERTY has negative impact on the speed of text entry.

Landscape
3 to 5 rows

Portrait
4 rows

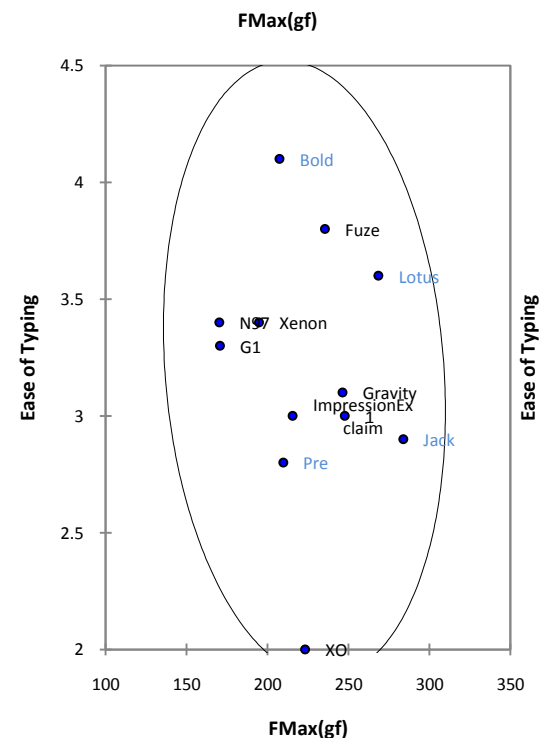
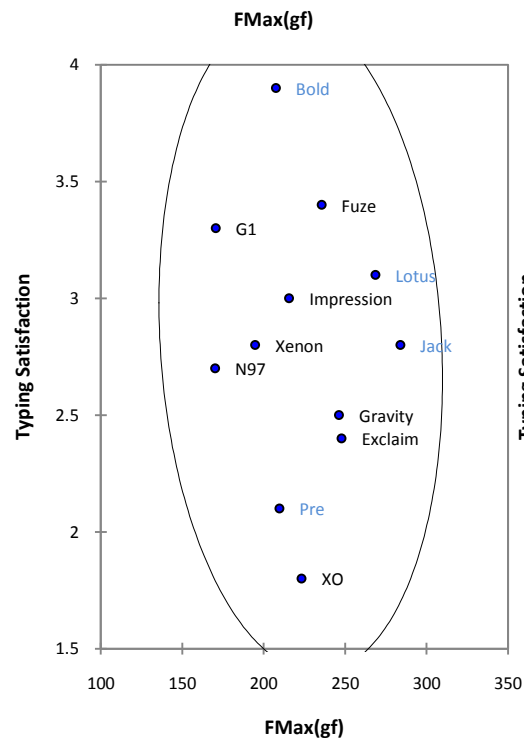


Trip Force

Analysis and Recommendations

Trip force is related positively to the accuracy but negatively to speed of text entry as well as satisfaction and ease of typing. This can be explained by the work done = force x displacement and less effort will be required if the force of activation is low.

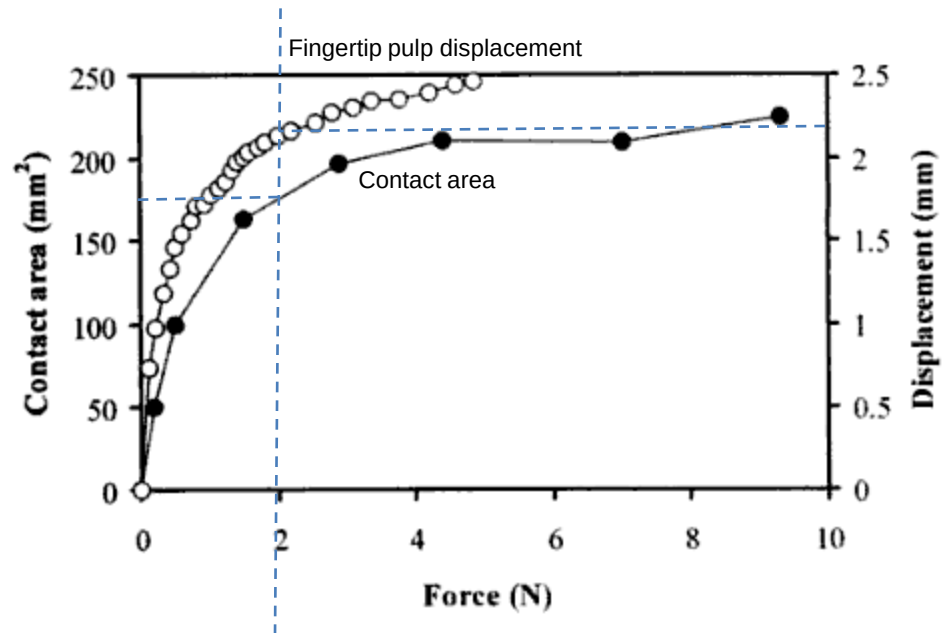
1. A minimum amount of force will be required to provide the necessary tactile feedback for accurate text input.
2. The trip forces of the devices used in the study did not provide enough variations to gauge the impact of trip forces lower than 170gf.
3. Comparing the best performing QWERTY devices Bold, Fuse and Lotus, a decreasing trend of satisfaction and ease of typing can be observed. The approximate trip force of Bold is used as the benchmark for reference design supported by existing knowledge about human fingers characteristics.



Trip Force

Analysis & Recommendations

1. The *maximum* trip force should be about 200gf (1.96N).
2. The mechanics of fingertips for tactile sensing and feedback are closely related to the contact area and the displacement of fingertip pulp. The fingertip functions effectively as a tactile sensor at low forces. Small changes of force can result in considerable enhancement of tactile mechanoreceptor activity due to increase in contact area (Serina,1997)
3. The relationship is nonlinear . At low forces, the fingertip contact areas increases sharply with increasing normal force. At 2N, contact area is approximately 80% of that at 5N.

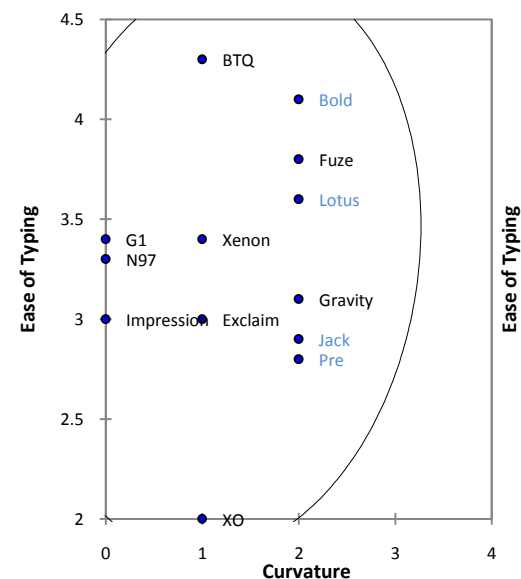
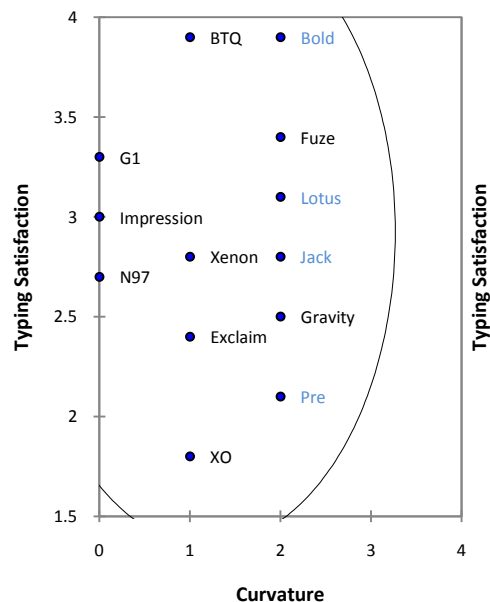


Curvature

Analysis and Recommendations

Curvature was estimated by the experimenters and was given discrete values of 0 (flat), 1 (slightly curved surface) and 2 (highly curved surface). Accuracy was found to be positively related to curvature.

1. For both portrait and landscape iconic premium QWERTY devices, custom design and development of the micro-geometry of the keys is recommended. This would address the complex interaction effects between curvature and various aspects of physical characteristics of the keys.
2. If custom design and development is not possible, keys with high curvature are recommended for Portrait type QWERTY devices.
3. For landscape type QWERTY devices, keys of at least moderate curvature are recommended for tactile sensing and feedback.



Landscape

Flat to Moderately Curved (or Micro-Geometry based design)

Portrait

Highly Curved or Micro-Geometry based design

Tactile Slope

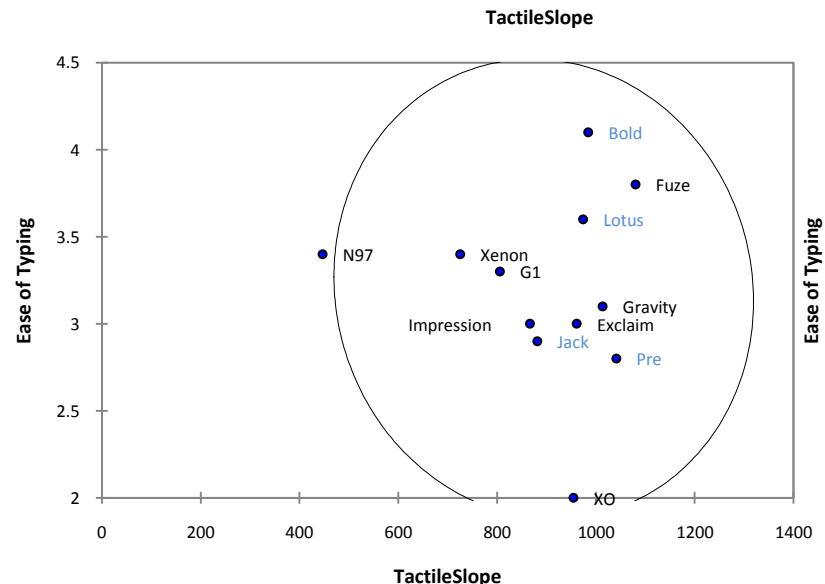
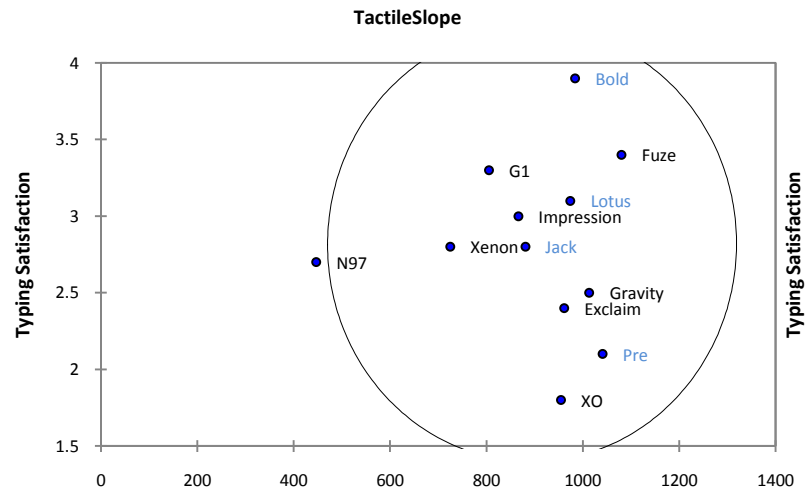
Analysis and Recommendations

The tactile response slope is a new physical parameter for tactile response defined by ASTM International F 2592-09 (Standard Test Method for Measuring the Force Displacement of a Membrane Switch, 2009). The tactile ratio used in the past has been deprecated.

The tactile response slope is defined as the rate of change of applied force with respect to displacement as measured by T_{fmax} and T_{fmin} where T_{fmax} is the displacement at F_{Max} and T_{fmin} is the displacement at F_{min} (see standard for detailed definition).

The top performing QWERTY devices (Bold, Fuze and Lotus) have tactile response slopes from around 980 g/mm to 1100 g/mm (recommended range).

The same recommendation applies to both portrait and landscape QWERTY devices.



Less important physical variables
driving Performance & User
Satisfaction

Key Height

Analysis and Recommendations

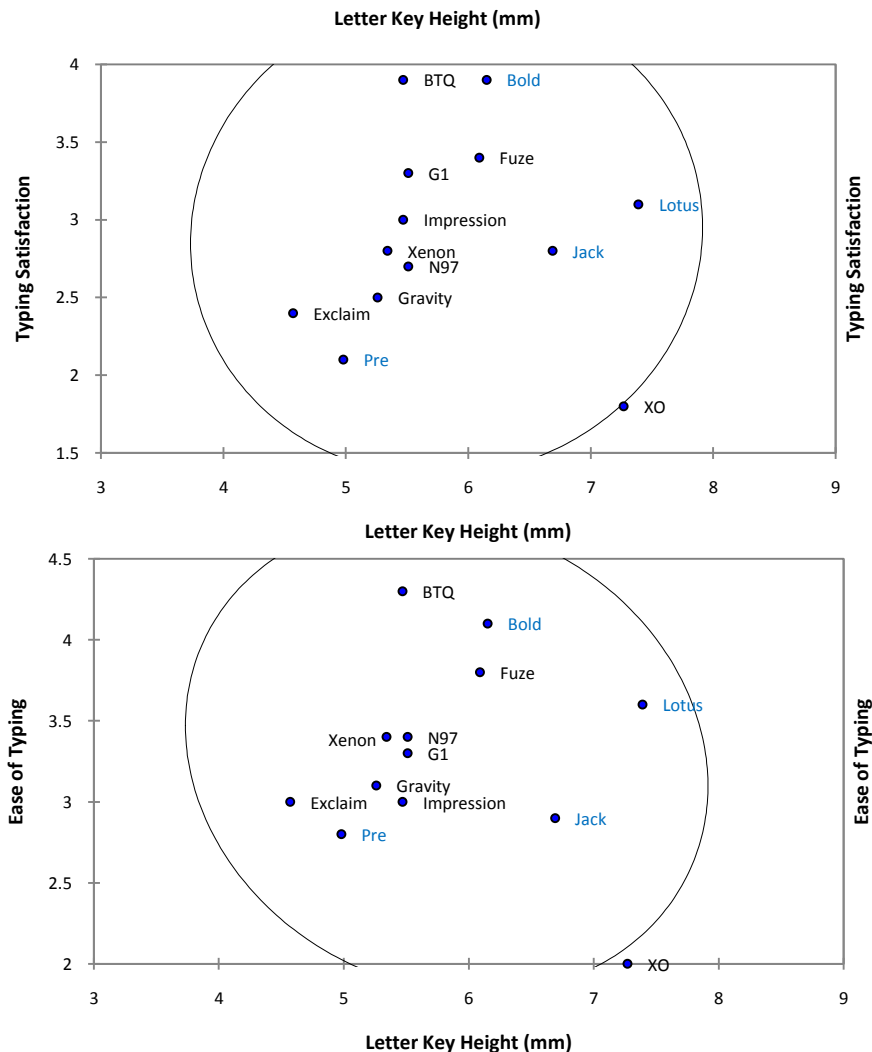
Key height is not identified as an important physical variable in affecting performance and satisfaction.

From a visual inspection of the scatter plot of the QWERTY devices tested, the range of letter key heights of best performing devices is between 5.4 mm and 6.2 mm.

As expected, keys with smaller heights are harder to locate and acquire as target (Fitts' Law). The same recommendation applies for both portrait and landscape type QWERTY.

Landscape
>6.5mm

Portrait
> 5.5mm



Key Width

Analysis and Recommendations

Key width is not identified as an important physical variable in affecting performance and satisfaction.

From a visual inspection of the scatter plot of the QWERTY devices tested, the range of letter key widths of best performing devices is between 5.6 mm (Portrait) and 7.2 mm (Landscape)

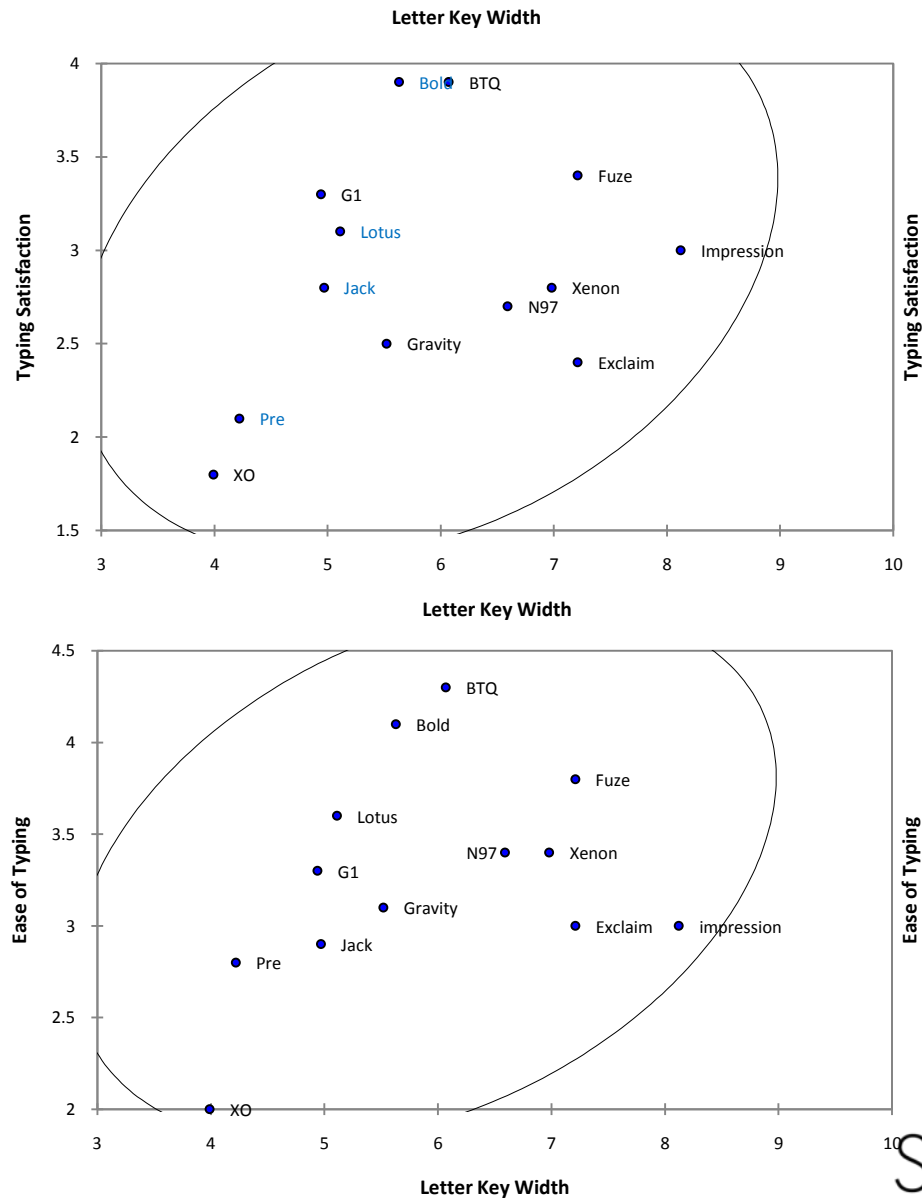
As expected, keys with smaller widths are harder to locate and acquire as target (Fitts' Law). The recommendation is based on the keyboard width of a QWERTY device is partially driven by whether the QWERTY orientation is Portrait or Landscape.

Landscape

>6.5 mm

Portrait

> 5mm



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C-C Key Horizontal Distance

Analysis and Recommendations

The Center-to-Center key horizontal distance is not identified as an important physical variable in affecting performance and satisfaction.

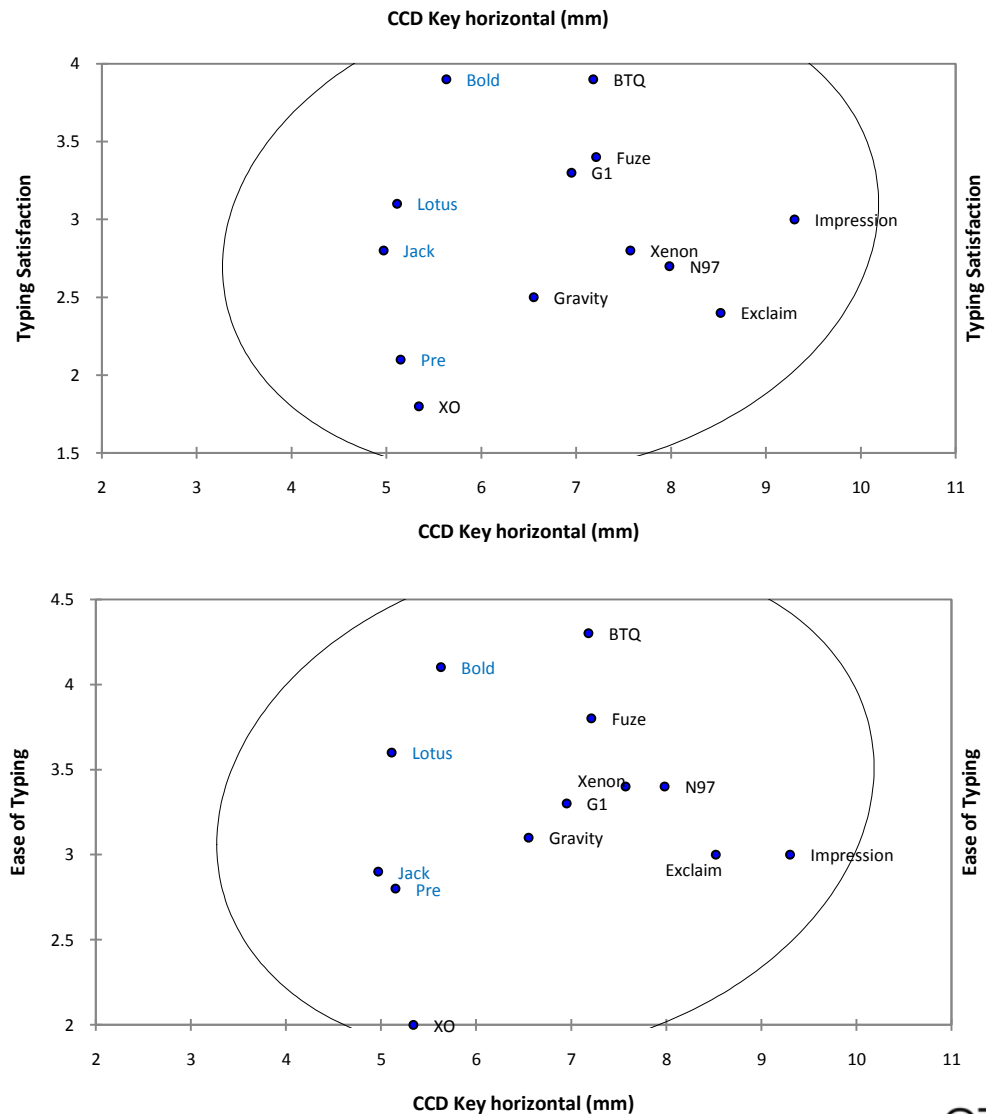
From a visual inspection of the scatter plot of the QWERTY devices tested, the portrait QWERTY devices tend to cluster around a narrow area of separation (between 5mm and 5.7mm). The landscape QWERTY devices have a wider spread and the distance is larger, between 6.55mm to 9.3mm excluding the outlier XO. This can be explained from the wider form factor of Landscape QWERTY design.

The recommended range for Portrait devices is 5mm to 5.7mm.

The recommended range for Landscape QWERTY is 6.55mm to 8mm. Devices with horizontal distance wider than 8mm may be too wide for the hand to move between keys efficiently and effectively.

Landscape
6.5 to 8 mm

Portrait
5 to 5.7 mm



Number of Keys

Analysis and Recommendations

The number of keys of Bold, Lotus and Pre is the same (35). Given the width constraints of Portrait type devices, it is recommended that the number of keys should be kept at around 35 and no more than 37.

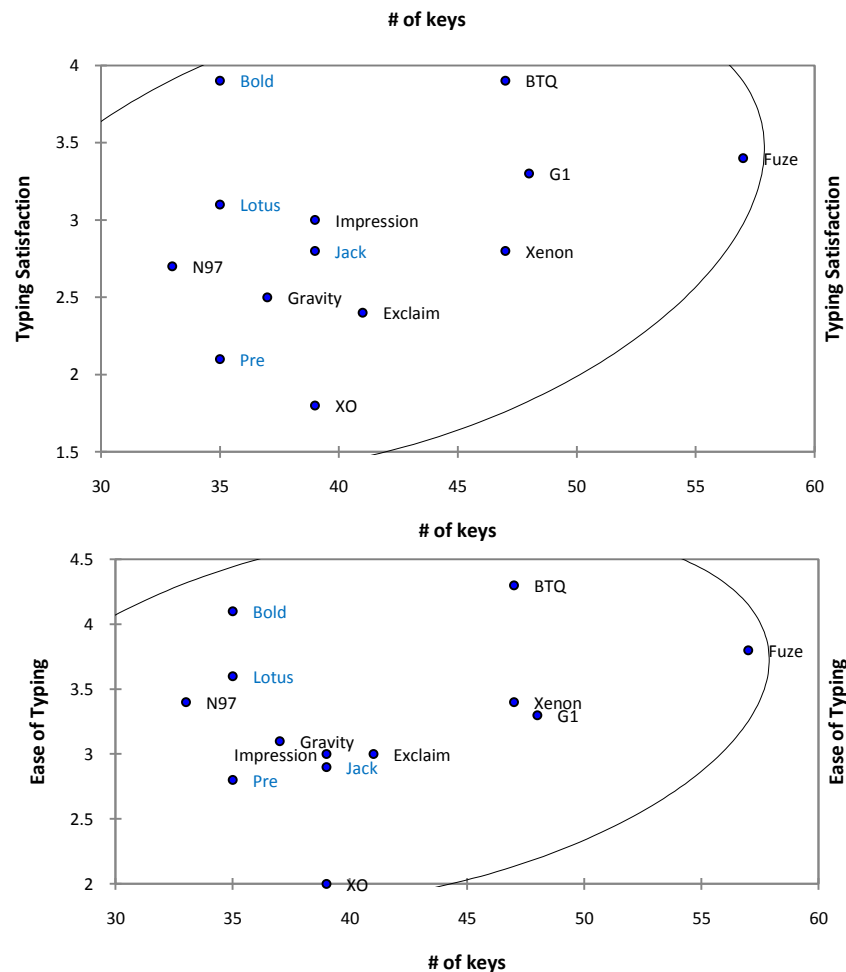
The number of keys for Landscape style devices depends on the number of rows and the width of the device. The number keys per row can vary from 10 to 13. Thus, the total number of keys will vary according to physical configurations and constraints of the device. Particular attention should be given to possible usability problems of the keys at the lower left and lower right corner.

Landscape

33 to 57

Portrait

35 to 37



After-thoughts and Input driving Design Excellence

Research & Analysis

Typing Model – Mobile QWERTY

In the two-thumb typing model adopted, the mobile QWERTY keyboard is designed around a center line split with space key at the center as anchor. The symmetry will support both left and right thumb interactions with the keyboard naturally.

The spacebar is activated by the alternate thumb to that used for the last letter in a word. The design driven by the “alternate” thumb policy of using spacebar, which is optimal (MacKenzie, 2002) . It was found that keying time is less when the preceding key is pressed by a finger on the opposite hand for 2-handed touch typing.

The design of the keyboard and the micro-geometry of the keys should be optimized for the differential direction of the respective thumb interactions. The keys can be constructed in a “sweeping direction” with customized ridges on the keys designed for tactile sensing and feedback.



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Research & Analysis

Typing Model – Mobile QWERTY

The mobile QWERTY keyboard, especially for Portrait type devices, will need to support both two-hand and one-hand operation. This is an important consideration in mobile usage situations. Ideally, landscape type keyboard should also be designed with the support of one-hand usage as a secondary goal.

The support for both one-hand and two-hand operations require the keys to be designed in such a way that they can be actuated easily with either thumb. The exemplary design for portrait QWERTY is the Blackberry Bold and Tour.



The narrow form factor of Portrait keyboard helps to support one-hand operation.



The wider Landscape keyboard and the weight distribution of the device makes it difficult to use the extreme corner keys.



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Research & Analysis

Typing Model – Mobile QWERTY

From the dual thumb typing model, the direction of the thumb hitting most of the keys lie between 0 and +/- 90 degrees to the horizontal axis of the keys.

A rough estimate of the angle* is 45 degrees. Thus the diagonal size of each key can serve as a rough guide the “width” for the thumb tip to land. A better approximation of the “thumb guide” can be seen in the micro-geometry of Blackberry keys.

The number of keys should be kept low, especially for portrait QWERTY because of the limited area of the keyboard. Increasing the number of keys (like many of our previous models) for more features is likely to have negative impact on typing experience as it also increases the attention effort to locate the appropriate keys and the smaller size of the keys will make typing more difficult and negatively impact on speed and accuracy. This observation may also extend to landscape type QWERTY if the keys increases are not related to improve text entry efficiency (e.g., dedicated punctuation keys).



An unnatural position of the thumbs for typing. Many QWERTY devices in the market seem to have been designed to support this model of interaction as the geometry of each key is symmetrical around the top of the key. Example (Blackberry Bold) shown here is an exception.



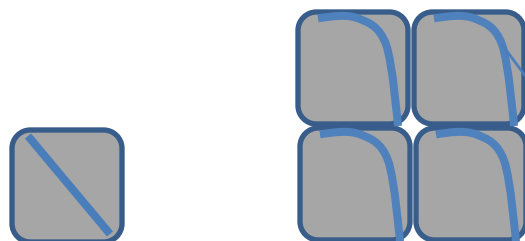
Blackberry Bold has an exemplary best-in-class QWERTY keyboard optimized for 2-thumb typing and supporting one-hand (thumb) typing reasonably well.

Research & Analysis

Typing Model – Mobile QWERTY

Maximizing the width of thumb contact area and tactile sensing of the button is one of the primary design goal to optimize the typing experience. Blackberry QWERTY devices like Bold or Tour, for instance, use a “sweeping” curve on each key to create a ridge with orientation optimized for the appropriate thumbs.

This is the most effective method observed among all the different devices as this can keep the keys small (5mm horizontal width in Tour) but the overall typing experience best in the Portrait devices. There are still rooms for improvement with this type of approach as physical design characteristics of the keys may be tweaked to enhance accuracy and usability.



Optimizing the thumb guide area and length between left and right thumb interactions (example above shows a key left of the device).

Left Thumb Guide maximizes the area of contact by providing a ridge with different slopes and enclosed areas on either side. The cusp on the ridge provides a natural focal point for the right thumb to actuate the key for one hand (right hand) but accommodate the natural shape of the left thumb in 2-hand usage.



Key Micro-Geometry

Analysis and Recommendations

The detailed design of Blackberry Bold or Tour should be studied more thoroughly with three dimensional modeling to gain more insight and knowledge of the possible configuration variations of each key or rows of keys.

The current Bold or Tour design can likely be tweaked to improve typing accuracy and speed from a detailed understanding of the micro-geometry of each key and the relationship of the keys to each other (beyond the observation that they are likely to be mirror image along the center split and the keys are sloping outwards slightly).



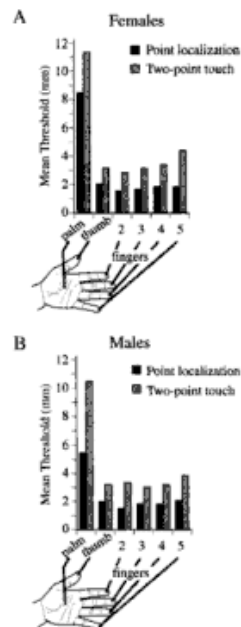
Research & Analysis

Human Psychophysics

The two-point touch threshold of fingertips is between 2mm and 4mm (Weinstein 1968). Thus, the minimum “center-to-center” distance between 2 buttons must be at least 4mm assuming the buttons have very distinct and pointed “peaks”. Since the crest of each button is not a needle, the practical minimum distance depends on the changes of the gradient between the adjacent peaks defined by the buttons.

The recommended minimum length of the diagonal line of the key is 7mm with the optimal size to be around 10mm (a better curve to use is a “sweeping curve” across the between the diagonal and the lengths formed by the width and height of the key (see previous page).

The recommended length ranges of the “thumb guide” are extrapolated from touchscreen button size study (Karlson, 2007; Katre, 2008)



Research

Secondary Research

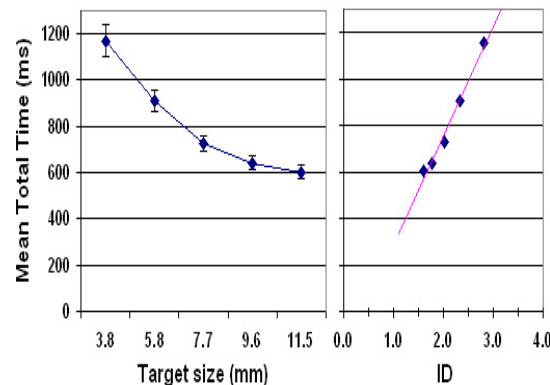
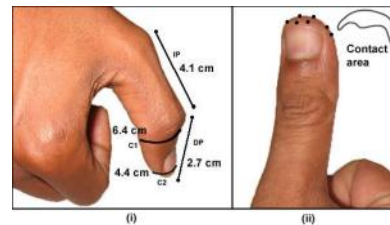
The following studies, together with the primary research and measurements of physical characteristics of the best performing devices (Blackberry Bold and Tour) is used for the recommendations of button sizes and the micro-geometry of the asymmetrical button layout.

Karlson, A. K. (2007) Interface and Interaction Design for One-handed Mobile Computing. Ph.D dissertation.

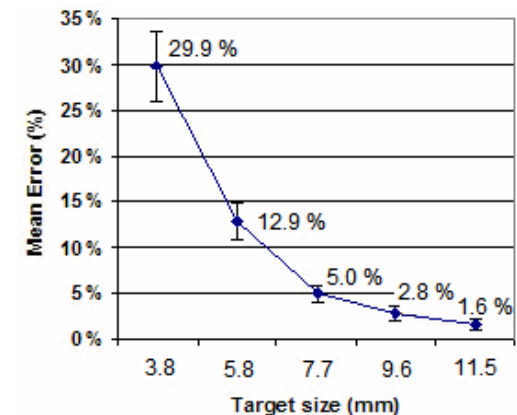
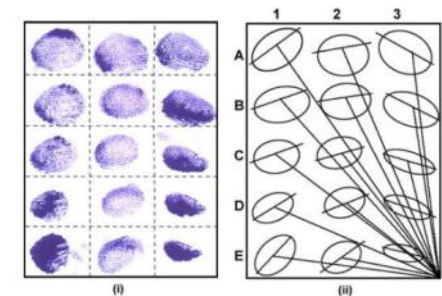
Katre, D. (2008) One-handed thumb use on smart phones by semi-literate and illiterate users in India: A usability report with design improvements for precision and ease, Workshop on Cultural Usability and Human Work Interaction Design, NordiCHI Conference, Lund, Sweden, 2008.

Serina, E.R. Mote C.D. & Rempel (1997). Force response of the fingertip pulp to repeated compression: Effects of loading rate, loading angle and anthropometry. Journal of Biomechanics 30, 1035-1040.

Weinstein S. (1968). Intensive and extensive aspects of tactile sensitivity as a function of body part, sex, and laterality. In D.R. Kenshalo (ed.). The Skin Senses.



Target Size Study: The mean task time between the release of the start button and release of the target 'x' for each target size in the discrete target phase (left), and the relationship between movement time and task index of difficulty (right).



Target Size Study: The mean error rate for each target size

QWERTY Key-Map Layout

Key Positions

Summary:

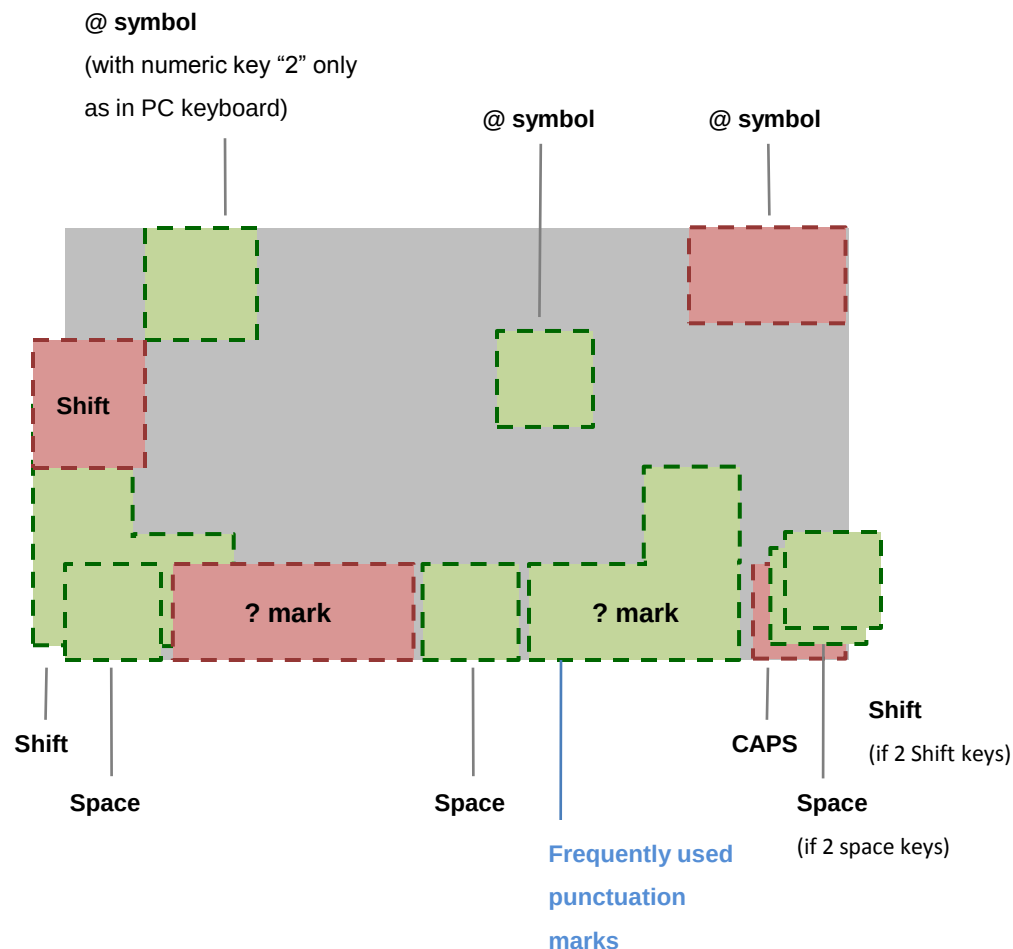
The recommended placements of different function and punctuation keys are depicted based on the analysis and study from the respondents' behavioral responses. Data from a number of different usability studies also corroborated with the recommendations.

This provides a device-independent overview of the qwerty elements that were targeted during this test.

Although other frequently used punctuation marks were not tested explicitly, the recommended spatial location was shown (in blue).

This research can be built upon by:

- Follow-up testing of refined prototypes
- Similar testing of alternative targets (i.e. punctuation, symbols, shortcuts, etc.)



effective/preferred problematic

Grouping of Symbols & Numbers

Portrait Devices

The recommended placements of different function and punctuation keys are depicted based on recommendations from STA Industrial Design.

The recommendations corroborate with the findings from both primary and secondary research.

The keys should be made from matte non-glossy non-slippery material .

Grouping - Symbols/Punctuation

- Always group symbols to the right side.
- Use your left thumb for alt. Right thumb for symbol.
- Build consistency in where you find these.



Grouping - Numbers/Phone Pad

- Keys grouped in cluster to the left.
- High contrast text for easy use.

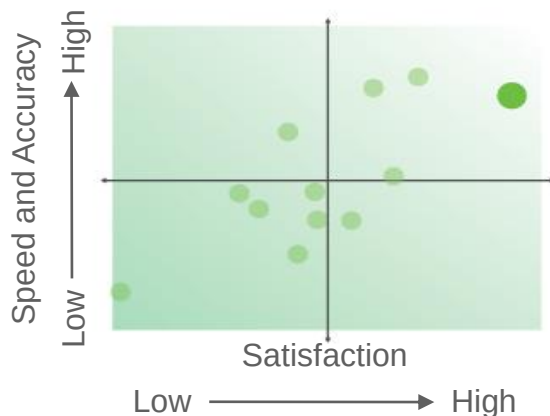


QWERTY Device Examples

Physical QWERTY

Phone P (Blackberry Bold)

- Rated highest in overall user satisfaction
- Performed well in terms of typing speed and accuracy.
- Special sweeping key curvature helped users distinguish among keys. Space on center line of symmetry
- Characteristics:
 - Portrait layout
 - Average size keys with special micro-geometry and sweeping shape.
 - Wide spacebar
 - Little to no key gap



The sloping geometry of the keys (defined by the line of symmetry into 2 different direction with regular changes in slopes across the keys) with the curving ridge enhance kinesthetic feedback and usability of the QWERTY.

The changing geometry of adjacent keys provides the distinctive tactile feedback.

Consistent interaction design for punctuations and symbols.



Small but strong visual separation between curvilinear rows of keys.

Bottom corner is clear of key possible usability problems of accessing keys at extremity.

Bottom corner is clear of key and hence possible usability problems of accessing keys at extremity

Dual Shift keys for convenience

Space spans 3 regular keys.

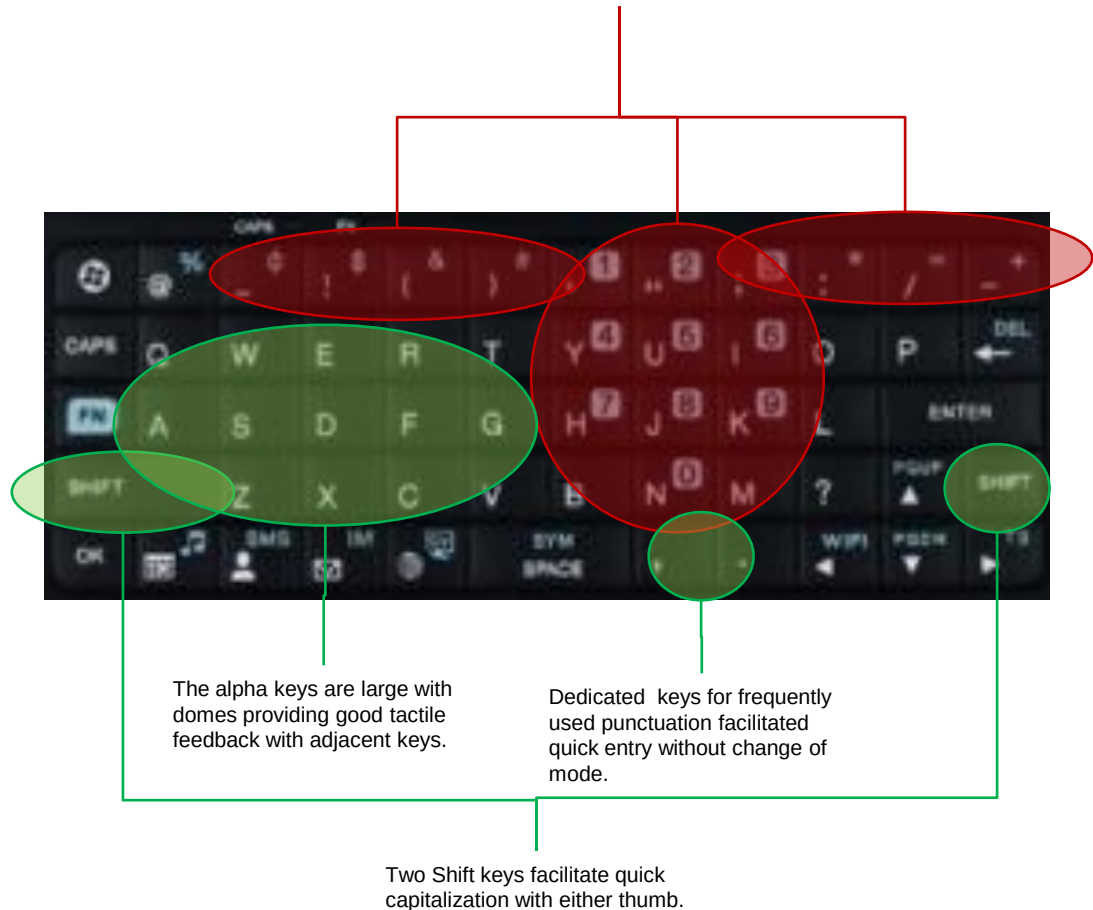
Physical QWERTY

Phone B (HTC Fuze)

- Rated well in overall user satisfaction
- High degree of speed and accuracy when typing
- Large keys with some curvature and ridge between keys and clear labels allowed for fast and accurate typing
- Characteristics:
 - Landscape layout
 - Wide, 5 line keyboard
 - Large keys with moderate curve and a ridge between the keys



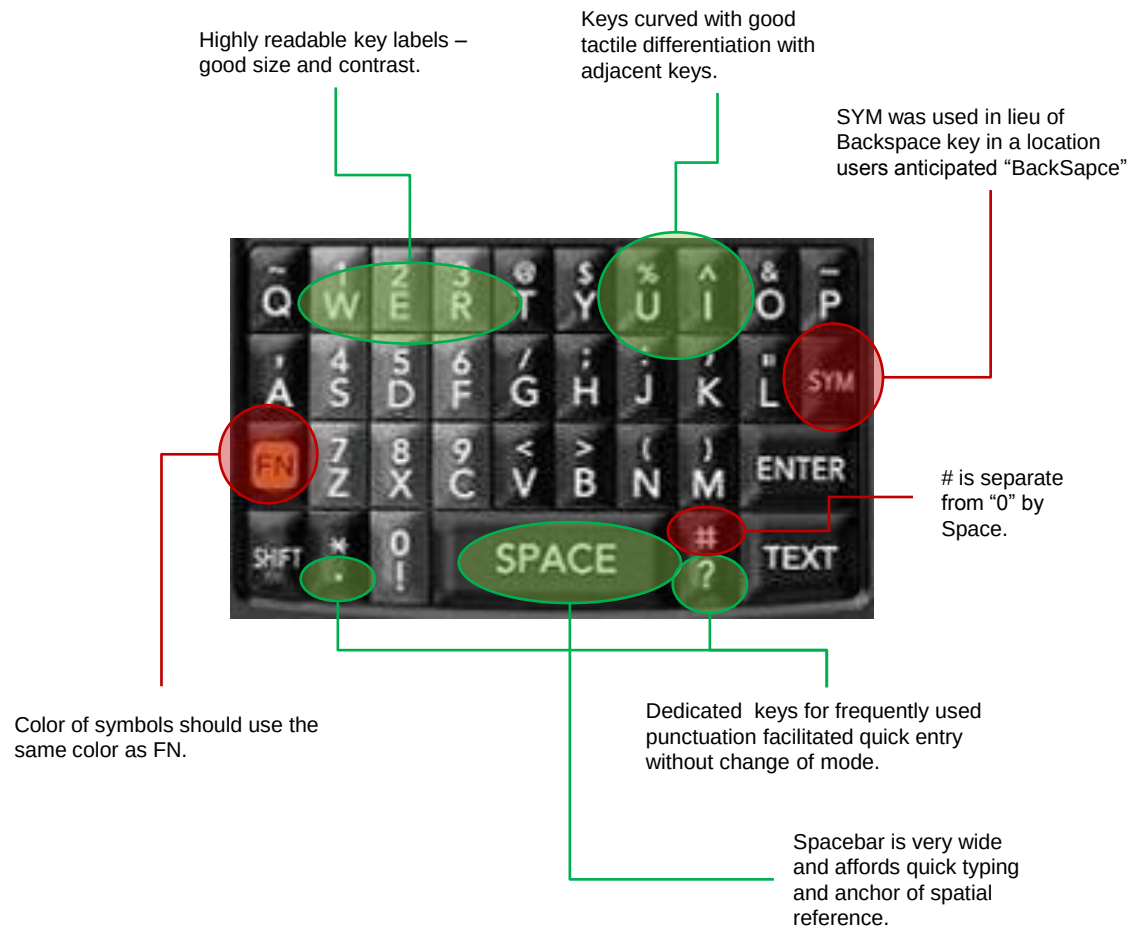
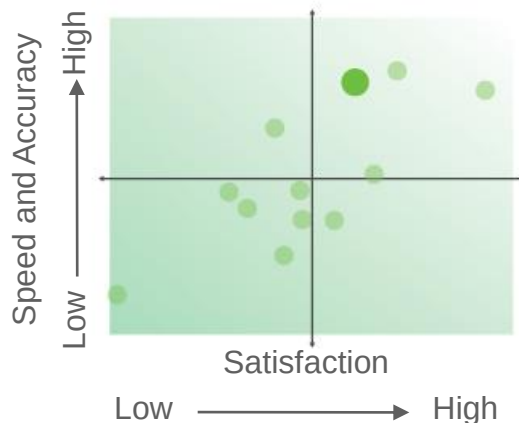
Symbols were mapped haphazardly and the overall layout of symbols and numbers conflict with traditional QWERTY layout, despite the overall similarity.



Physical QWERTY

Phone W (LG Lotus)

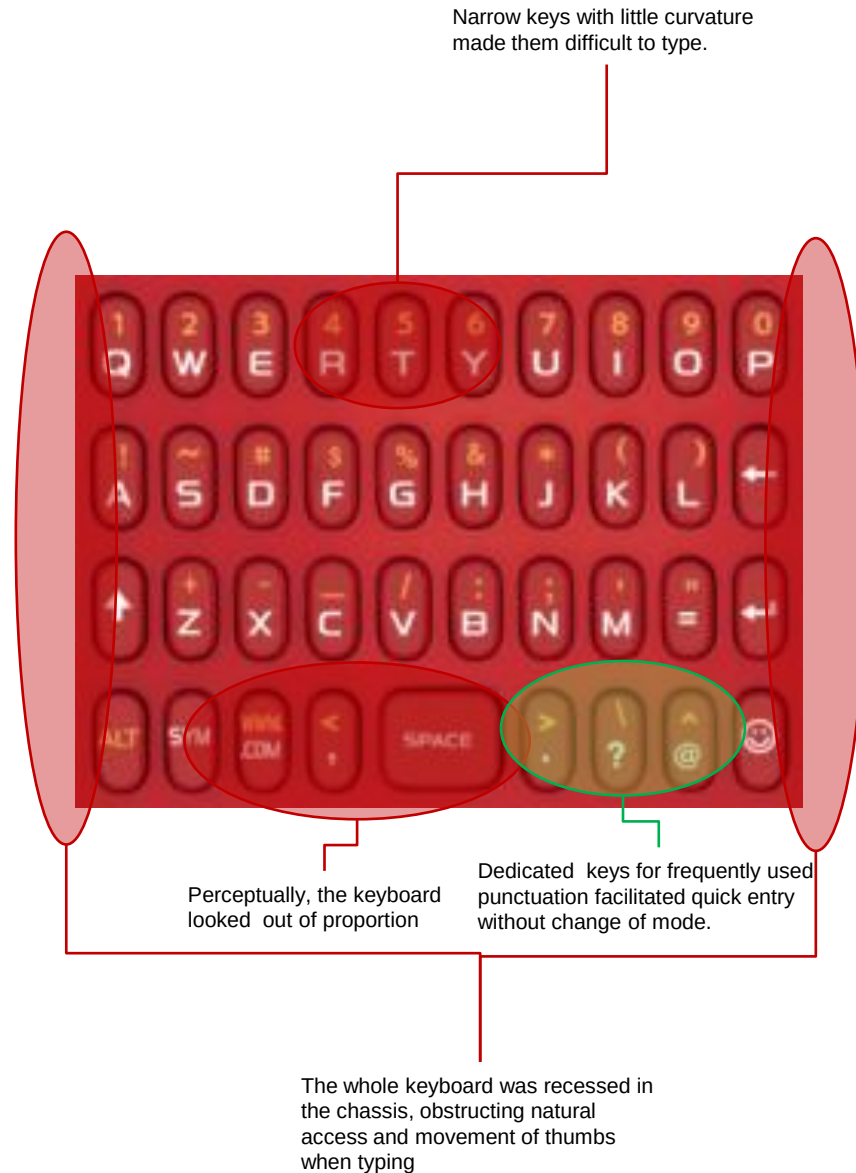
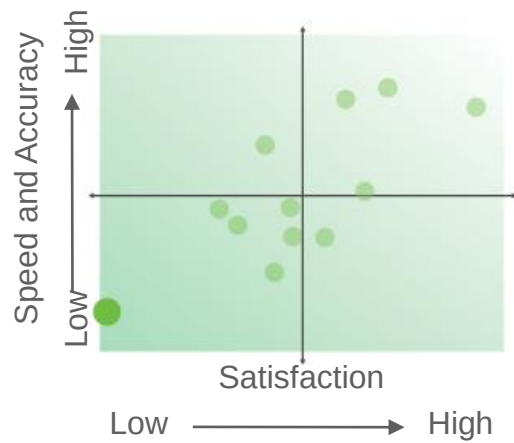
- Rated well in overall user satisfaction
- Performed well in speed and accuracy when typing
- Keys with large depth and strong curvature allowed for ease of typing
- Characteristics:
 - Portrait layout
 - Tall keyboard
 - Large key depth
 - Average size key surface with a high degree of curvature
 - No key gap



Physical QWERTY

Phone T (samsung XO)

- Rated lowest in overall user satisfaction
- Users had a low degree of speed and accuracy during typing.
- Narrow keys and tall keyboard made it difficult for users to type
- Characteristics:
 - Landscape layout
 - Tall keyboard
 - Large key depth
 - Keys with small width
 - Large key gap size

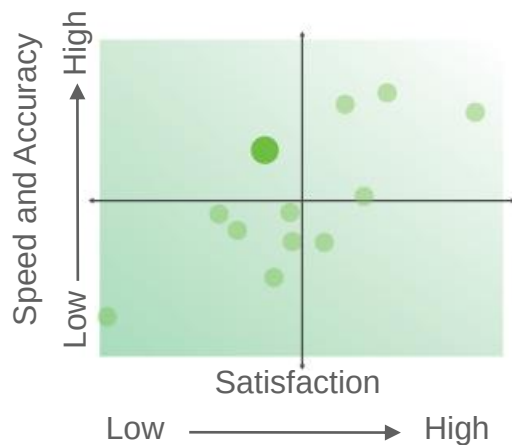


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Physical QWERTY

Phone A (Palm Pre)

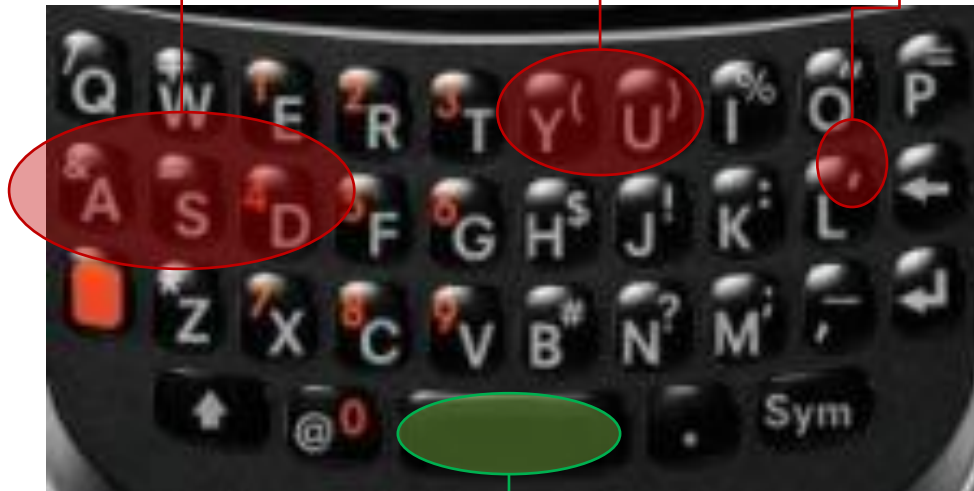
- Did not rate well among users for typing satisfaction.
- Performed well in speed and accuracy when typing
- Small, curved, rubbery keys allowed for moderate typing speed and accuracy. However, users generally did not like the size and feel of the keys; almost all were forced to use their thumbnails to type
- Characteristics:
 - Portrait layout
 - Large key depth
 - Very small key surface with moderate curvature



Key were too small and sticky.

Keys suffered from secular reflection/glare in many directions.

Color of punctuation key was not consistent with orange "function" key.



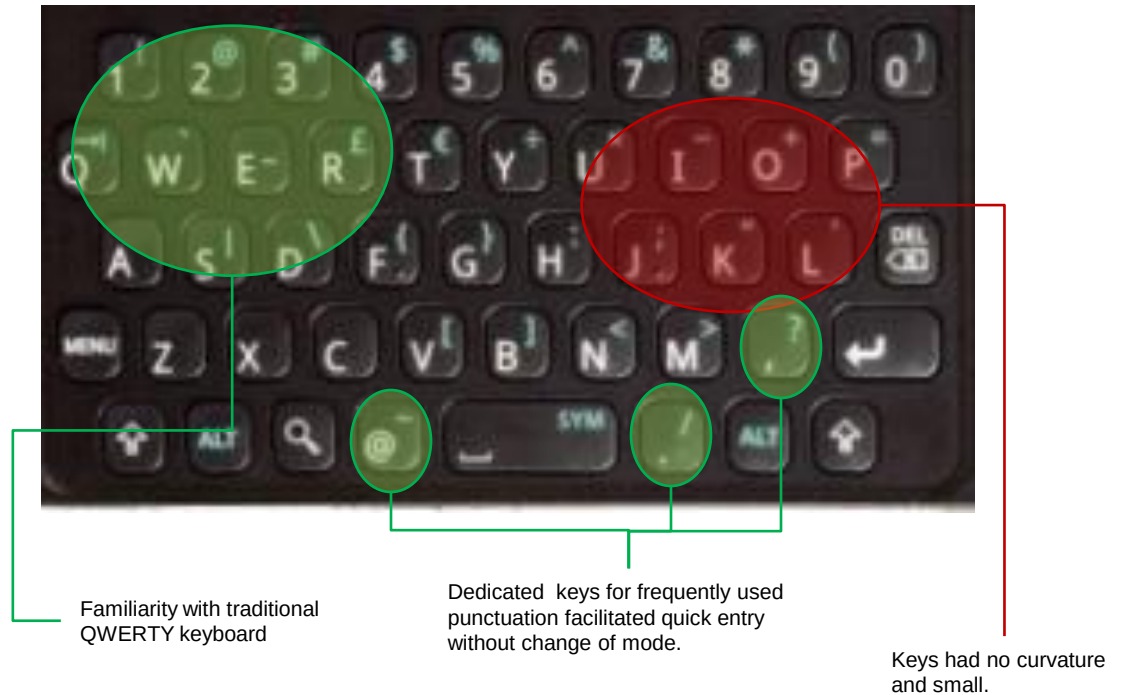
Space is along the line of symmetry with a relatively large width.

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Physical QWERTY

Phone C (HTC G1)

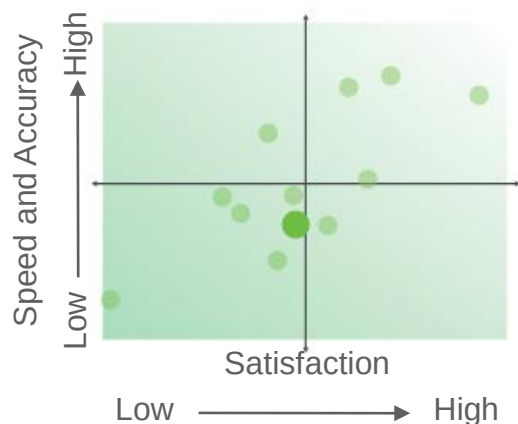
- Rated well in terms of user satisfaction.
- Average speed and accuracy performance when typing
- Small keys with no curvature were offset by a tall keyboard with large key gaps, allowing users moderate speed and accuracy during typing.
- Characteristics:
 - Landscape layout
 - Tall, 5-line keyboard
 - Moderate key depth
 - Small key surface size with no key curvature



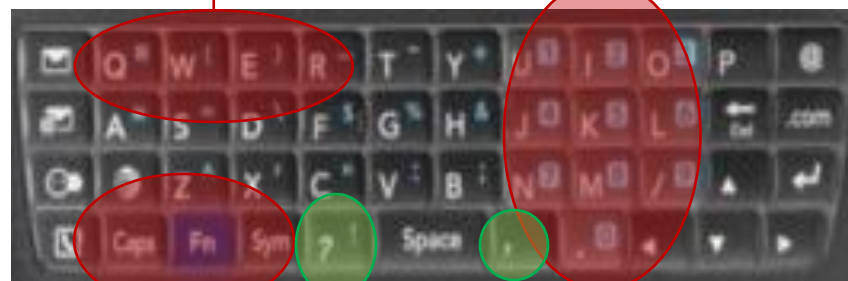
Physical QWERTY

Phone D (LG Xenon)

- Received average ratings in terms of user satisfaction
- Low to average speed and accuracy performance when typing
- Small key depth and size with little gap or curvature and a short keyboard hindered typing speed and accuracy.
- Characteristics:
 - Landscape layout
 - Short keyboard
 - Small key depth
 - Small key surface size with some key curvature
 - Small key gap



The key only "curved" between rows.
Tactile feel between horizontal keys were problematic.



The mapping of numeric keys was different for traditional landscape QWERTY keyboard. They also tend to distract alpha entry because of the contrast and special emphasis of the number keys in blue boxes.

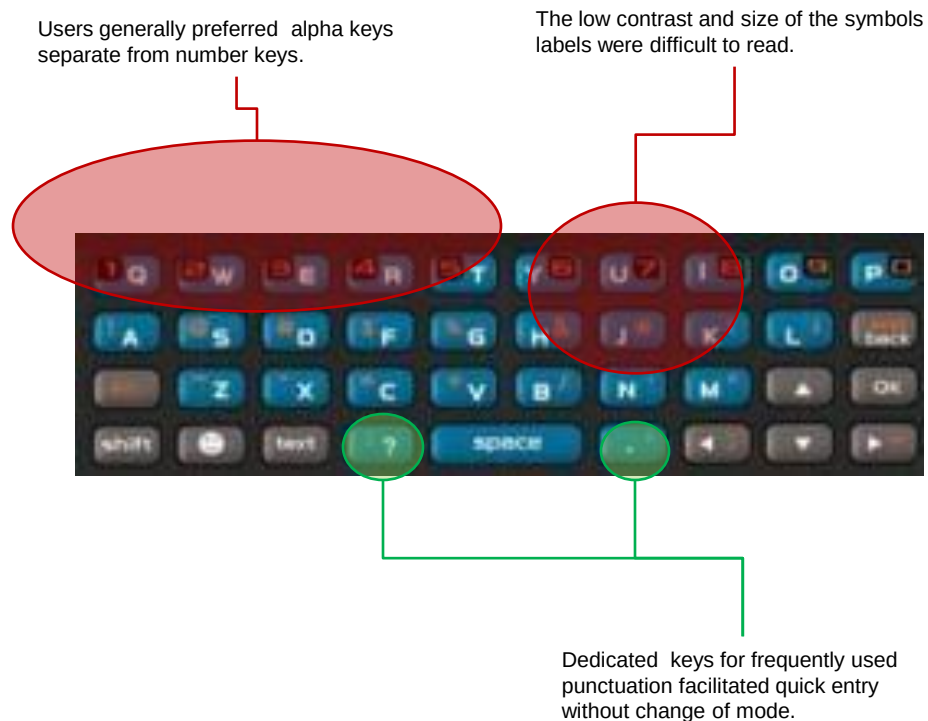
Keys were relatively small with unconventional label "Caps".

Dedicated keys for frequently used punctuation facilitated quick entry without change of mode.

Physical QWERTY

Phone E (Samsung Exclaim)

- Received low ratings in terms of user satisfaction
- Average performance in speed and accuracy when typing
- The positive effects of typing with wide, curved keys with a large gap was hindered by short keys without depth and some unclear, possibly distracting, labels.
- Characteristics:
 - Landscape layout
 - Short, wide keyboard
 - Low key depth
 - Wide key surface with some key curvature
 - Fairly large key gap



Physical QWERTY

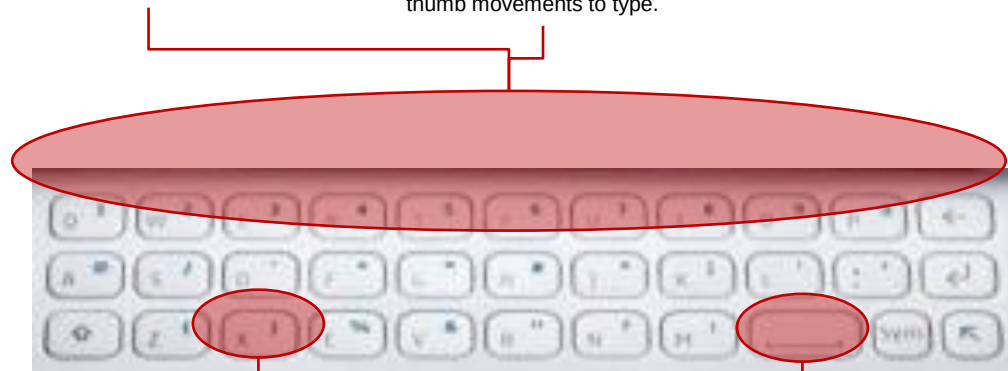
Phone H (Nokia N97)

- Received low to average ratings in terms of user satisfaction
- Speed and accuracy performance during typing was low with this phone.
- Flat keys with low key depth and a short keyboard made it difficult for users to type on this phone.
- Characteristics:
 - Landscape layout
 - Short, wide 3 line keyboard
 - Average key surface size
 - No key curvature, little depth
 - Large key gap



Users generally preferred alpha keys separate from number keys.

Short but wide Keyboard required disproportionate thumb movements to type.



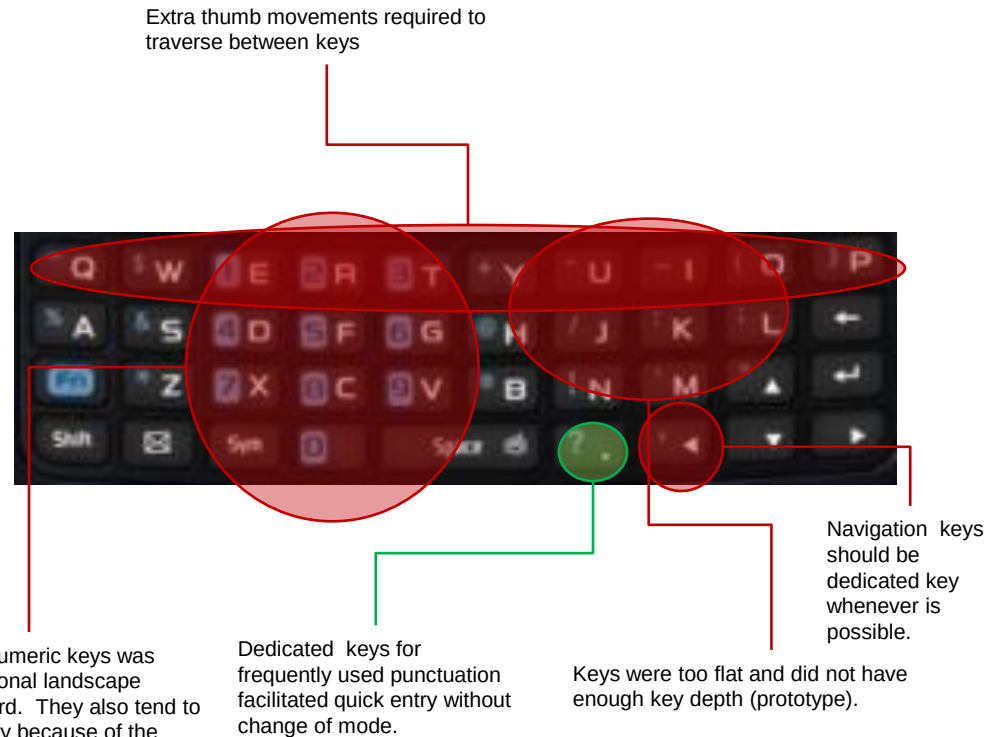
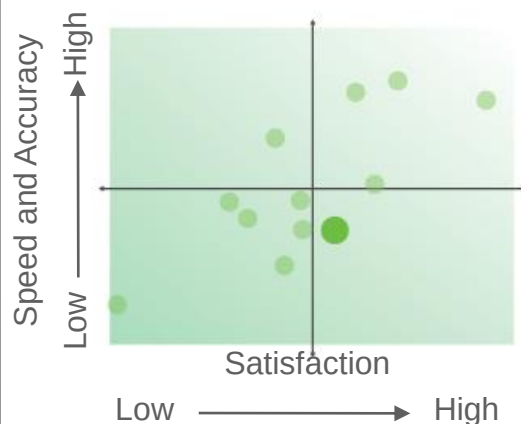
Keys were flat with low key depth.

The space key was in an unconventional location

Physical QWERTY

Phone K (Samsung Impression)

- Received average ratings in terms of user satisfaction
- Typing performance was low to average in terms of speed and accuracy
- A wide keyboard with flat, very low depth keys made it difficult for users to type
- Characteristics:
 - Landscape layout
 - Very wide keyboard
 - Very short key depth
 - Wide key surface
 - No key curvature



Physical QWERTY

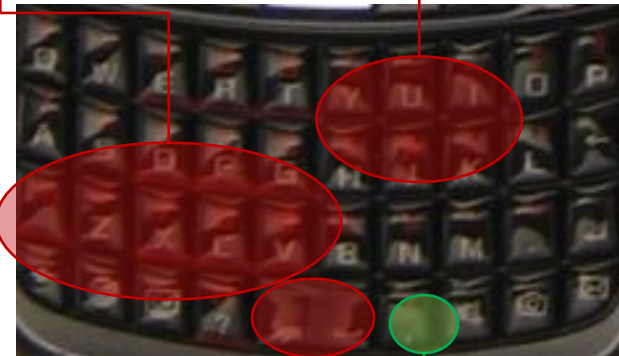
Phone M (Samsung Jack)

- Received average ratings in terms of user satisfaction
- Average typing performance in terms of speed and accuracy
- Key glare and poor label contrast led to average speed and accuracy typing performance
- Characteristics:
 - Portrait layout
 - Very Tall key depth
 - Small key size with strong curvature
 - No key gap



Keys were small with difficult to see symbol labels.

Keys suffered from secular reflection/glare in many directions.



Space key was small and keys did not enough friction.

Dedicated keys for frequently used punctuation facilitated quick entry without change of mode.

Physical QWERTY

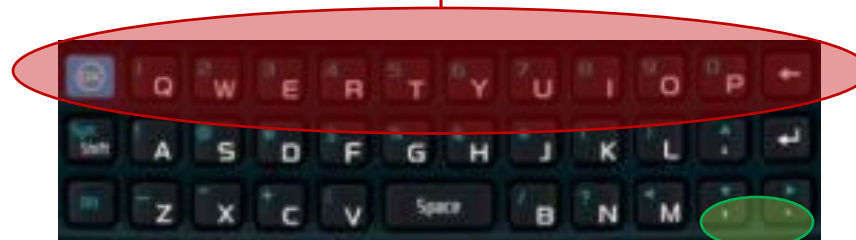
Phone S (Samsung Gravity)

- Received low user satisfaction ratings
- Low to average typing performance in terms of speed and accuracy
- A wide, short keyboard with small keys contributed to poor typing performance.
- Characteristics:
 - Landscape layout
 - Short, wide keyboard
 - Shorter key depth
 - Small key size with strong curvature
 - Moderate key gap



Users generally preferred alpha keys separate from number keys.

Short but wide Keyboard required disproportionate thumb movements to type.



Dedicated keys for frequently used punctuation facilitated quick entry without change of mode.

Recommendations

Guiding Principles for QWERTY Keyboard Design

- **Time to tap a target follows Fitts' Law**
- $\text{Time} = a + b * \log(1 + D / W)$
 - D = distance to target
 - W = target width in direction of movement
- Larger keys/small gap will be faster
- For phones with no key gap, smaller keypads will be faster

For mouse clicking, contrast affects time

- High contrast is faster
- Low contrast is slower

Corollary for QWERTY phones:

ability to distinguish individual keys will affect time

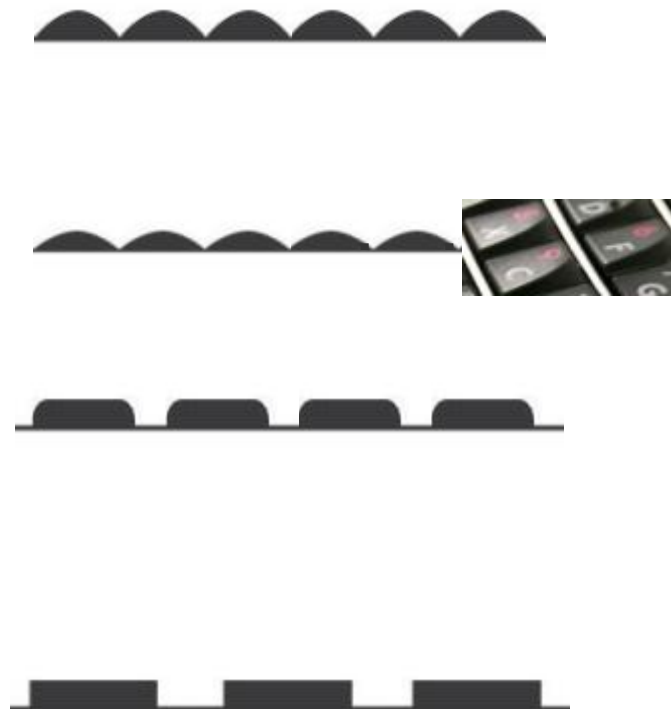
- Keys that are easy to distinguish are faster
- Use curvature, ridges and dots to increase tactile contrast between adjacent keys
- Gap increases contrast but also increases distance and decreases target size
- **Best phones use**
 - Small gap to maximize key size (target size)
 - Curves and ridges to increase tactile contrast between adjacent keys
- **Weak phones use**
 - Large gap decreases target size
 - Flat, shallow keys decrease tactile contrast



Recommendations

Discussion of the Interaction among Key Curvature, Gap and Depth

- Tradeoffs
- Domed keys can be small and adjacent to each other – no gap needed
 - Curvature helps distinguish adjacent keys
 - Curvature creates gap between keys
 - Curvature helps users center their finger on a key
- Ridges can be used to distinguish keys as well
 - Example: Fuze has a ridge between keys, each ridge slopes down from the center of the key.
 - Example: Blackberry devices have ridges on the keys with asymmetrical surface sloping in a direction optimized for thumb press.
- Flatter keys need to be bigger, taller and have a gap
 - Flat keys make it more difficult to feel the separate keys
 - Flat keys make it more difficult to avoid pressing the adjacent key
- Design Recommendations
 - Keyboards with a small width should use domed keys
 - Wide keyboards can use domed, rounded or flat keys
 - Keyboards with flatter keys will need larger key size, tallness and gap



Recommendations

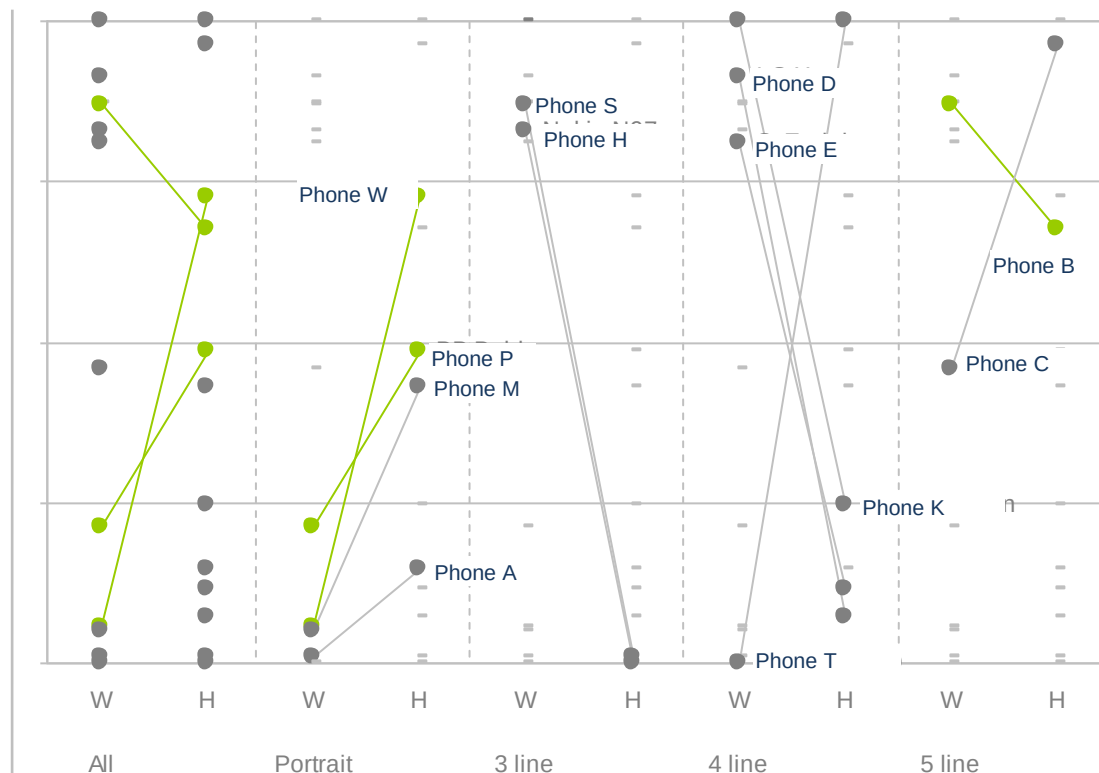
Keypad Size

Top Phones

- | | |
|---------|---|
| Phone P | 59 x 28 mm |
| Phone W | 53 x 32 mm |
| • | Small travel distance, small gap and strong curvature maximize target size and tactile contrast |
| Phone B | 86 x 31 mm |
| • | Large keys, small gap and inter-key ridge maximize target size and tactile contrast |

Examples of Poor Design Due to Keypad Size

- | | |
|-------------------|--|
| Phone A | 51 x 23 mm |
| • | Keys are not tall enough |
| • | Many users were force to use their fingernails due to the small key size |
| Phone M | 53 x 27 mm |
| • | Keys are not big enough for the tactile contrast used in the keycap design |
| Phone H | 85 x 20 mm |
| if it had 4 lines | 85 x 27 mm |
| • | Travel distance is long, but target size is small. |



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Recommendations

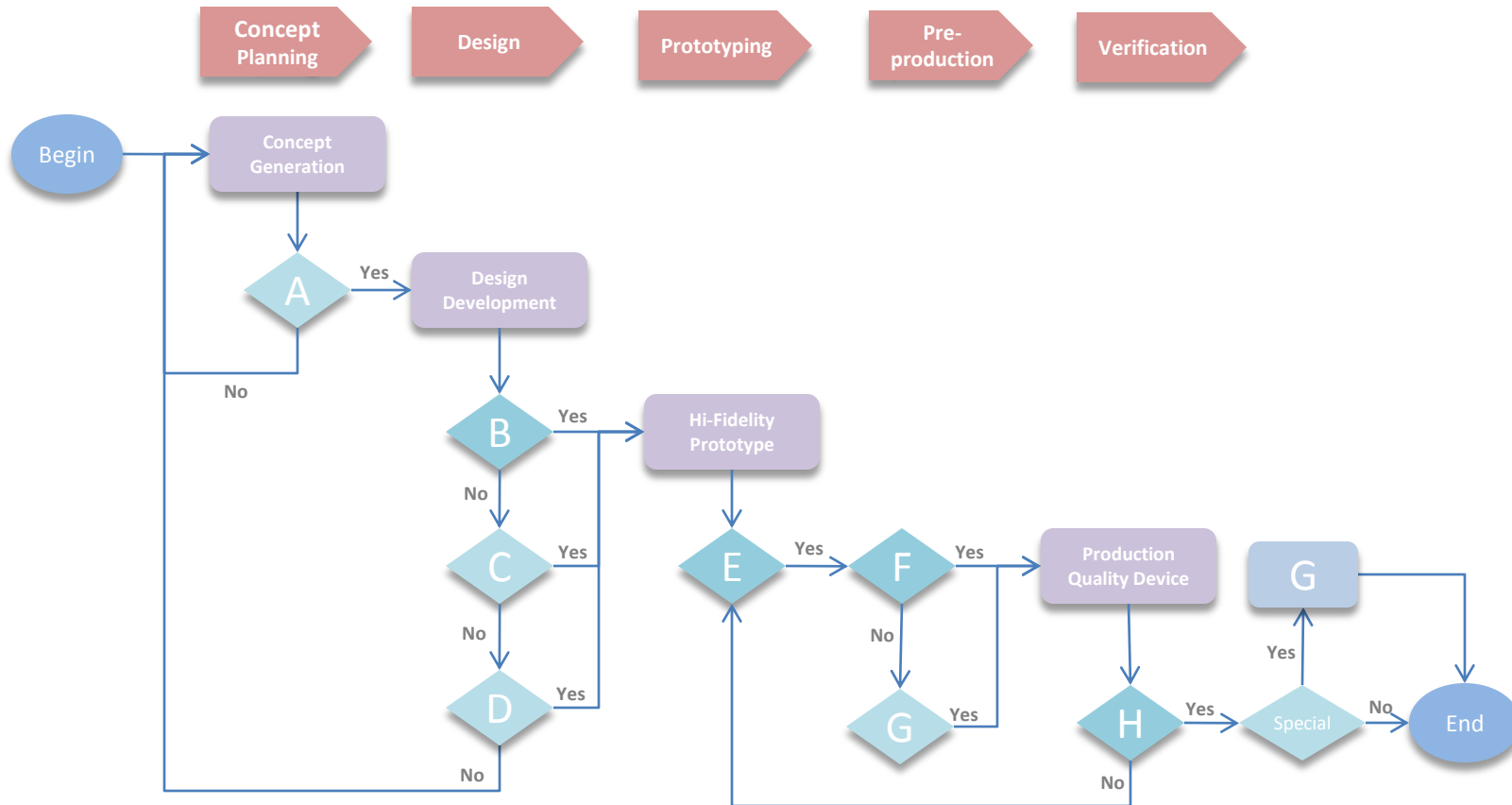
Next Steps

- Build a database for storing results of performance testing of phones
 - Build a tool that allows designers to search for studies of phones that are similar to new phones they are considering. For example, allow the designer to specify a range of key sizes and forces, and the tool would return all studies of phones that fall into this range
- Conduct performance testing of a wide range of phones
 - Gather performance on a wide range of phones
 - Add a brief performance benchmarking activity into studies conducted for other goals
 - Every study should use a common methodology so that performance data can be compared across studies
 - Explore the interactions among key size, gap, height and curvature; test additional phones that represent other combinations of these parameters so that statistical models can be developed that describe these complex interactions
- Conduct additional user research to uncover their preference for specific phone and keyboard design



Proposed Mobile QWERTY Design Development and Certification Process

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- A. Concept Evaluation: Lo-Fidelity QWERTY prototype research
- B. Compliance CheckList (HQ Human Factors Lab, STA UX)
- C. Special Compliance Review (HQ Human Factors Lab, STA UX, HQ PP)
- D. Special QWERTY User Research (HQ Human factors Lab, STA UX as needed)
- E. Compliance Testing for North America (STA Design/UX) including Accessibility
- F. Heuristic Usability Evaluations – HQ Human Factors Lab, STA Design/UX and/or external experts
- G. QWERTY Human Performance & Usability Study (Speed, Accuracy, Satisfaction & Ease) – Target Consumers in North America (STA Design/UX)
- H. QWERTY Performance & Usability Certification (HQ & STA PP, HQ Human factors Lab, STA Design/UX)

QWERTY Design Resources

1. Physical QWERTY Design Guidelines (HQ Human Factors Lab)
2. Mobile QWERTY Design Best Practice for North America (STA UX)
3. Design Review: HQ Human factors Lab and STA Design/UX

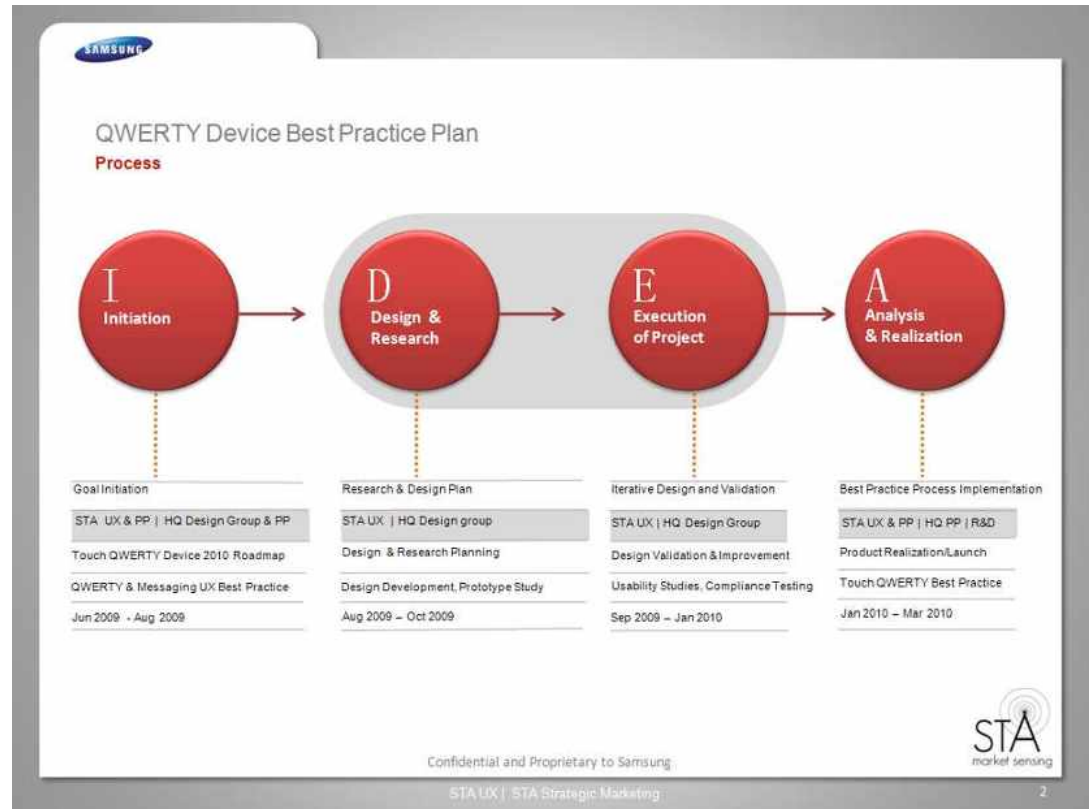
	Test ID	Design Development Gate Testing	Stage	Decision Criteria	Responsible Party
Concept Planning	A	Concept Evaluation: Lo-Fidelity QWERTY prototype research (upon request)	Concept Generation	Product team request	HQ Human Factors, STA Design/UX or HQ Design Group
	B	QWERTY design Compliance CheckList (non-compliance requires Special Compliance Review)		Compliance to Ergonomics Guidelines of HQ Human Factors Lab	HQ or SDA Product Concept or UX teams
Design	C	Special Compliance Review	Design Development	Strategic or non-compliant design	HQ Human Factors, STA Design/UX
	D	Special QWERTY User Research		Strategic or non-compliant design	HQ Human Factors, STA Design/UX & STA Strategic Marketing Research
Prototyping	E	Compliance Testing for North America	Hi-Fidelity Prototype	Mandatory compliance testing of pre-production	STA Design/UX
	F	Heuristic Usability Evaluations		Mandatory heuristic usability evaluation (including accessibility)	HQ Human Factors Lab, STA Design/UX or External Experts
Pre-production	G	QWERTY Human Performance & Usability Study (Speed, Accuracy, Satisfaction & Ease): User Validation & Competitive Benchmarking	Production Quality Device	Strategic or special QWERTY devices requiring extra consumer validation	STA Design/UX & STA Strategic Marketing Research
Verification	H	QWERTY Performance & Usability Certification		Results from E , F and G	HQ & STA PP, HQ & STA Design/UX, HQ Human Factors Lab.
		Core process: QWERTY Performance & Usability Certification (North American Products)			

Mobile QWERTY Development

Objectives

The objectives of this process are as follows:

1. To deploy the best practice QWERTY Design Practice based on user-centered research and grounded on human perception and performance analysis.
2. To establish an effective and efficient QWERTY keyboard design development process where design can be evaluated rapidly and feedback can be utilized by Design Team and R&D for improvement.
3. To conduct QWERTY design sensing research and benchmark devices under development against the best competitive QWERTY devices and continually build our knowledge and competence.



Appendix

Data Analyses and Comparisons

Phone Physical Measurement

ID	Speed	Accuracy	F max (grams)	Height (.001)	Travel (.001)	Tactile Ratio	Tactile Slope	Key Width	Key Height	Keyboard Width	Keyboard Height	Spacebar Width	Gap Width	Gap Height	Curvature
A	127.7	0.9	209.63	31.45	18.06	16.35	1041.09	4.22	4.98	51.02	22.51	12.82	0.93	1.11	2
B	135.2	0.9	235.48	5.38	11.80	22.58	1080.08	7.21	6.09	86.36	30.75	14.3	0	0	2
C	125.8	0.9	170.47	12.03	9.81	15.94	805.16	4.94	5.51	69.48	35.24	13.51	2.01	1.9	0
D	119.8	0.9	194.75	8.80	11.82	21.69	725.05	6.98	5.34	88.04	21.33	12.53	0.59	0.47	1
E	117.4	0.9	247.80	5.32	16.20	22.69	961.07	7.21	4.57	83.89	22	15.62	1.31	1.3	1
H	116.7	0.8	170.12	10.02	8.82	12.20	446.61	6.59	5.51	84.69	20.22	9.84	1.39	1.92	0
K	111.2	0.9	215.49	2.97	16.68	18.36	866.12	8.12	5.47	91.65	24.07	17.31	1.18	0.9	0
M	116.6	0.9	283.93	36.53	21.40	13.13	881.01	4.97	6.69	52.79	26.9	10.75	0	0	2
P	135.6	0.9	207.39	30.72	14.00	13.24	984.09	5.63	6.15	59.42	27.79	19.5	0	0.87	2
S	111.2	0.9	246.23	8.72	15.25	19.65	1013.28	5.52	5.26	86.35	20.38	12.79	1.03	1.57	2
T	102.9	0.9	223.18	16.82	15.67	23.58	954.74	3.99	7.27	69.48	35.24	13.51	2.01	1.9	0
W	139.6	0.9	268.54	15.47	11.14	24.21	974.00	5.11	7.39	53.02	31.52	20.98	0	0	2

Phone Subjective Measure

	Speed	Accuracy	Ease of Typing	Ease of Reading Key Labels	Typing Satisfaction	Force	Click	Gap	Keyboard Width	Keyboard Height	Feel Keys
A	127.7 (38.3)	0.9 (0.1)	2.8 (1.2)	2.8 (1.0)	2.7 (1.1)	4.3 (1.1)	3.5 (1.3)	2.6 (1.2)	2.2 (1.1)	2.7 (1.2)	3.4 (1.7)
B	135.2 (38.2)	0.9 (0.1)	3.8 (1.0)	3.6 (1.0)	3.4 (1.0)	4.0 (1.1)	3.8 (1.2)	3.2 (1.2)	4.1 (0.9)	3.9 (0.9)	3.6 (1.7)
C	125.8 (34.1)	0.9 (0.1)	3.3 (1.2)	3.6 (1.0)	3.3 (1.1)	3.7(1.4)	2.9 (1.3)	4.1 (1.0)	4.2 (1.2)	4.3 (1.0)	4.1 (1.8)
D	119.8 (34.8)	0.9 (0.1)	3.4 (1.1)	3.2 (1.0)	2.8 (1.1)	4.0 (1.0)	3.7 (1.1)	3.1 (1.1)	4.0 (1.0)	3.3 (1.0)	3.4 (1.5)
E	117.4 (32.8)	0.9 (0.1)	3.0 (1.1)	3.0 (1.0)	2.4 (0.9)	4.6 (1.6)	3.7 (1.5)	3.9 (1.1)	4.4 (1.2)	3.2 (1.1)	3.6 (1.7)
H	116.7 (35.1)	0.8 (0.1)	3.4 (1.1)	3.1 (1.1)	2.7 (1.0)	3.0 (1.3)	2.5 (1.4)	4.0 (1.1)	4.5 (1.2)	3.3 (1.1)	3.6 (1.7)
K	111.2 (30.3)	0.9 (0.1)	3.0 (1.1)	3.8 (0.9)	3.0 (1.0)	5.0 (1.5)	4.3 (1.4)	4.1 (1.2)	4.8 (1.1)	3.7 (1.1)	3.7 (1.7)
M	116.6 (37.2)	0.9 (0.1)	2.9 (1.2)	3.2 (1.0)	2.8 (1.0)	4.7 (1.2)	4.3 (1.1)	2.2(1.2)	2.9 (1.1)	3.6 (1.0)	3.4 (1.8)
P	135.6 (43.2)	0.9 (0.1)	4.1 (1.0)	4.0 (0.9)	3.9 (0.9)	4.0 (0.7)	4.1 (0.6)	3.3 (0.9)	3.7 (0.7)	3.9 (0.6)	4.6 (1.8)
S	111.2 (29.2)	0.9 (0.1)	3.1 (1.0)	2.8 (0.9)	2.5 (1.0)	5.2 (1.2)	4.4 (1.3)	3.8 (1.1)	4.7 (1.2)	3.1 (1.0)	3.6 (1.6)
T	102.9 (32.4)	0.9 (0.1)	2.0 (1.0)	2.6 (1.0)	1.8 (0.9)	5.2 (1.5)	4.1 (1.7)	3.5 (2.0)	3.5 (1.6)	4.6 (1.7)	3.2 (1.7)
W	139.6 (42.0)	0.9 (0.1)	3.6 (1.1)	3.5 (1.0)	3.1 (0.9)	4.6 (0.9)	4.6 (1.1)	2.7 (1.1)	3.1 (1.0)	4.0 (0.9)	3.9 (1.7)

Multiple Regression Analysis (HumanCentric)

Regression Analysis

Typing Speed

- 27% of the variability in the typing speed is explained by the model
- 73% remains unexplained
- The low percentage explained by the model suggests that there are complex interactions between design parameters which will require more than 12 phones to fully explore

Variable	Coefficient	People type faster when...
Key Curvature	.44	...keys are curved
Key Travel	-.43	...keys have smaller travel
Curvature x Key Depth	.25	...keys have large curve & depth or small curve & depth; having both large will be better than both small because people type faster on curved keys
Number of Rows	.16	...keyboard has more rows
Spacebar Width	.12	...spacebar is wide
Key Height	-.10	...keys have less height

The larger the absolute value of the coefficient the more that variable affects performance. The model was built on main effects and a few simple interactions. Presence in the model does not prove causality; it could be an artifact of the design of the specific phones chosen for the study.

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Regression Analysis

Typing Accuracy

- 4% of the variability in the typing accuracy is explained by the model
- 96% remains unexplained
- The low percentage explained by the model suggests that there are complex interactions between design parameters which will require more than 12 phones to fully explore
- Additional analyses for specific errors (typing on adjacent keys, missed keys and double presses) should be investigated, since the different errors are affected by different design parameters

Variable	Coefficient	People type more accurately when...
Tactile Slope	.21	...keys have a strong force-to-travel ratio
Gap Width x Key Center to Center Distance	.07	...keys have a large gap and center-to-center distance, or a small gap and center-to-center distance; people will type less accurately when one is large and the other is small

The larger the absolute value of the coefficient the more that variable affects performance. The model was built on main effects and a few simple interactions. Presence in the model does not prove causality; it could be an artifact of the design of the specific phones chosen for the study.

Regression Analysis

Typing Speed + Accuracy

- 18% of the variability in the typing speed + accuracy is explained by the model
- 82% remains unexplained
- there may be complex interactions between design parameters which will require more than 12 phones to fully explore

Variable	Coefficient	People type faster and more accurately when...
Key Curvature	.33	...keys are curved
Gap Width x Curvature	-.22	...keys have gap and curvature that are both large; large curvature is better for performance
Spacebar Width	.22	...spacebar is wide
Key Height	-.14	...keys have less vertical height
Gap Width x Key CCD Width	-.09	...keys have gap and center to center distance that are either both large or both small

The larger the absolute value of the coefficient the more that variable affects performance. The model was built on main effects and a few simple interactions. Presence in the model does not prove causality; it could be an artifact of the design of the specific phones chosen for the study.

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Regression Analysis

Overall Satisfaction

- 21% of the variability in the user satisfaction is explained by the model
- 79% remains unexplained
- There may be complex interactions between design parameters which will require more than 12 phones to fully explore.

Variable	Coefficient	People like the phones more when...
Spacebar Width	0.34	...the spacebar is wide
Tactile Ratio	-0.33	...there is less "click"
Number of Keys	0.20	...there are more keys
Gap Width	-0.14	...there is less gap
Age Group	-0.11	...they are younger
Messaging Frequency	-0.10	...they send fewer messages
Force	-0.10	...the keys use less force
Curvature x Key Tallness Interaction	-0.08	...the keys are tall and curved or short and flat
Fingernail Length	0.05	...they have longer fingernails

The larger the absolute value of the coefficient the more that variable affects performance. The model was built on main effects and a few simple interactions. Presence in the model does not prove causality; it could be an artifact of the design of the specific phones chosen for the study.

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Regression Analysis

Ease of Typing Letters

- 20% of the variability in the user satisfaction is explained by the model
- 80% remains unexplained
- There may be complex interactions between design parameters which will require more than 12 phones to fully explore

Variable	Coefficient	People found it easier to type letters when...
Gap Width	-0.29	...the gap was small
Key Travel	-0.22	...the keys had low travel
Tactile Ratio	-0.19	...there was a small tactile ratio
Spacebar Width	0.17	...there was a large spacebar
Index of Difficulty	0.16	...there was a higher ratio of key CCD to key size
Messaging Frequency	-0.13	...they sent fewer messages
Typing Speed	0.10	...they were faster typists
Thumb Length	-0.09	...they had shorter thumbs
Number of Rows	0.07	...there were more rows of keys
Has a Pure Touch Phone	-0.06	...they currently have a phone with a keyboard or keypad

The larger the absolute value of the coefficient the more that variable affects performance. The model was built on main effects and a few simple interactions. Presence in the model does not prove causality; it could be an artifact of the design of the specific phones chosen for the study.

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Regression Analysis

Ease of Typing Letters

- 20% of the variability in the user satisfaction is explained by the model
- 80% remains unexplained
- The low percentage explained by the model suggests that there are complex interactions between design parameters which will require more than 12 phones to fully explore

Variable	Coefficient	People found it easier to type letters when...
Gap Width	-0.29	...the gap was small
Key Travel	-0.22	...the keys had low travel
Tactile Ratio	-0.19	...there was a small tactile ratio
Spacebar Width	0.17	...there was a large spacebar
Index of Difficulty	0.16	...there was a higher ratio of key CCD to key size
Messaging Frequency	-0.13	...they sent fewer messages
Typing Speed	0.10	...they were faster typists
Thumb Length	-0.09	...they had shorter thumbs
Number of Rows	0.07	...there were more rows of keys
Has a Pure Touch Phone	-0.06	...they currently have a phone with a keyboard or keypad

The larger the absolute value of the coefficient the more that variable affects performance. The model was built on main effects and a few simple interactions. Presence in the model does not prove causality; it could be an artifact of the design of the specific phones chosen for the study.

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Regression Analysis

User Impression of Force

- 20% of the variability in the user satisfaction is explained by the model
- 80% remains unexplained
- The low percentage explained by the model suggests that there are complex interactions between design parameters which will require more than 12 phones to fully explore

Variable	Coefficient	People felt that more force was required to press the keys when...
Key Travel	.33	...the keys had more travel
Tactile Ratio	.24	...the keys had a high ratio of force/travel
Key Height	.09	...the keys had more vertical height

The larger the absolute value of the coefficient the more that variable affects performance. The model was built on main effects and a few simple interactions. Presence in the model does not prove causality; it could be an artifact of the design of the specific phones chosen for the study.

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Regression Analysis

User Impression of Click

- 20% of the variability in the user satisfaction is explained by the model
- 80% remains unexplained
- The low percentage explained by the model suggests that there are complex interactions between design parameters which will require more than 12 phones to fully explore

Variable	Coefficient	People gave the keys a higher rating for “click” when...
Force	.33	...there was more force
Gap Width x Key Center-to-Center Width	-.22	...there was a large gap and small center-to-center distance, or large distance and small gap
Spacebar	.18	...the spacebar was larger
Curvature x Key Center-to-Center Width	-.15	...there was a strong curvature and small center-to-center distance or flat keys with large center-to-center distance

The larger the absolute value of the coefficient the more that variable affects performance. The model was built on main effects and a few simple interactions. Presence in the model does not prove causality; it could be an artifact of the design of the specific phones chosen for the study.

Regression Analysis

User Impression of Key Distinctiveness

- 20% of the variability in the user satisfaction is explained by the model
- 80% remains unexplained
- Curvature appears in multiple terms, all of which need to be considered together to understand the relation between curvature and key distinctiveness.

Variable	Coefficient	People rated the phones as easier to feel individual keys when...
Spacebar Width	.24	...the spacebar was wide
Tactile Ratio	.17	...there was a lower ratio of force to travel
Curvature x Key Center-to-Center Width	.13	...there was strong curvature and big width or flat keys with small width
Mean Accuracy for the Participant	.11	...they type more accurately on all the phones
Fingernail Length	.08	...they have longer fingernails; note that people with long fingernails tend to type with their nails
Curvature	-.07	...the keys are flatter
Typing Accuracy	.07	...they type more accurately on a computer
Typing Speed	.07	...they type faster on a computer
Has a phone with a touchscreen only	.06	...they own a phone with a keyboard or keypad

The larger the absolute value of the coefficient the more that variable affects performance. The model was built on main effects and a few simple interactions. Presence in the model does not prove causality; it could be an artifact of the design of the specific phones chosen for the study.

Regression Analysis

User Demographic and Speed

- 40% of the variability in the average speed across all 12 phones is explained by the model
- A similar analysis was performed for typing accuracy, but none of the demographic factors were significant

Variable	Coefficient	People typed faster across all phones when...
Age Group	-.41	...they are younger
Typing Speed	.29	...they type faster on a computer
Has a Portrait QWERTY Phone	.29	...they currently own a QWERTY phone in portrait orientation
Fingernail Length	-.21	...they have shorter fingernails
Has Landscape QWERTY Phone	.17	...they currently own a QWERTY phone in landscape orientation

The larger the absolute value of the coefficient the more that variable affects performance. The model was built on main effects and a few simple interactions. Presence in the model does not prove causality; it could be an artifact of the design of the specific phones chosen for the study.

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Typing Speed

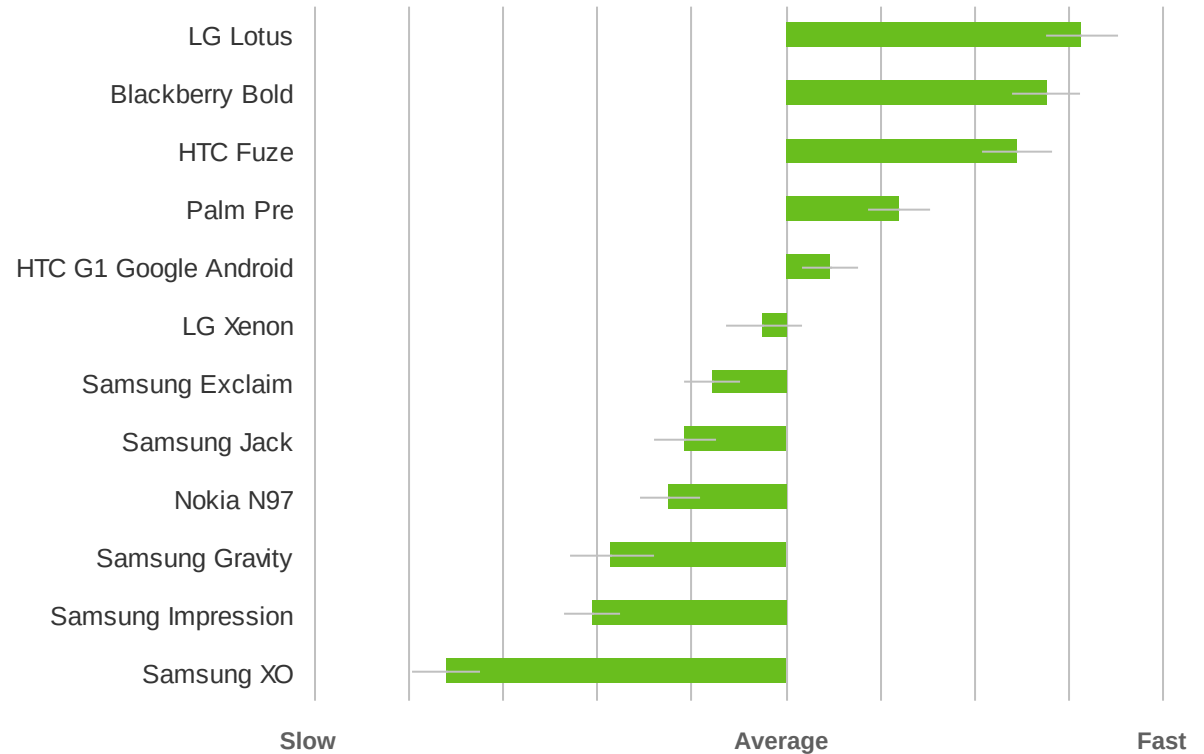
Speed Comparison

•Top phones

- LG Lotus
- Blackberry Bold
- HTC Fuze

•Bottom phones

- Samsung Impression
- Samsung XO



Lists of top and bottom phones are the phones that are statistically tied with the top or bottom phone, as determined by an ANOVA

Typing Accuracy

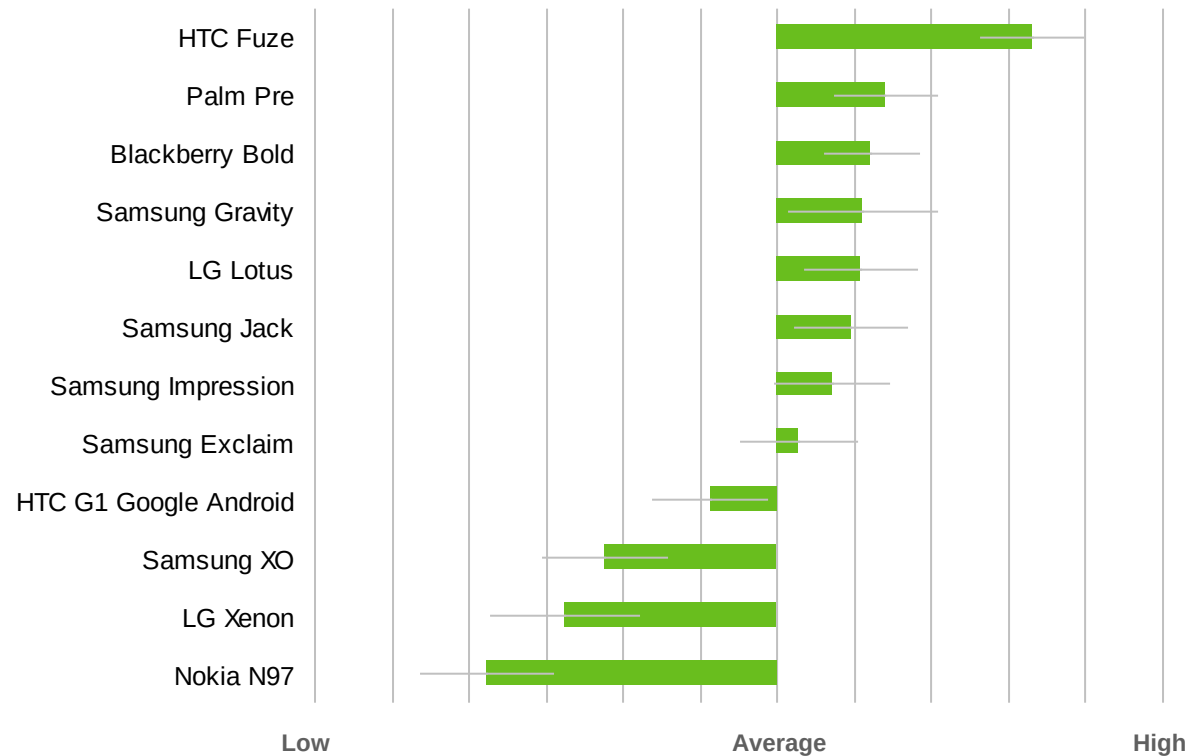
Speed Comparison

•Top phones

- HTC Fuze
- Palm Pre
- Blackberry Bold
- Samsung Gravity
- LG Lotus
- Samsung Jack
- Samsung Impression
- Samsung Exclaim

•Bottom phones

- HTC G1
- Samsung XO
- LG Xenon
- Nokia N97



Lists of top and bottom phones are the phones that are statistically tied with the top or bottom phone, as determined by an ANOVA

Speed + Accuracy

Speed + Accuracy Comparison

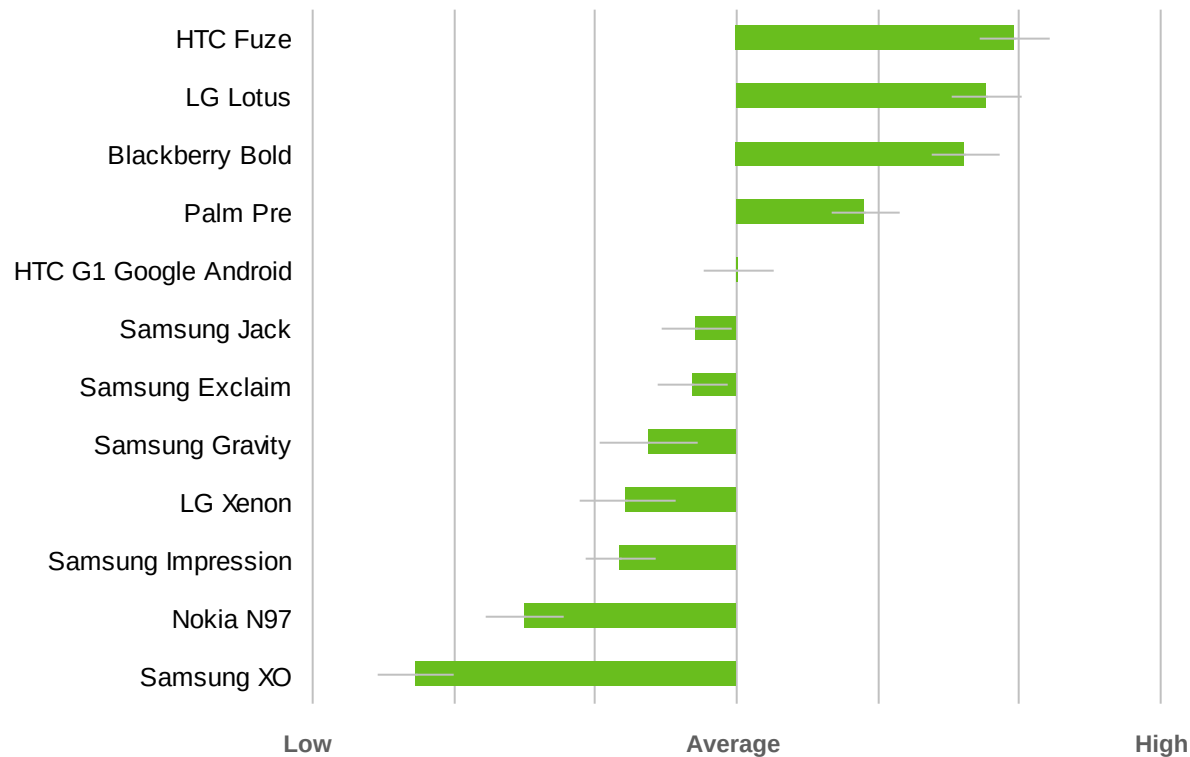
• Combined speed and accuracy scores for a balanced view of performance on the phones

•
Top phones

- HTC Fuze
- LG Lotus
- Blackberry Bold
- Palm Pre

• Bottom phones

- LG Xenon
- Samsung Impression
- Nokia 97
- Samsung XO



Lists of top and bottom phones are the phones that are statistically tied with the top or bottom phone, as determined by an ANOVA

Ease of Typing Text

Ease of Typing Comparison

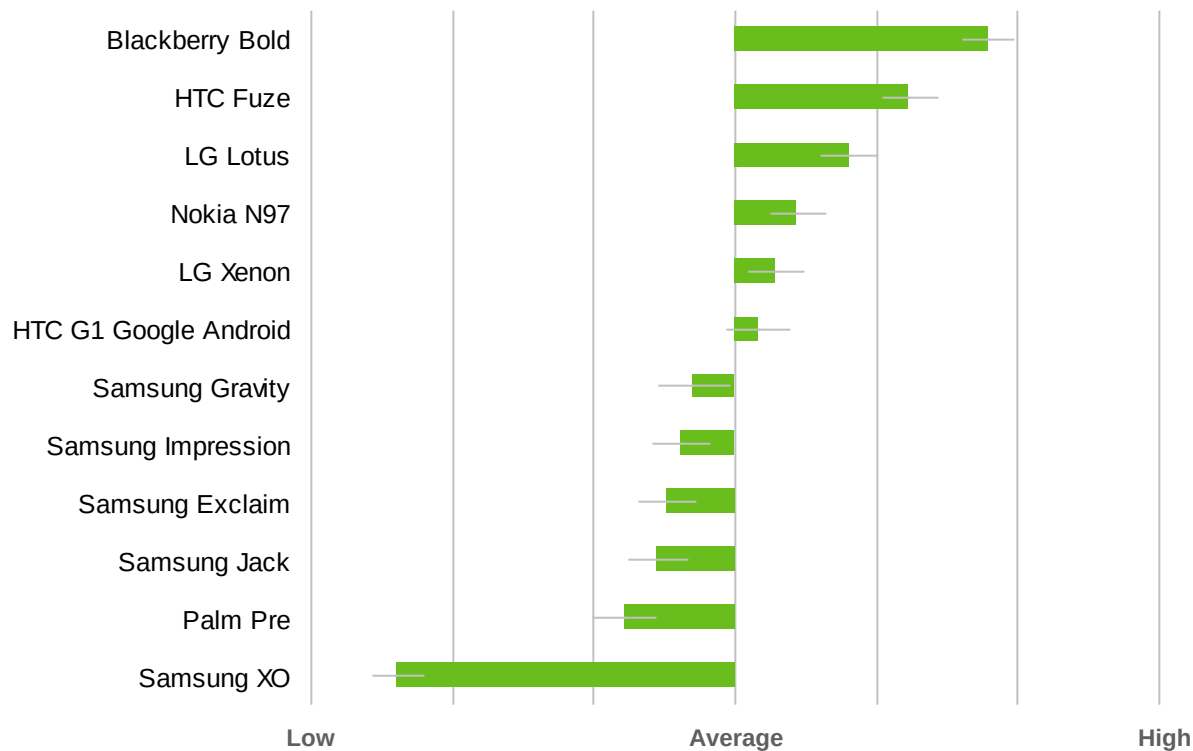
•Users rated the easy of typing letters using the QWERTY keypad

•Top phones

- Blackberry Bold
- HTC Fuze
- LG Lotus

•Bottom phone

- Samsung XO



Lists of top and bottom phones are the phones that are statistically tied with the top or bottom phone, as determined by an ANOVA

Overall Satisfaction

Satisfaction Comparison

•Users rated their overall satisfaction with typing on the phone

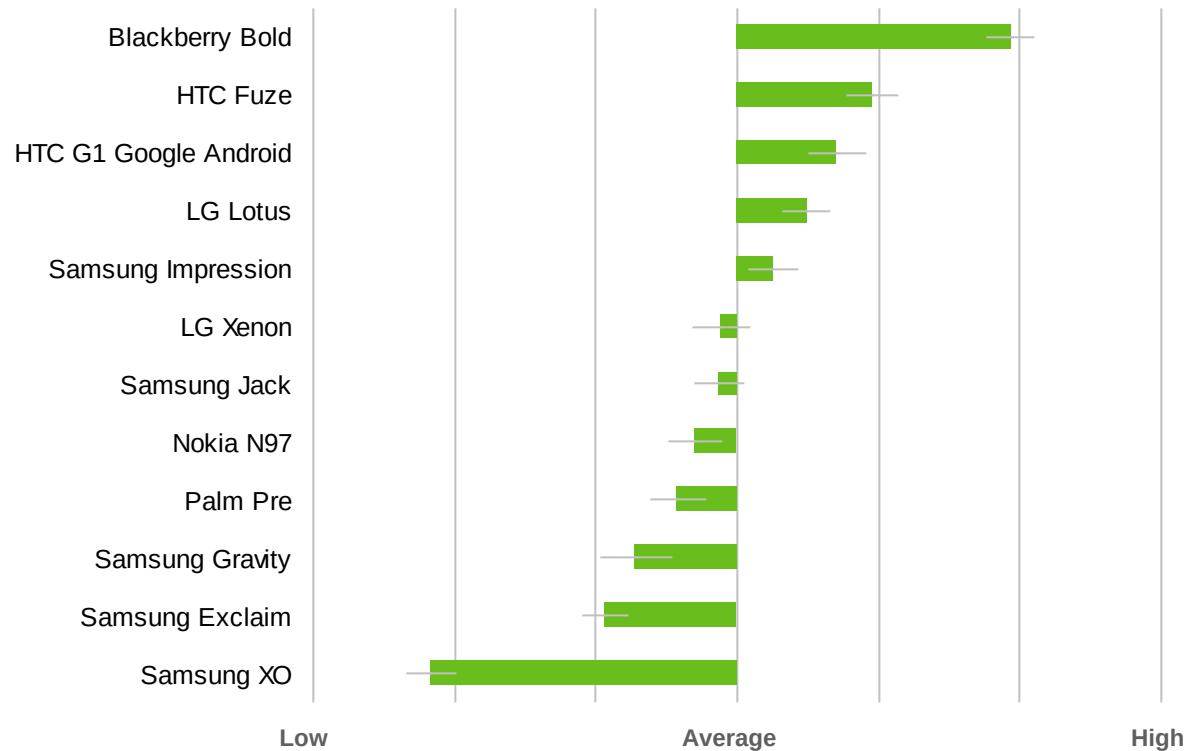
•

Top phone

— Blackberry Bold

•Bottom phone

— Samsung XO



Lists of top and bottom phones are the phones that are statistically tied with the top or bottom phone, as determined by an ANOVA

Confidential and Proprietary to Samsung

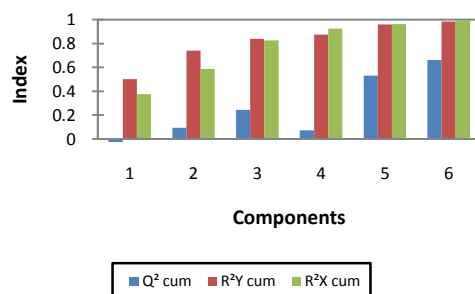
Partial Least Square Regression (STA UX)

PLS Regression

2 Components

- The cumulative model quality is at 0.2 which was relatively low, suggesting there were complex interaction effects between the variables or there were other underlying factors to be uncovered in further investigations.
- Since the number of devices used was relatively low compared to the number of independent variables in the model, increasing the number of components in the analysis would likely to over-fit the model. The problem of over-fitting could be seen in the comparison between the different models of PLS analyses (e.g., Portrait devices only 3 components or Landscape devices 6 components). The model quality increased dramatically to over 0.66 (example below) with almost 100% account of X and Y variances, which we know are symptomatic of over-fitting.

Model quality by number of components



Variable	Observations	Obs. with missing data	Obs. without missing data	Minimum	Maximum	Mean	Std. deviation
Speed	12	0	12	102.900	139.600	121.642	11.264
Accuracy	12	0	12	0.849	0.919	0.888	0.021
Ease of Typing	12	0	12	2.000	4.100	3.200	0.539
Typing Satisfaction	12	0	12	1.800	3.900	2.817	0.580
FMax(grams)	12	0	12	170.118	283.933	222.750	35.585
Height(.001)	12	0	12	2.967	36.533	15.351	11.386
Travel(.001)	12	0	12	8.820	21.397	14.221	3.683
TactileSlope	12	0	12	446.610	1080.07	894.359	173.397
TactileRatio	12	0	12	12.196	24.212	18.633	4.383
Letter Key Width	12	0	12	3.990	8.120	5.874	1.317
Letter Key Height	12	0	12	4.570	7.390	5.853	0.885
CCD Key horizontal	12	0	12	4.970	9.300	6.690	1.469
CCD Key vertical	12	0	12	5.810	9.700	6.892	1.067
Keyboard Width	12	0	12	50.690	91.650	71.450	16.902
Keyboard Height	12	0	12	20.220	35.810	26.543	5.671
Spacebar Width	12	0	12	9.010	20.980	14.080	3.700
Gap Width	12	0	12	0.000	2.010	0.816	0.687
Gap Height	12	0	12	0.000	2.430	1.039	0.821
Curvature	12	0	12	0.000	2.000	1.250	0.866
# of keys	12	0	12	33.000	57.000	40.417	6.986
# of rows	12	0	12	3.000	5.000	4.000	0.603

Model quality:

Index	Comp1	Comp2
Q² cum	0.070	0.208
R²Y cum	0.469	0.705
R²X cum	0.232	0.461

PLS Regression

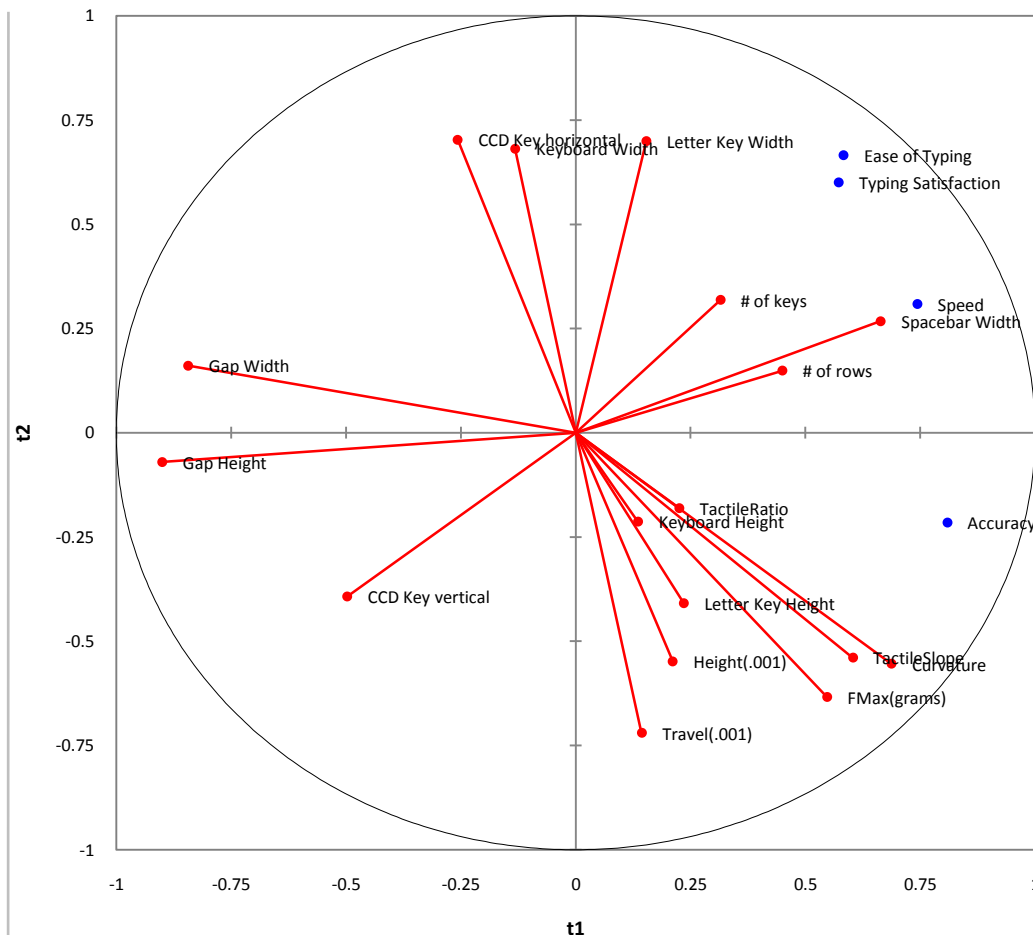
2 Components

- The correlations of the dependent and independent variables were shown.

Observations:

- Component t1: high negative correlations of gap width and gap height and strong positive correlation of curvature, spacebar width, tactile slope and trip force with t1. The dependent variables speed and accuracy also correlated highly with t1.
- Component t2: The correlation of accuracy was negative while the other dependent variables (speed, ease of typing and typing satisfaction) were positive. Thus, the linear variate of the independent variables affected accuracy in an opposite direction than the speed, ease of typing and typing satisfaction. Travel, Trip force, height, tactile slope, letter key height and curvature varied negatively with speed, ease of typing and satisfaction.

Correlations with t on axes t1 and t2



PLS Regression

2 Components

- The variables importance in the projection (>0.8) are identified as follows:
 - 1.) spacebar width
 - 2.) gap height
 - 3.) travel
 - 4.) gap width
 - 5.) tactile slope
 - 6.) curvature
 - 7.) CCD key vertical
 - 8.) Trip force
 - 9.) # of rows

Variable Importance in the Projection (VIP):

Variable	VIP	Standard deviation	Lower bound(95%)	Upper bound(95%)	VIP	Standard deviation	Lower bound(95%)	Upper bound(95%)
Spacebar Width	1.977	0.647	0.709	3.245	1.688	0.577	0.558	2.819
Gap Height	1.817	0.686	0.472	3.162	1.482	0.609	0.289	2.675
Travel(.001)	0.558	0.701	-0.816	1.931	1.399	0.322	0.767	2.030
Gap Width	1.703	0.507	0.709	2.697	1.389	0.471	0.466	2.312
TactileSlope	0.993	0.526	-0.039	2.024	1.205	0.226	0.762	1.648
Curvature	1.234	0.493	0.268	2.200	1.159	0.222	0.724	1.594
FMax(grams)	0.470	0.574	-0.655	1.594	1.129	0.361	0.421	1.837
CCD Key vertical	1.240	0.901	-0.526	3.006	1.020	0.419	0.198	1.842
# of rows	1.082	0.520	0.062	2.102	0.887	0.421	0.061	1.712
TactileRatio	0.237	0.778	-1.287	1.762	0.799	0.620	-0.415	2.014
Letter Key Width	0.542	0.694	-0.818	1.902	0.653	0.371	-0.074	1.379
# of keys	0.664	0.655	-0.621	1.949	0.552	0.561	-0.547	1.652
CCD Key horizontal	0.310	0.780	-1.219	1.839	0.518	0.582	-0.624	1.659
Keyboard Width	0.006	1.125	-2.200	2.212	0.502	0.532	-0.541	1.546
Height(.001)	0.347	0.588	-0.806	1.499	0.387	0.490	-0.574	1.348
Keyboard Height	0.316	1.263	-2.159	2.790	0.257	1.115	-1.929	2.443
Letter Key Height	0.190	1.153	-2.070	2.451	0.255	0.878	-1.465	1.976

PLS Regression

2 Components

- Speed of text entry was related to the following factors (with decreasing standardized coefficients>1):

- 1.) spacebar width
- 2.) gap height (-)
- 3.) gap width (-)
- 4.) travel (-)
- 5.) CCD Key vertical (-)
- 6.) number of rows

- Accuracy of text entry was related to the following factors (with decreasing standardized coefficients>1):

- 1.) gap height (-)
- 2.) gap width (-)
- 3.) spacebar width
- 4.) curvature
- 5.) tactile slope
- 6.) CCD Key Vertical (-)
- 9.) trip force

Speed – standardized coefficients

Variable	Coefficient	Std. deviation	Lower bound (95%)	Upper bound (95%)
FMax(grams)	-0.027	0.124	-0.270	0.215
Height(.001)	0.020	0.086	-0.148	0.188
Travel(.001)	-0.165	0.133	-0.426	0.095
TactileSlope	0.046	0.053	-0.059	0.151
TactileRatio	-0.087	0.178	-0.435	0.262
Letter Key Width	0.099	0.101	-0.099	0.297
Letter Key Height	0.006	0.168	-0.323	0.336
CCD Key horizontal	-0.001	0.073	-0.145	0.143
CCD Key vertical	-0.153	0.253	-0.648	0.342
Keyboard Width	0.039	0.099	-0.155	0.234
Keyboard Height	0.036	0.172	-0.301	0.373
Spacebar Width	0.266	0.174	-0.076	0.608
Gap Width	-0.193	0.095	-0.380	-0.005
Gap Height	-0.206	0.125	-0.451	0.039
Curvature	0.098	0.057	-0.015	0.210
# of keys	0.085	0.126	-0.162	0.332
# of rows	0.131	0.066	0.002	0.261

Accuracy – standardized coefficients

Variable	Coefficient	Std. deviation	Lower bound (95%)	Upper bound (95%)
FMax(grams)	0.101	0.062	-0.020	0.222
Height(.001)	0.047	0.057	-0.064	0.158
Travel(.001)	0.018	0.126	-0.229	0.265
TactileSlope	0.142	0.058	0.029	0.255
TactileRatio	0.019	0.099	-0.174	0.212
Letter Key Width	0.026	0.106	-0.182	0.234
Letter Key Height	0.029	0.068	-0.105	0.163
CCD Key horizontal	-0.054	0.059	-0.168	0.061
CCD Key vertical	-0.110	0.089	-0.285	0.065
Keyboard Width	-0.026	0.085	-0.193	0.141
Keyboard Height	0.030	0.092	-0.151	0.211
Spacebar Width	0.160	0.121	-0.077	0.397
Gap Width	-0.164	0.093	-0.346	0.018
Gap Height	-0.174	0.093	-0.356	0.007
Curvature	0.148	0.041	0.068	0.228
# of keys	0.057	0.128	-0.194	0.308
# of rows	0.098	0.100	-0.099	0.294

PLS Regression

2 Components

- Typing Satisfaction was related to the following factors (with decreasing standardized coefficients>1):

- 1.) spacebar width
- 2.) travel (-)
- 3.) gap height (-)
- 4.) gap width (-)
- 5.) CCD Key vertical (-)
- 6.) tactile ratio (-)
- 7.) letter key width
- 8.) number of rows
- 9.) trip force (-)

- Ease of Typing was related to the following factors (with decreasing standardized coefficients>1):

- 1.) spacebar width
- 2.) travel (-)
- 3.) gap height (-)
- 4.) gap width (-)
- 5.) CCD Key vertical (-)
- 6.) tactile ratio (-)
- 7.) letter key width
- 8.) number of rows
- 9.) trip force (-)

Typing Satisfaction – standardized coefficients

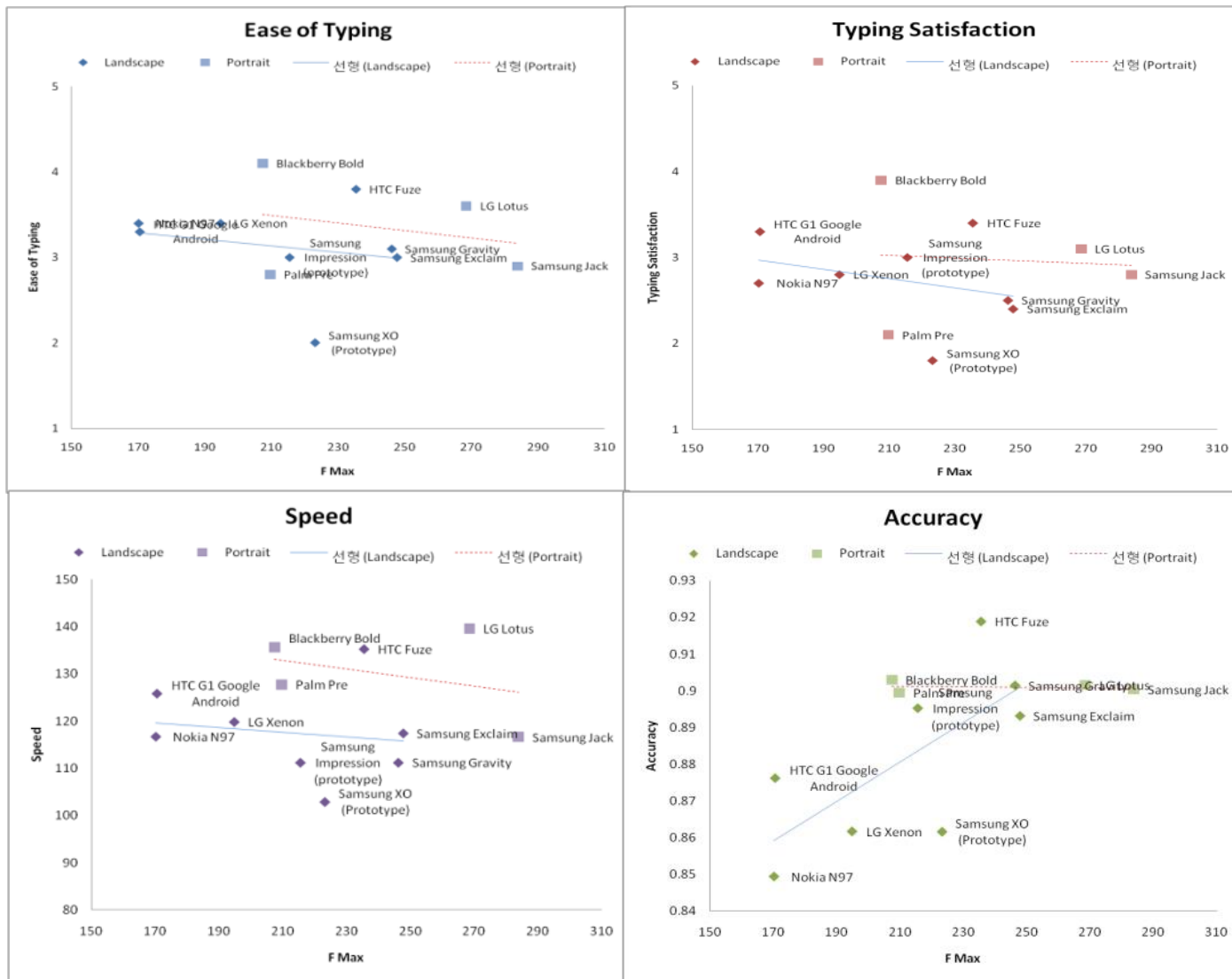
Variable	Coefficient	Std. deviation	Lower bound (95%)	Upper bound (95%)
FMax(grams)	-0.107	0.123	-0.348	0.134
Height(.001)	-0.002	0.119	-0.235	0.232
Travel(.001)	-0.258	0.127	-0.506	-0.009
TactileSlope	-0.025	0.068	-0.158	0.107
TactileRatio	-0.141	0.200	-0.534	0.252
Letter Key Width	0.130	0.098	-0.061	0.322
Letter Key Height	-0.010	0.183	-0.369	0.350
CCD Key horizontal	0.034	0.077	-0.117	0.185
CCD Key vertical	-0.155	0.300	-0.743	0.434
Keyboard Width	0.075	0.103	-0.125	0.276
Keyboard Height	0.034	0.198	-0.354	0.421
Spacebar Width	0.290	0.187	-0.076	0.656
Gap Width	-0.178	0.113	-0.399	0.043
Gap Height	-0.191	0.144	-0.473	0.091
Curvature	0.048	0.085	-0.118	0.214
# of keys	0.089	0.127	-0.160	0.337
# of rows	0.131	0.097	-0.058	0.321

Ease of Typing – standardized coefficients

Variable	Coefficient	Std. deviation	Lower bound (95%)	Upper bound (95%)
FMax(grams)	-0.122	0.131	-0.380	0.135
Height(.001)	-0.004	0.128	-0.254	0.246
Travel(.001)	-0.282	0.131	-0.538	-0.026
TactileSlope	-0.035	0.082	-0.195	0.125
TactileRatio	-0.155	0.215	-0.576	0.266
Letter Key Width	0.141	0.112	-0.079	0.361
Letter Key Height	-0.012	0.177	-0.359	0.335
CCD Key horizontal	0.039	0.086	-0.129	0.208
CCD Key vertical	-0.163	0.299	-0.750	0.424
Keyboard Width	0.084	0.111	-0.134	0.302
Keyboard Height	0.035	0.185	-0.328	0.398
Spacebar Width	0.308	0.219	-0.121	0.736
Gap Width	-0.186	0.125	-0.431	0.059
Gap Height	-0.199	0.161	-0.514	0.116
Curvature	0.045	0.090	-0.131	0.221
# of keys	0.094	0.134	-0.169	0.356
# of rows	0.138	0.102	-0.063	0.339

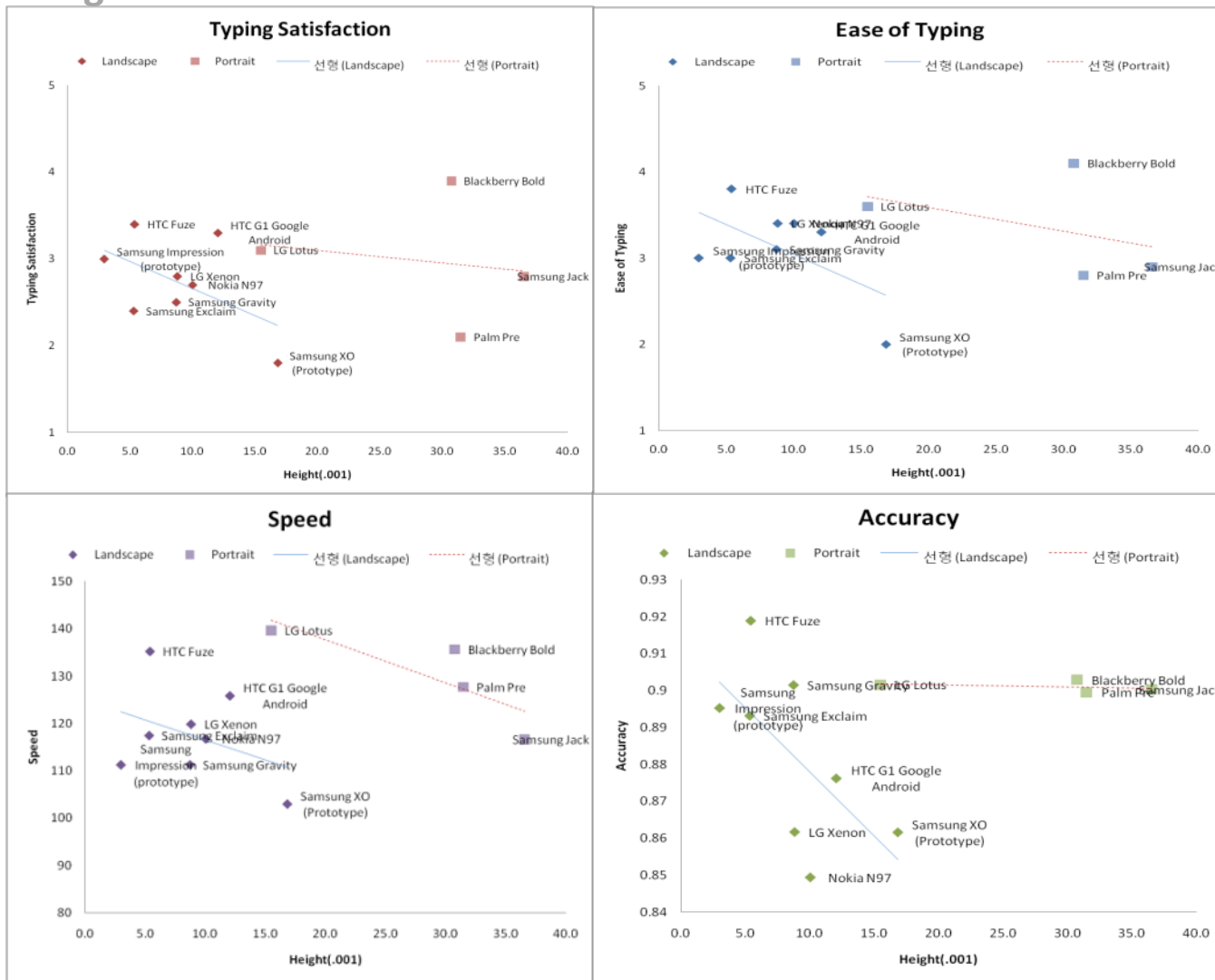
Linear Regression (HQ Human Factors Lab)

F Max



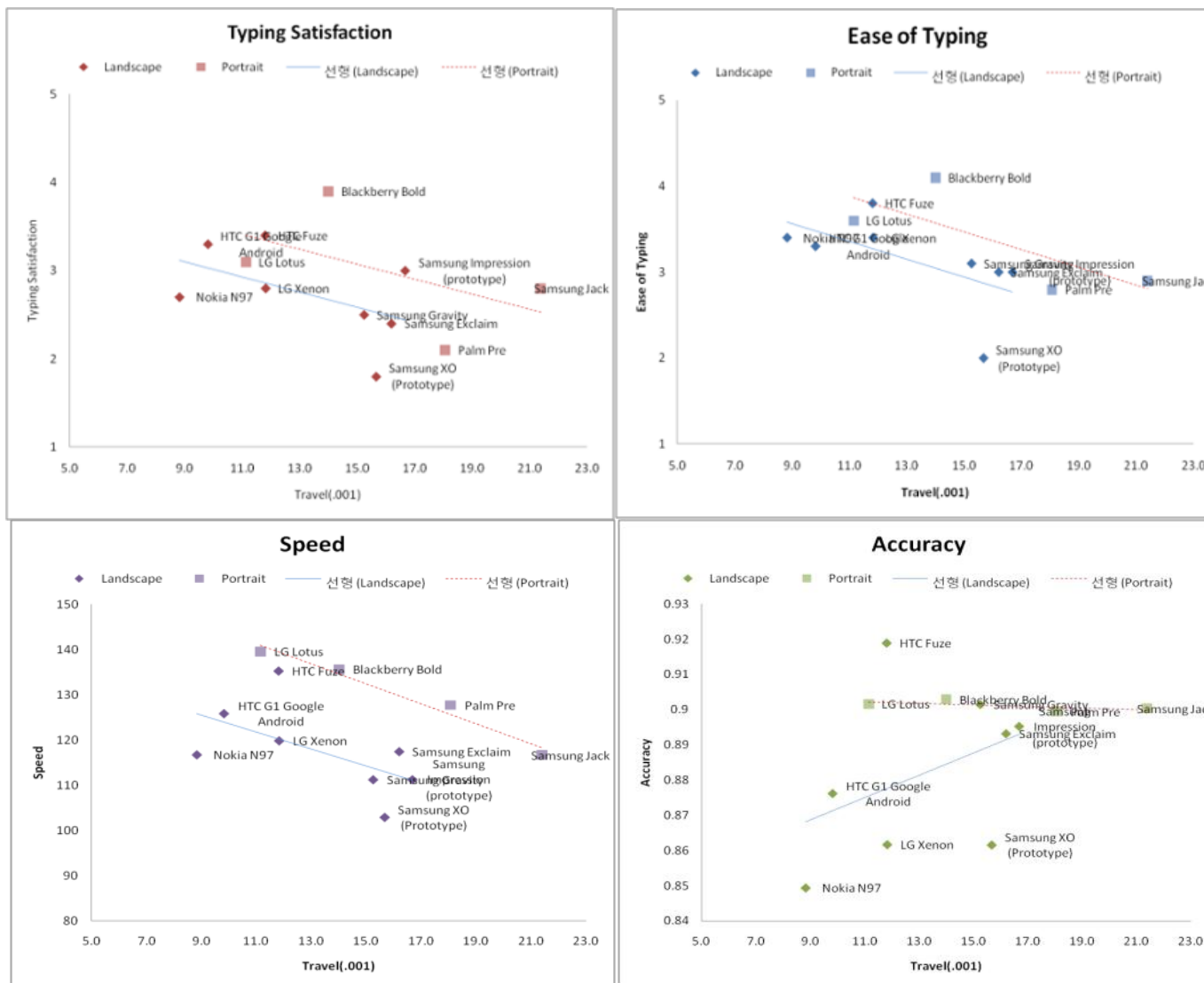
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Height



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Travel



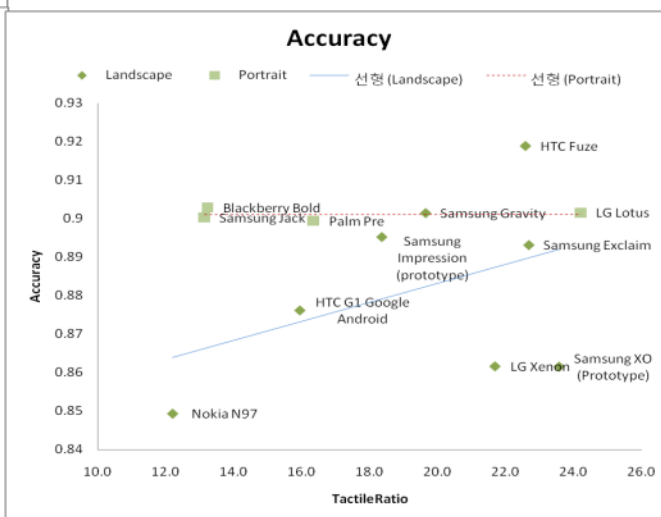
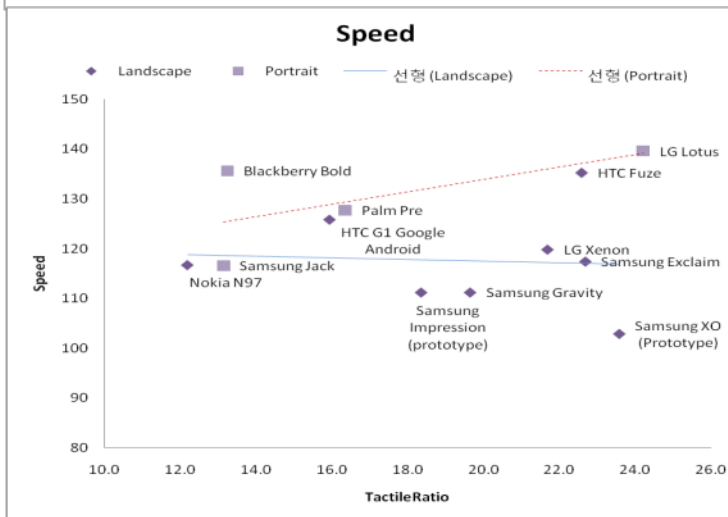
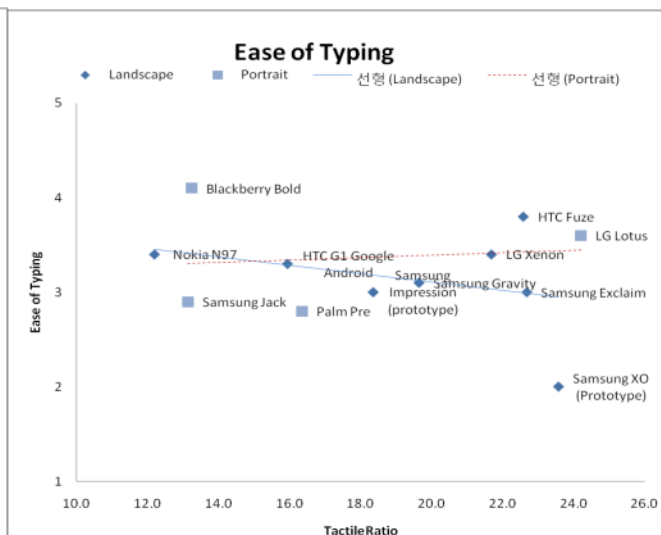
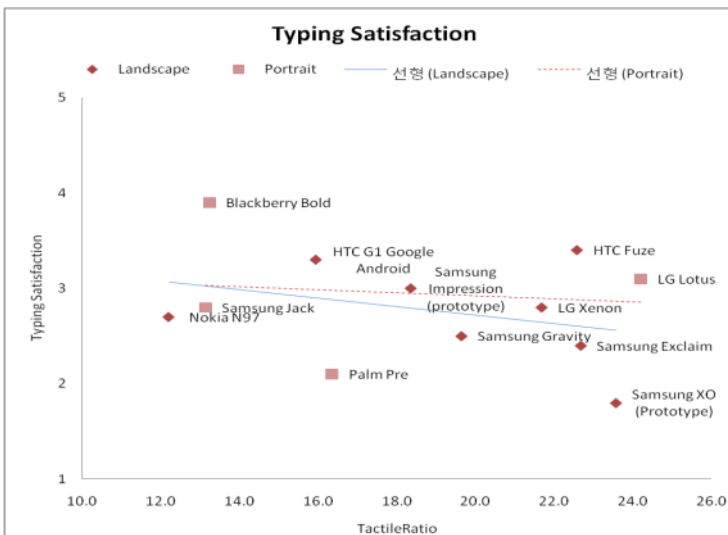
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Tactile Slope

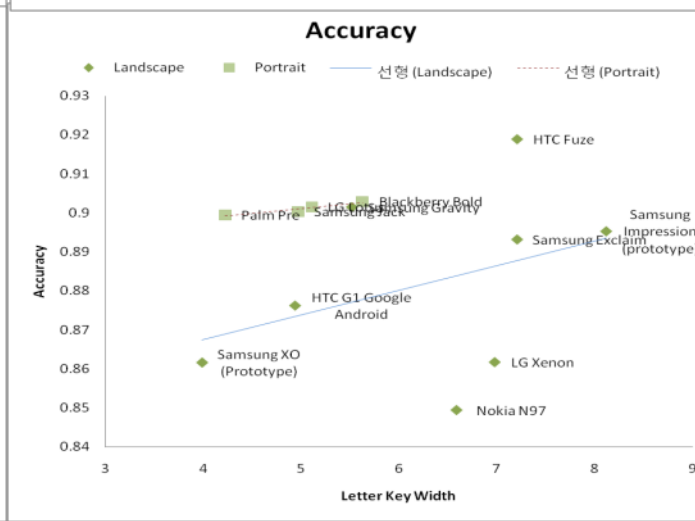
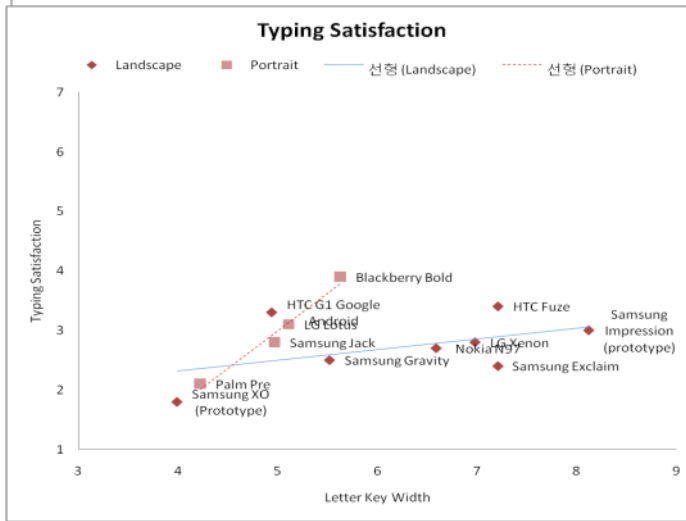
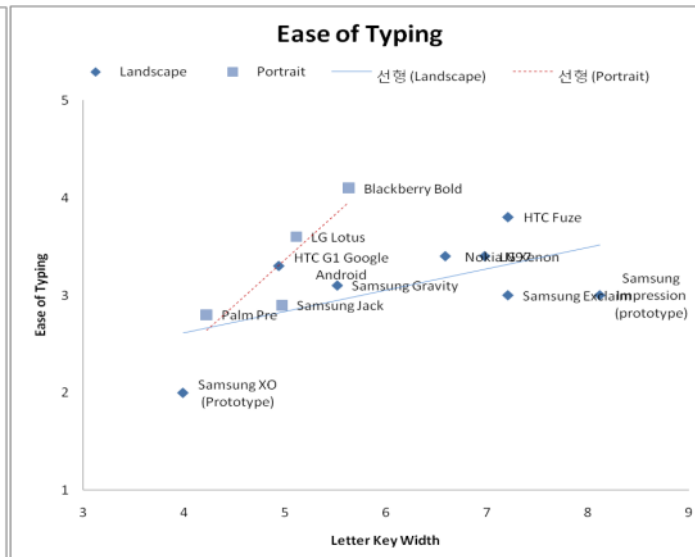
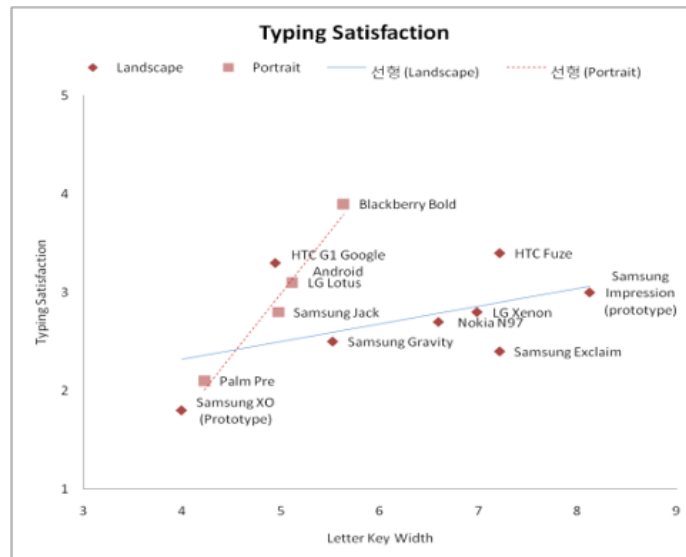


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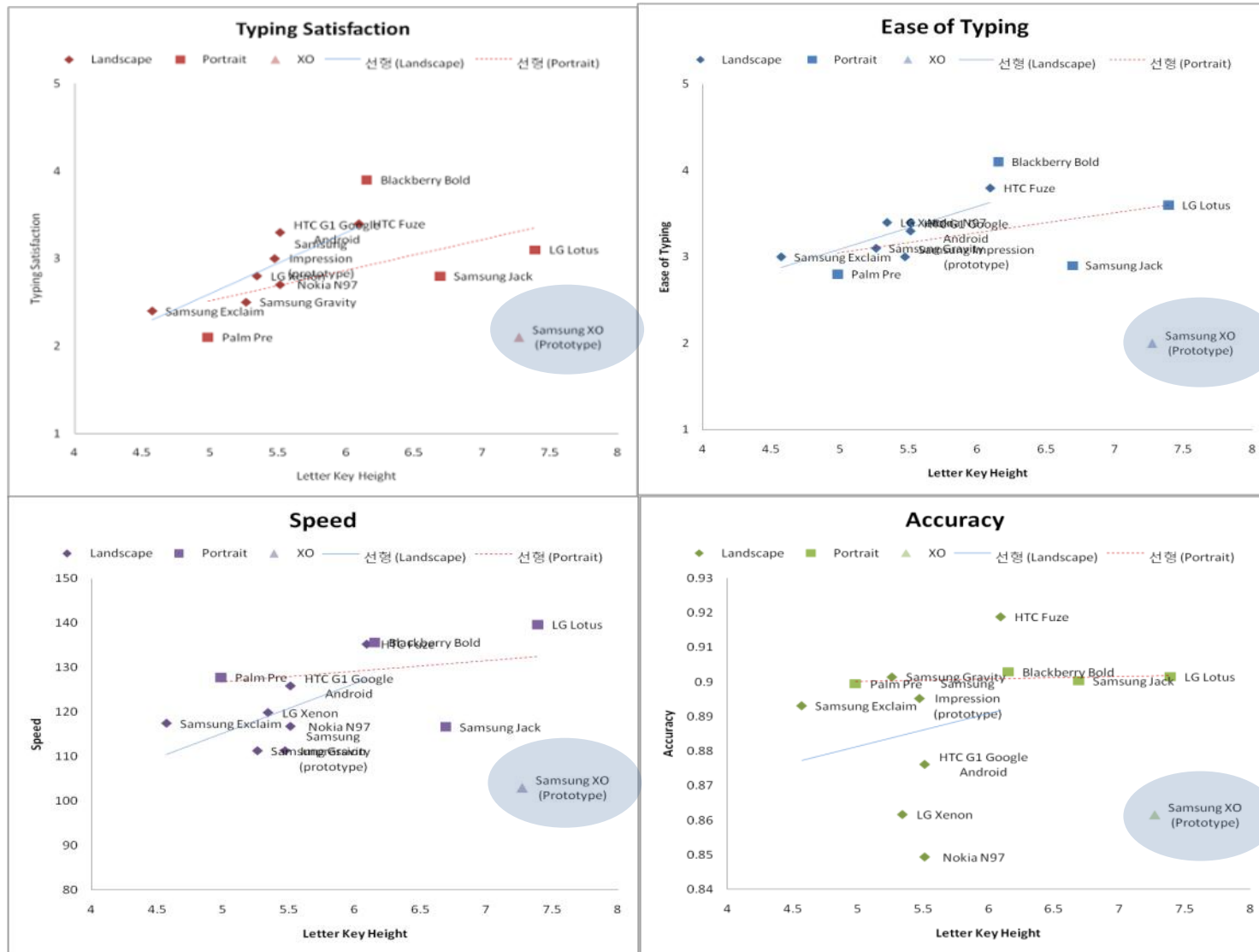
Tactile Ratio



Letter Key Width

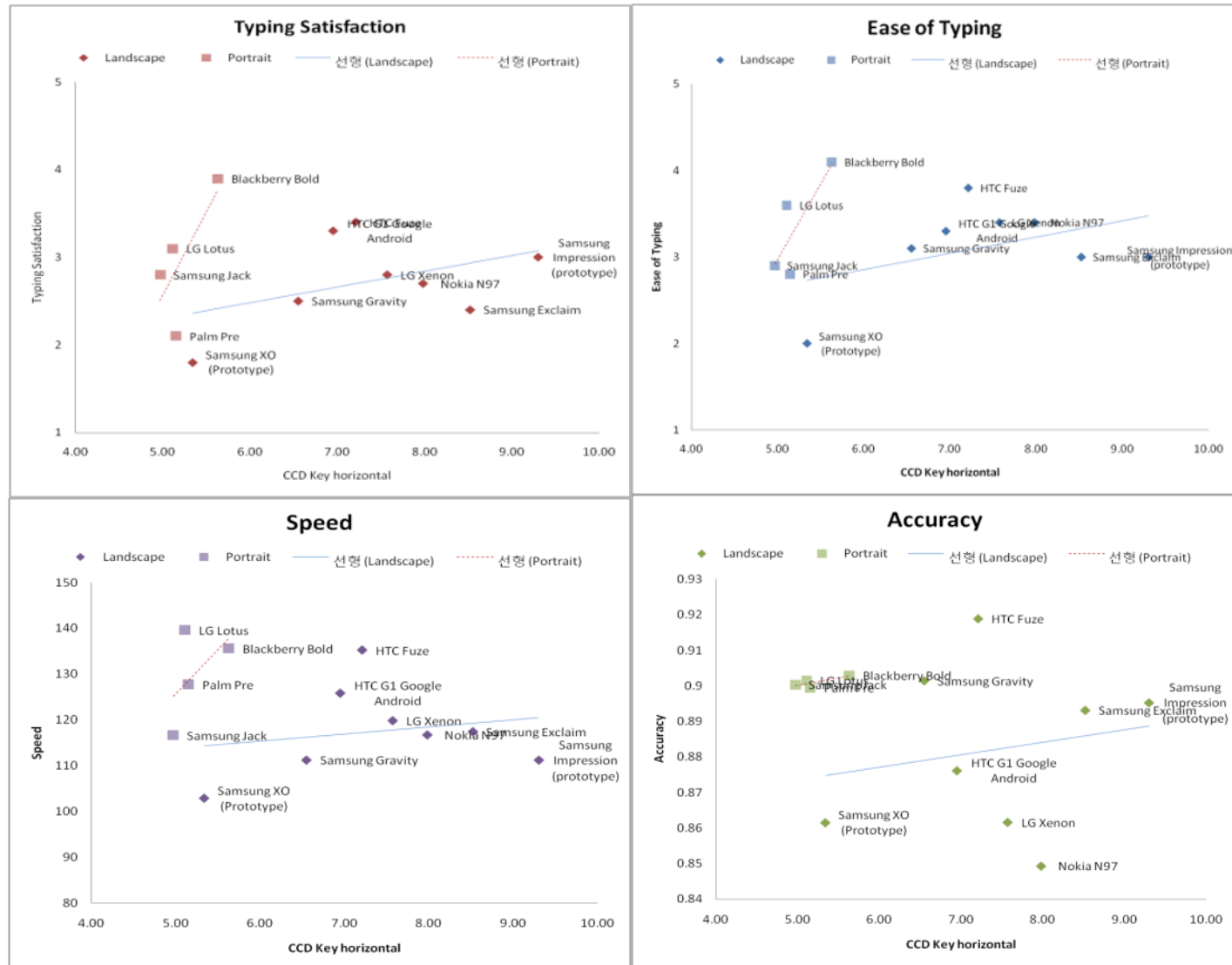


Letter Key Height



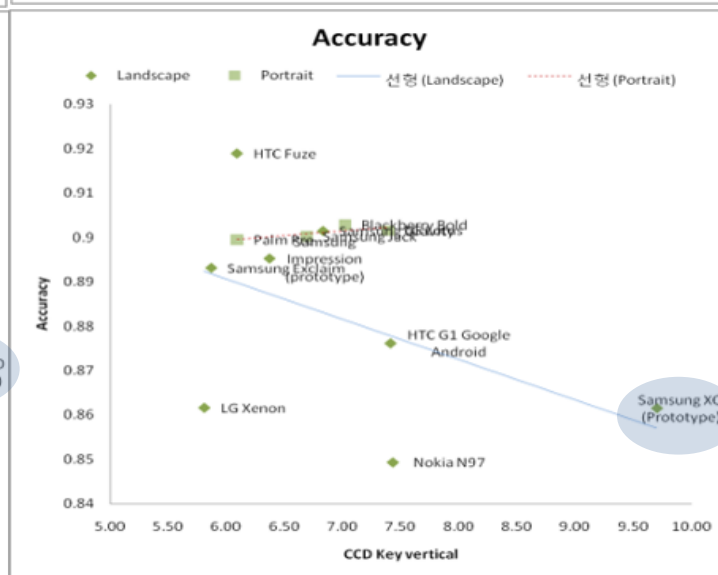
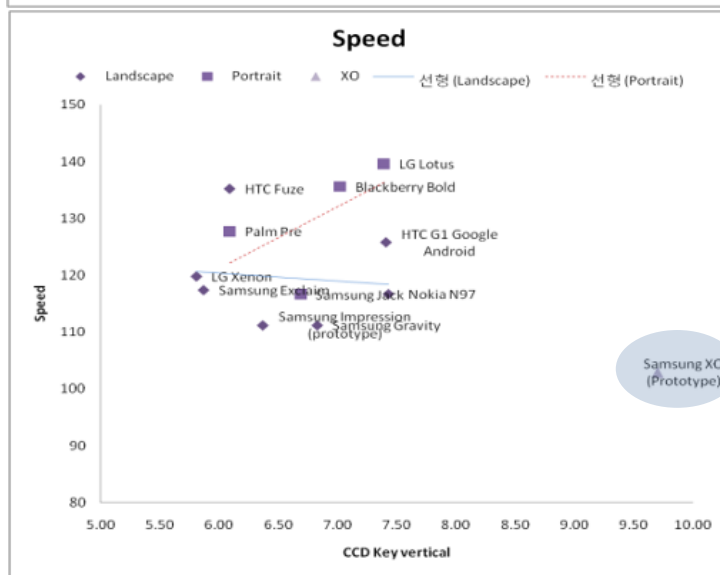
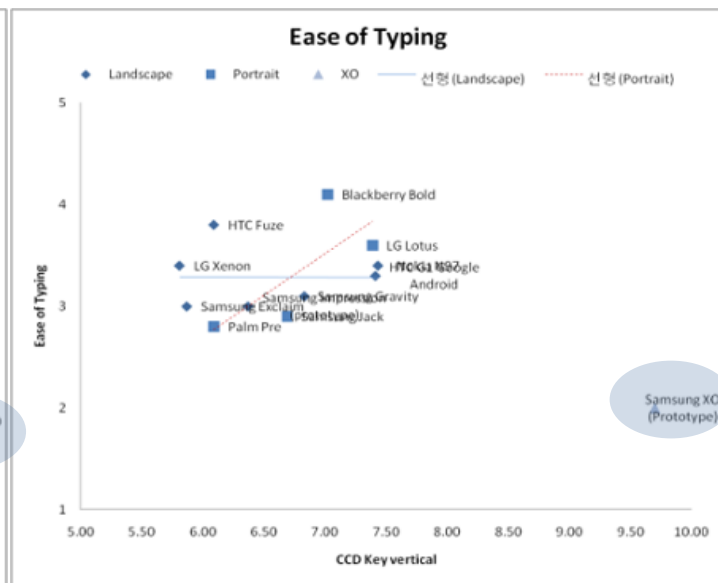
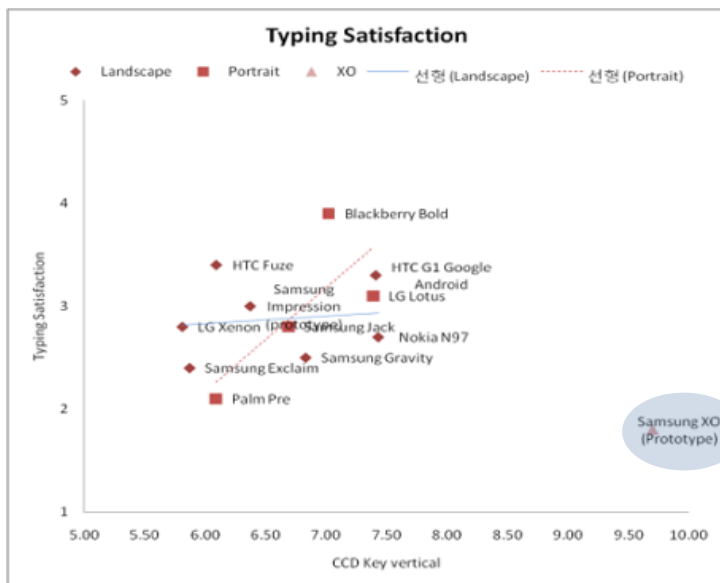
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CCD Key Horizontal



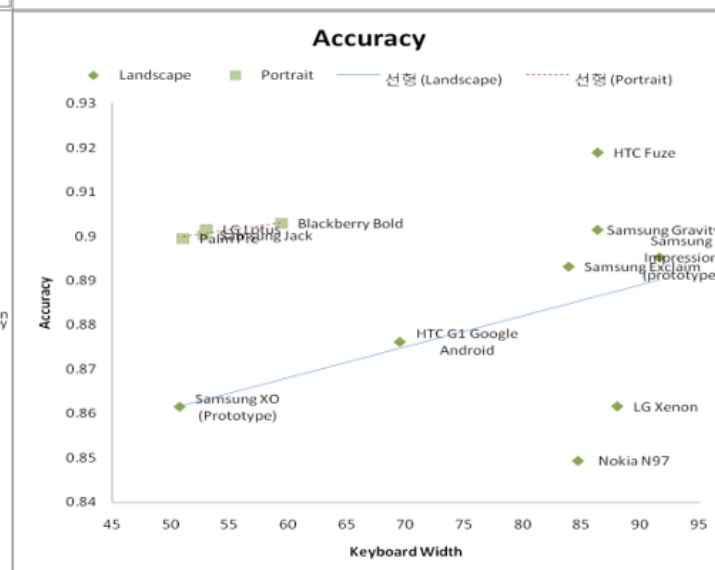
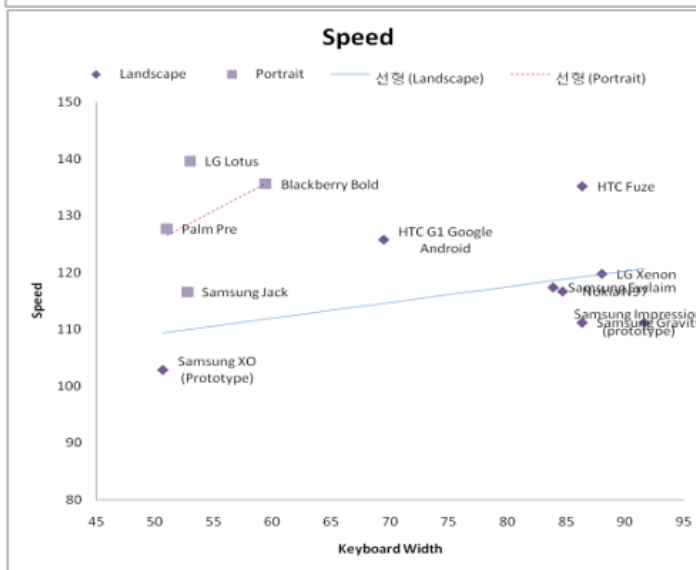
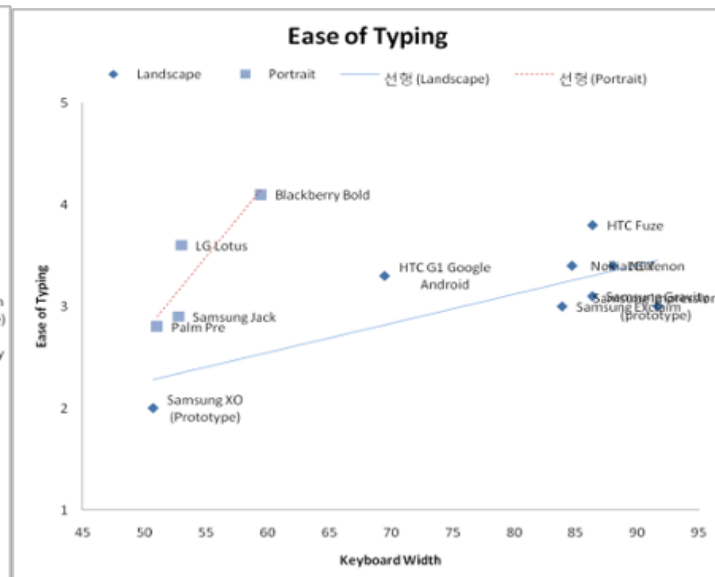
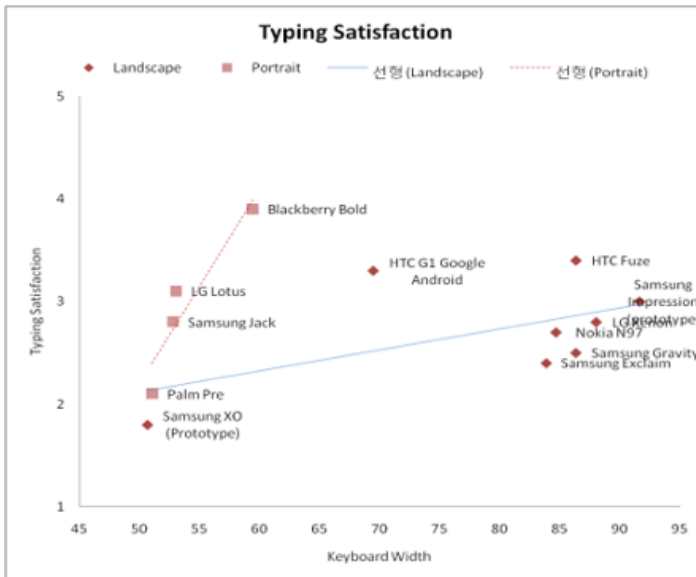
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CCD Key Vertical



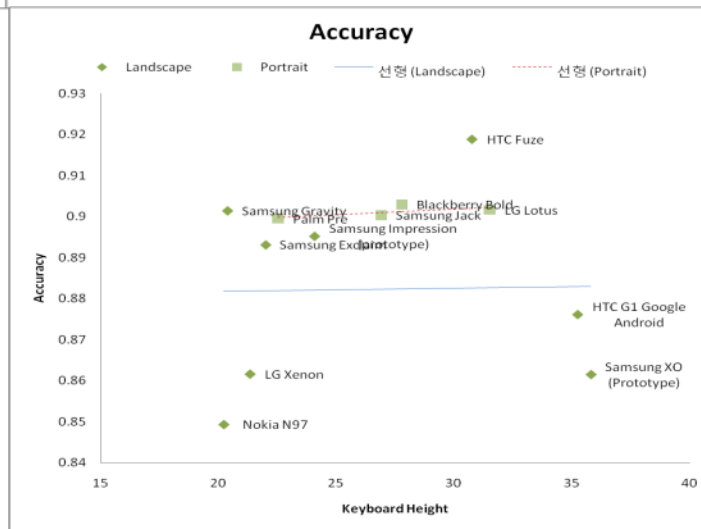
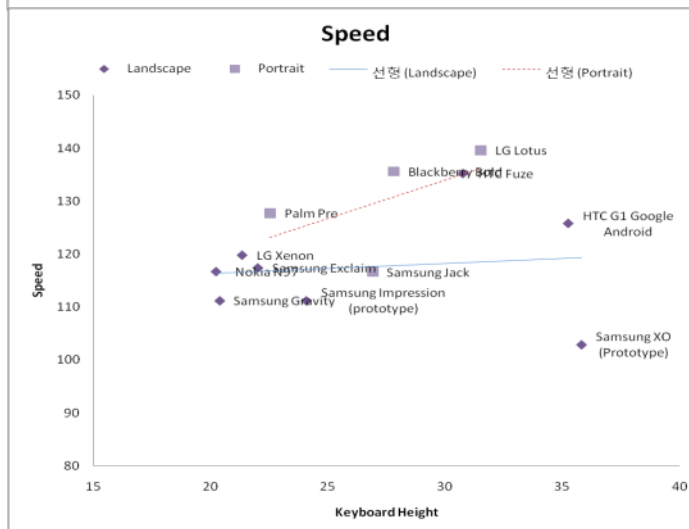
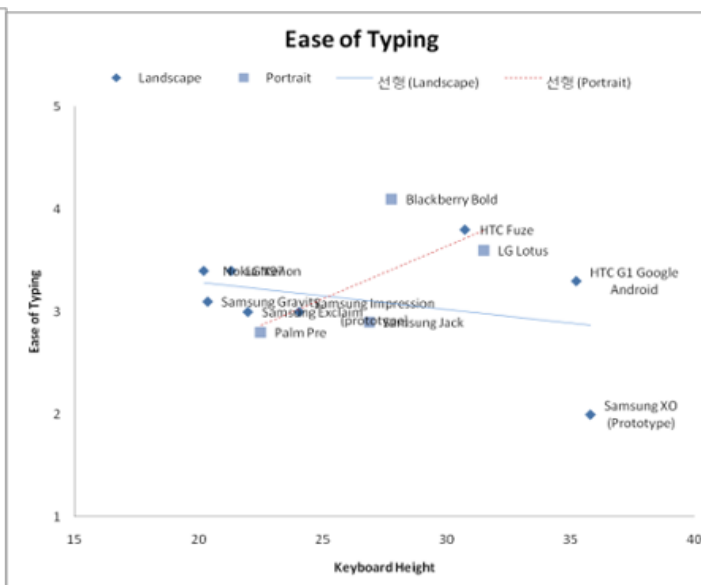
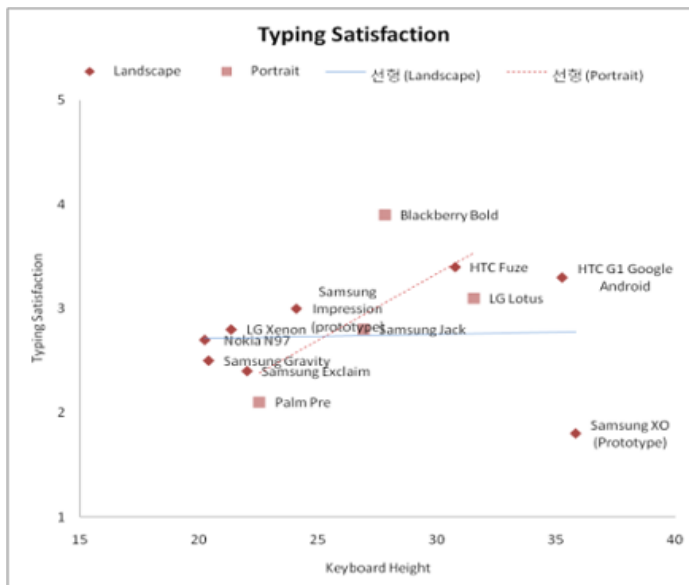
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Keyboard Width

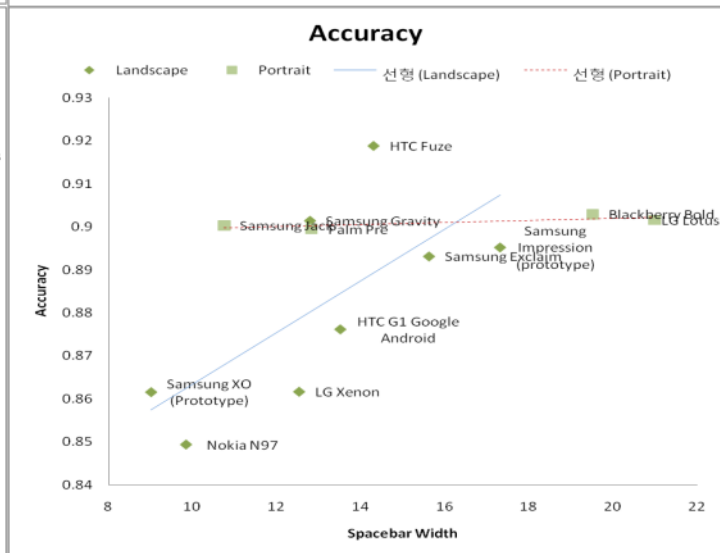
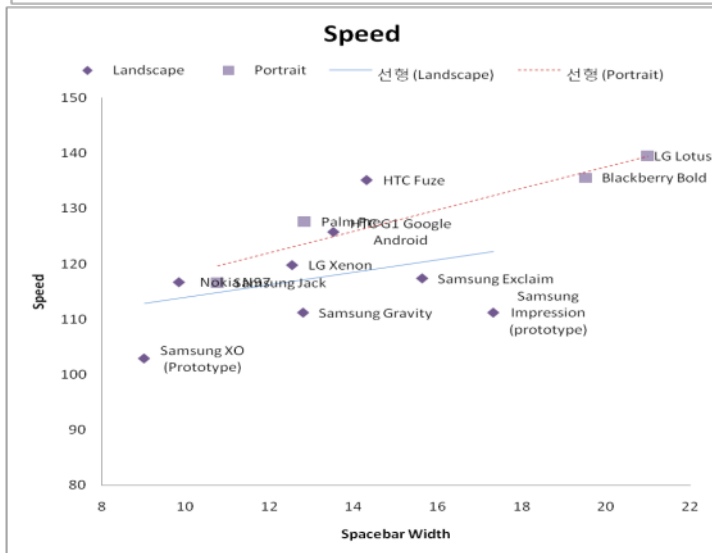
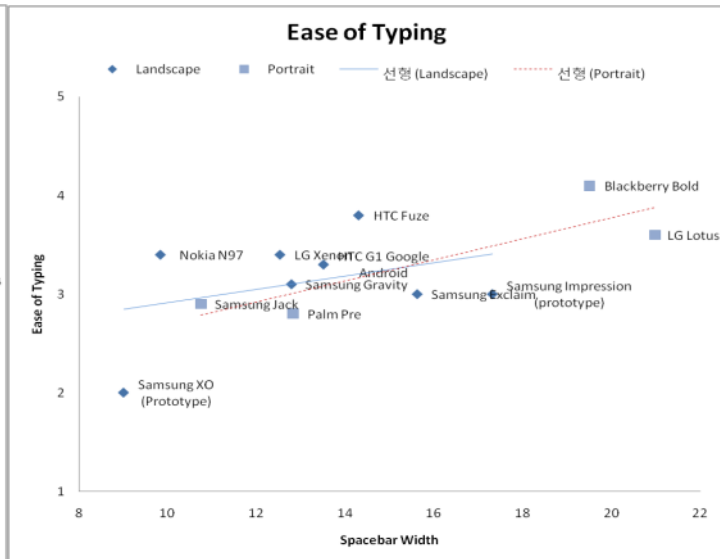
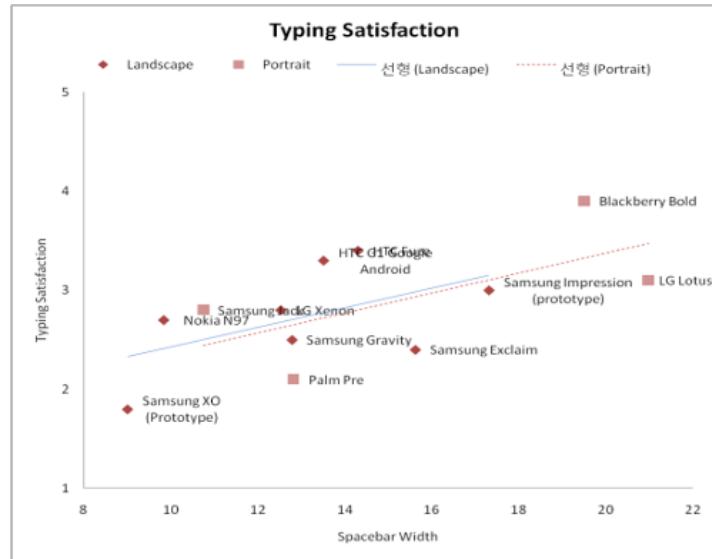


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Keyboard Height

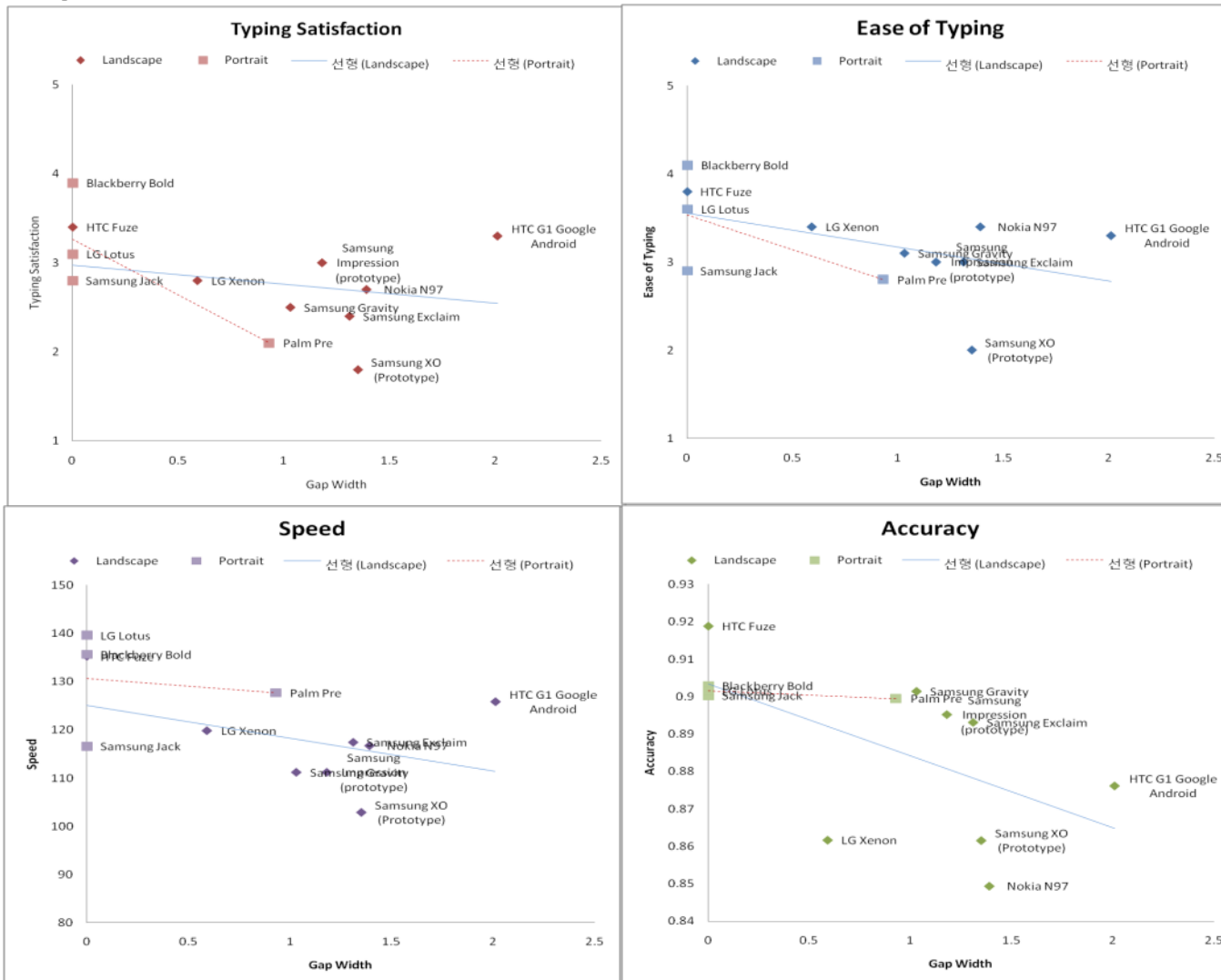


Spacebar Width



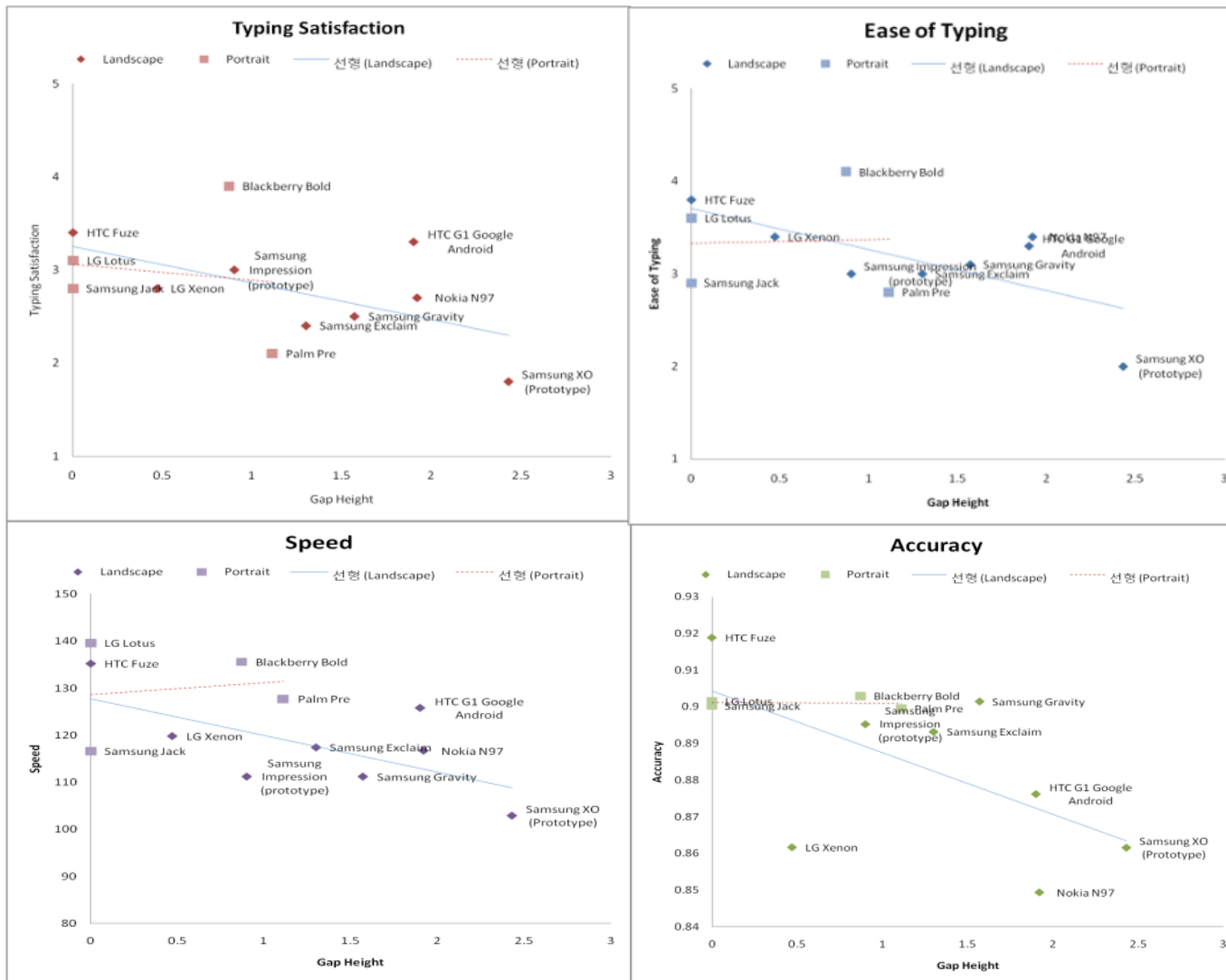
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Gap Width



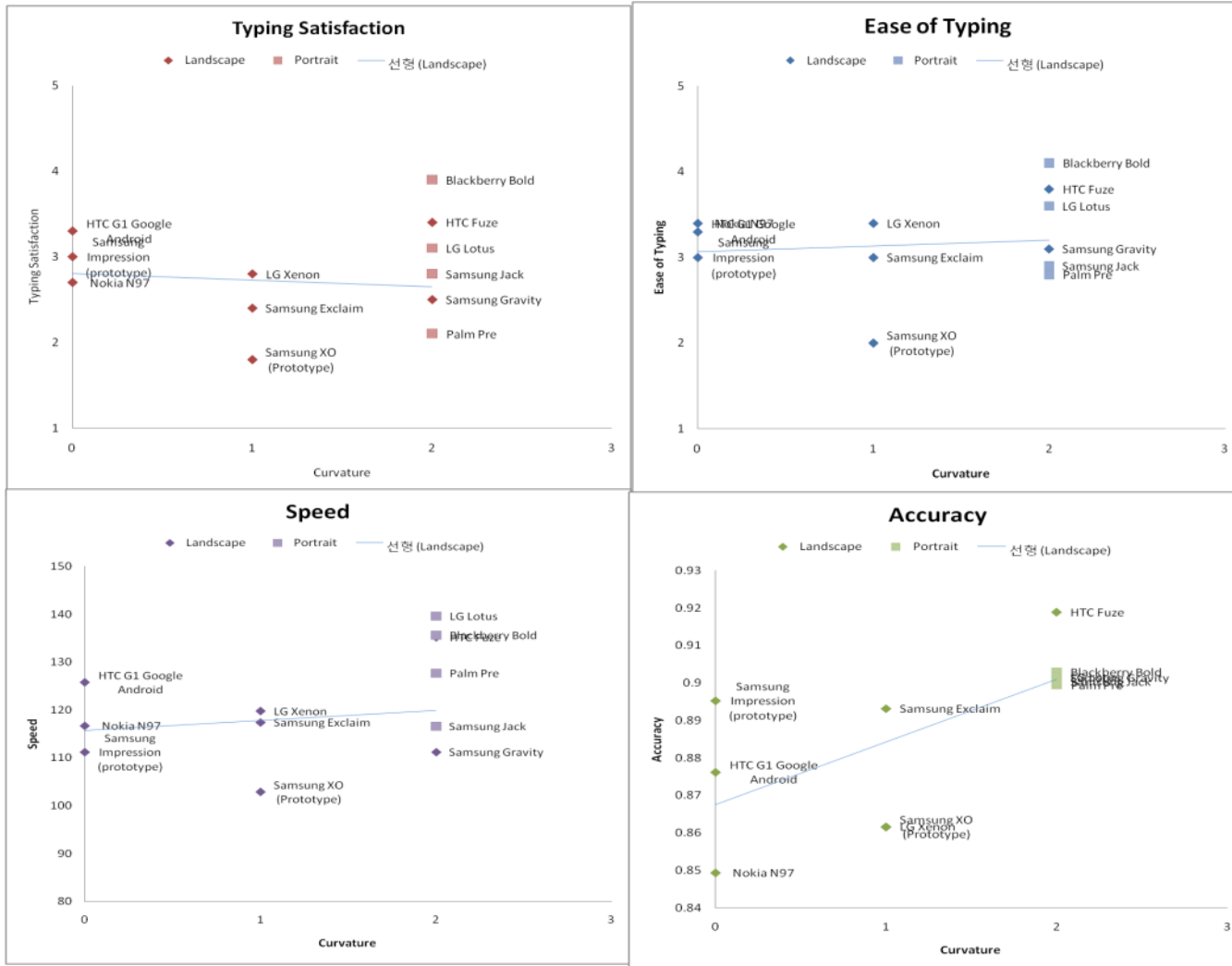
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Gap Height



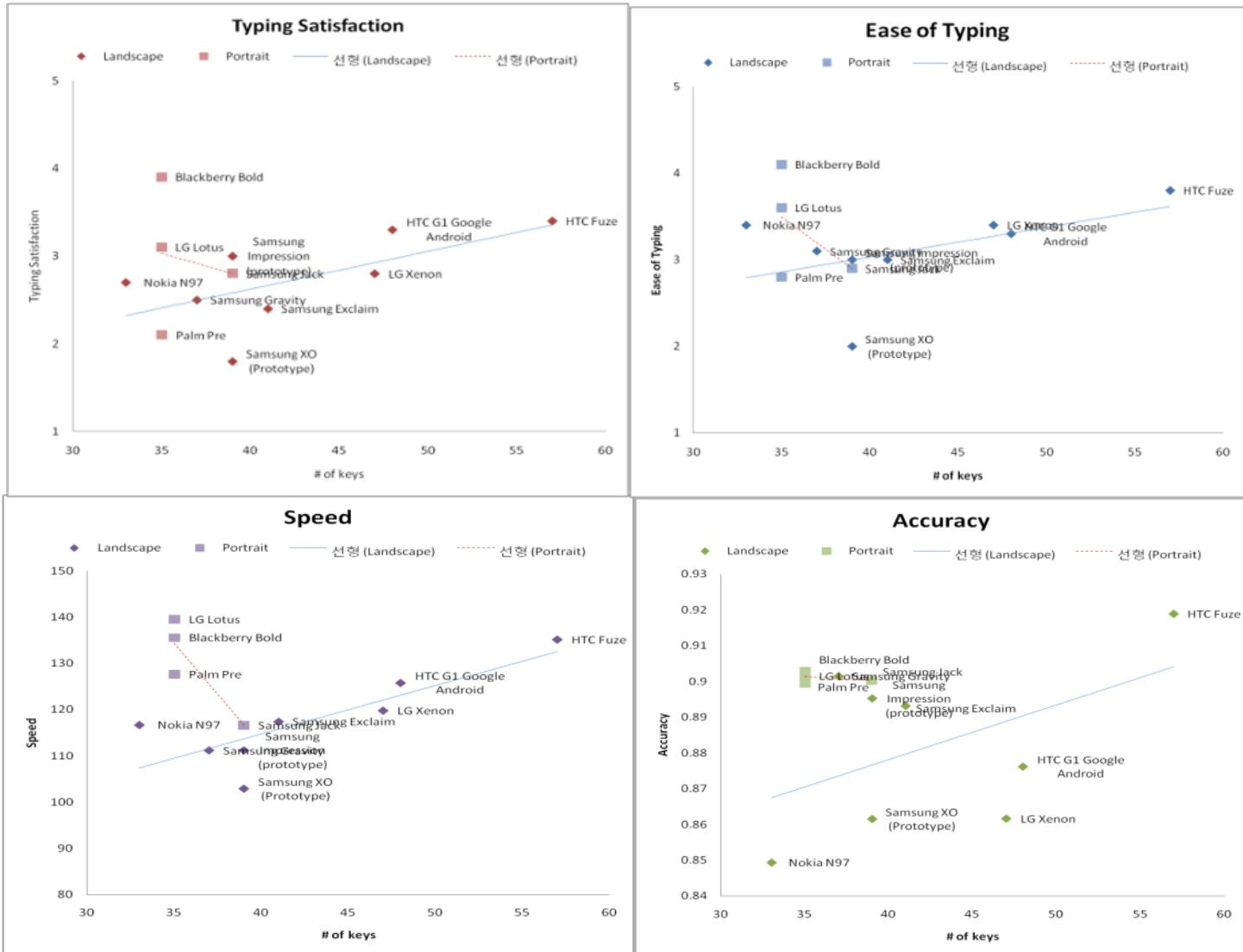
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Curvature



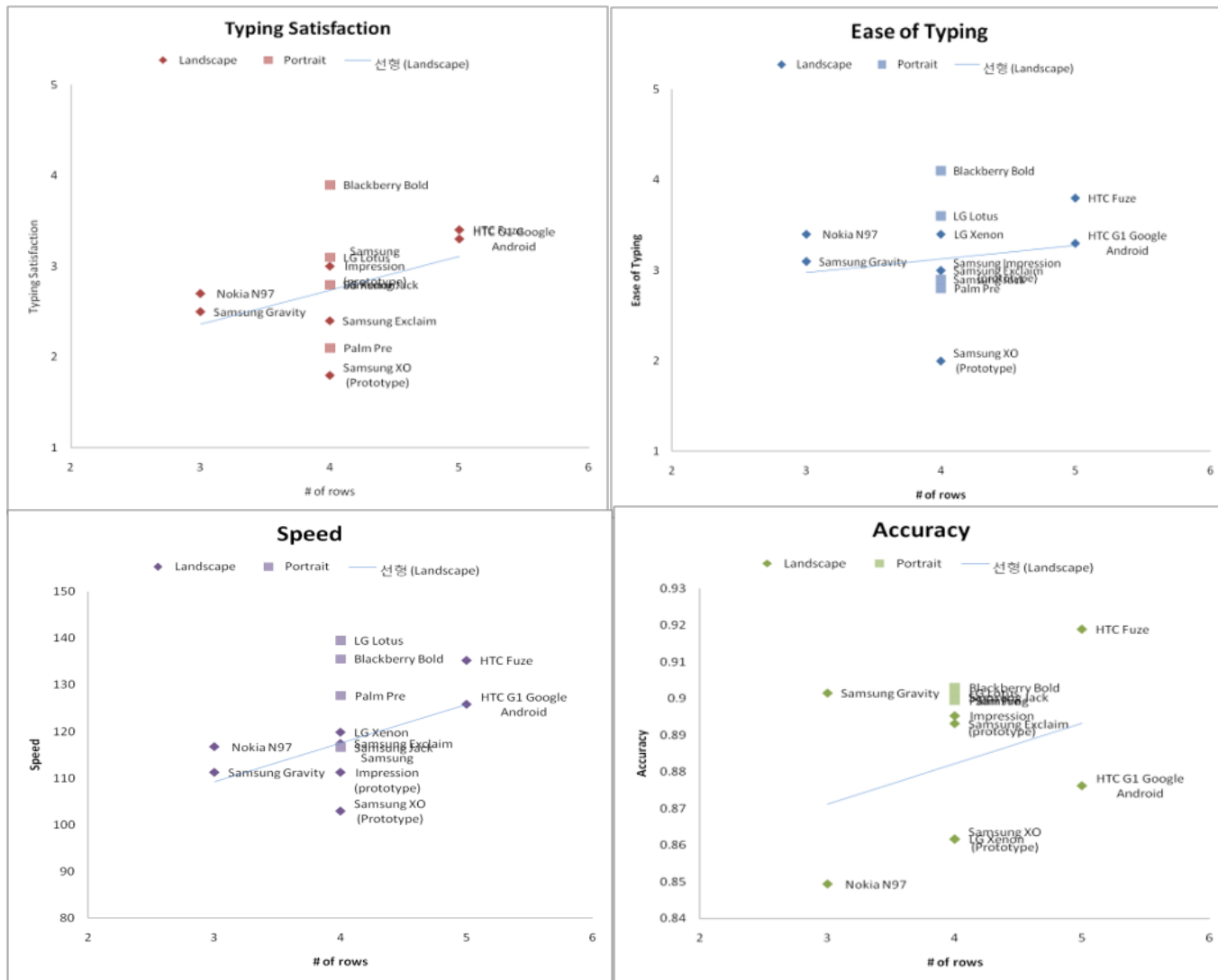
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of keys



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of rows



Comparing Different Analyses

Comparing Analyses

Multiple Regression, PLS Regression

- 40% of the variability in the average speed across all 12 phones is explained by the model
- A similar analysis was performed for typing accuracy, but none of the demographic factors were significant

Stepwise Regression Model used or PLS with/without Type	Multiple Regression (HC - SPSS) (VF < 4)					Default (type not used)				with Landscape & Portrait as Type				Default			
	7	2	6	10	10	2				2				3			
Model Quality (cum)						0.208				0.192				0.378			
R-square X (cum)						0.705				0.681				0.813			
R-square Y (cum)						0.461				0.506				0.63			
Adjusted R square	0.274	0.042	0.186	0.202	0.209												
Speed																	
Accuracy																	
Speed + Accuracy																	
Ease of Typing																	
Typing Satisfaction																	
Notes																	
	Beta																
FMax(grams)					-0.101	-0.027	0.101	-0.122	-0.107	-0.023	0.052	-0.097	-0.084	-0.13	0.144	-0.184	
Tallness(.001)																	
Travel(.001)	-0.427			-0.218		-0.165	0.018	-0.282	-0.258	-0.153	-0.025	-0.246	-0.224	-0.329	0.086	-0.381	
TactileSlope		0.211				0.046	0.142	-0.035	-0.025	0.05	0.094	-0.01	-0.003	-0.016	0.169	-0.074	
TactileRatio			-0.185	-0.327				-0.155	-0.141					-0.219	0.074	-0.235	
Letter Key Width								0.141	0.13	0.102	0.035	0.148	0.135	0.037	0.051	0.103	
Letter Key Height	-0.1		-0.14														
CCD Key horizontal																	
CCD Key vertical						-0.153	-0.11	-0.163	-0.155	-0.146	-0.1	-0.159	-0.149	-0.109	-0.129	-0.136	
Keyboard Width																	
Keyboard Height															0.105		
Spacebar Width	0.123		0.225	0.169	0.339	0.266	0.16	0.308	0.29	0.243	0.158	0.273	0.255	0.356	0.123	0.362	
Gap Width				-0.29	-0.141	-0.193	-0.164	-0.186	-0.178	-0.172	-0.142	-0.159	-0.151	-0.242	-0.143	-0.216	
Gap Height						-0.206	-0.174	-0.199	-0.191	-0.188	-0.151	-0.18	-0.17	-0.208	-0.173	-0.201	
Curvature	0.445		0.331			0.098	0.148	0.045	0.048	0.088	0.111	0.04	0.042	0.113	0.141	0.054	
# of keys					0.195									0.113	0.105		
# of rows	0.163			0.068		0.131	0.098	0.138	0.131	0.127	0.088	0.136	0.127	0.154	0.088	0.151	
Type Landscape										0.101	0.104						
Type Portrait										-0.101	-0.104						
Age Group					-0.105												
Typing Accuracy																	
Typing Speed				0.096													
Thumb Diameter																	
Thumb Length				-0.087													
Fingernail Length					0.052												
Has Portrait																	
Has Landscape																	
Has TouchScreen				-0.057													
Messaging Frequency				-0.129	-0.105												
Round in Study	0.228			0.159	0.054												
Gap width x Key CCD width		0.073		-0.091													
Gap width x Curvature				-0.239													
Curvature x Key CCD width																	
Curvature x Key Tallness	0.245				-0.076												
Key Surface Area																	
Index of Difficulty					0.162												
Index of Difficulty = Log2(A/W) + 2																	
A: amplitude of movement																	
W: target width																	

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Thank You!